Persistent Drift and Point Selection of Nesterov Flows over Semialgebraic Convex Valleys

Abstract

We study point selection and persistent drift for inertial gradient flows with damping α/t near valleys exactly homogeneous of degree $p > 2$ in the normal variables. Above $\alpha = 1+2/p$, actual compact selected confinement and uniform normal coercivity imply finite full physical length. At equality, scalar, multiplicative and separable normal actions enforce point selection in jointly convex valleys. A more general force-corrector theorem derives its integrating factor from phase volume and its effective convexity from Brunn–Minkowski; an explicit rotating nonseparable quartic satisfies the hypotheses. Equality in volume-root concavity forces the entire frozen potential to be constant along the corresponding selected segment, identifying the geometry of flat effective minima. On excited zero-residual-energy tails, an energy-normalized joint-measure limit retains conditional normal invariance, velocity stationarity, a variance contribution to dissipation, and transported first-integral constraints. Singleton invariant force means give qualitative closure without a corrector; joint convexity then forces attraction to the effective minimizer set and excludes reciprocal-logarithmic rescaled energy decay. These necessary laws are not realization statements. In contrast, for one sufficiently large finite even p and every $m \geq 1$, we construct a fixed coercive convex semialgebraic C^2 objective on $\mathbb{R}^{3m}$ with globally Lipschitz gradient and a compact minimizer box. Every continuum in a fixed open minimizer sub-box is the omega-limit set of a regular cold trajectory, varying only m transverse initial coordinates. The same three-dimensional objective has a nonconvergent regular cold trajectory with exactly zero residual rescaled normal energy and mechanical energy of order $1/(t^2 \log t)$. Its proof combines energy-normalized coding, two slow clocks and an adaptive mesh. The constructions retain singular-time cold access and an exact global convex extension. The results distinguish strict excess damping, constrained critical transport and accessible drift under explicit tail hypotheses.

Keywords: Nesterov flow; vanishing damping; semialgebraic convex optimization; homogeneous valleys; adiabatic action; omega-limit sets; cold initial data; effective transport; phase volume; zero residual energy

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1 Introduction and main results

Consider the inertial gradient equation

$$X''(t) + \frac{\alpha}{t}X'(t) + \nabla f(X(t)) = 0, \qquad t > 0. \tag{1.1}$$

The continuous-time model of accelerated gradient dynamics introduced by Su, Boyd and Candès [16] has led to extensive work on objective decay and trajectory convergence. These are distinct questions when the minimizer set is not a singleton. A trajectory

may approach that set while continuing to move along it. The damping, the geometry of the normal restoring force, and the ability of the normal motion to transfer energy into tangential displacement all enter the selection problem.

This paper compares point selection and persistent drift in valleys that are exactly homogeneous of degree $p > 2$ in their normal variables. Strict excess damping above $\alpha = 1+2/p$ makes the actual selected force absolutely integrable in the displacement equation. At equality, scalar actions can still force point selection, even when the frozen force hull is not a singleton. A different critical normal dynamics, with accessible transitions between periodic branches, permits persistent drift and arbitrary connected positional limit sets. Such drift can also persist while the residual rescaled normal energy tends exactly to zero. A normalized transport law describes necessary limits in that regime, while phase-volume geometry and verified correctors give a broader structural selection mechanism.

Our strongest global conclusion concerns regular data imposed at the singular time. A *regular cold solution* is a function $X \in C^2([0,\infty);\mathbb{R}^n)$ satisfying (1.1) for $t > 0$ and

$$X(0) = X_0, \qquad X'(0) = 0. \tag{1.2}$$

We write $\omega(X)$ for its positional omega-limit set, and call a nonempty compact connected set a *continuum*, including a singleton. A function is semialgebraic if its graph is a finite Boolean combination of polynomial equalities and inequalities with real coefficients.

Theorem 1 (One fixed objective realizes every continuum in a sub-box)**.** *There are one finite even integer $p > 4$, a damping exponent*

$$\alpha = 1 + \frac{2}{p} \in (1, 3/2),$$

a nondegenerate compact interval I, and a nonempty open interval $J \Subset \operatorname{int} I$, such that for every integer $m \geq 1$ there is a fixed function $F_m : \mathbb{R}^{3m} \to \mathbb{R}$ with the following properties:

1. *F_m is nonnegative, coercive, convex, semialgebraic and C^2, and $\operatorname{Lip}(\nabla F_m) \leq 1$.*
2. *Writing $X = (s, q) \in \mathbb{R}^m \times \mathbb{R}^{2m}$, its minimizer set is exactly $I^m \times \{0\}$.*
3. *For every continuum $K \subset J^m$, some regular cold solution of*

$$X'' + \frac{\alpha}{t} X' + \nabla F_m(X) = 0$$

is bounded and satisfies

$$q(t) \to 0, \qquad \omega(X) = K \times \{0\}. \tag{1.3}$$

The objective, p, α, both boxes, and the cold family are fixed before K is chosen. More precisely, there are fixed $s_a, q_* > 0$ such that all selected cold coordinates equal s_a, and, in each two-dimensional normal block,

$$q_j(0) = (q_*, \zeta_j).$$

Only the m transverse cold parameters $\zeta_1, \dots, \zeta_m$ vary with K. The same p and α work simultaneously for every m.

The degree is chosen by a finite steep-potential persistence argument and remains existential. No limiting objective or infinite series is used. The theorem does not assert drift for every prescribed degree or every damping coefficient in an interval. Semialgebraicity concerns the fixed objective; neither a target continuum nor its encoding cold datum must be semialgebraic. When $m = 1$, the theorem yields a nonconvergent bounded cold orbit with an interval of minimizing limits.

For the same three-dimensional objective F_1, Theorem 11 constructs a further regular cold trajectory with a nondegenerate interval omega-limit set and zero residual rescaled energy. With the exact variables of Section 2, it satisfies

$$H(\sigma) \longrightarrow 0, \qquad E(\sigma) \asymp (1+\sigma)^{-1}, \qquad \sigma = \log t.$$

Its physical mechanical energy is therefore of order $1/(t^2 \log t)$, despite failure of point selection. This is an interval realization at zero residual energy, not an assertion that every continuum in Theorem 1 is realized with zero residual energy.

1.1 The local selection results

The positive results concern an exact homogeneous model

$$X = (s, q) \in U \times \mathbb{R}^n, \qquad P(s, \lambda q) = \lambda^p P(s, q), \quad \lambda \geq 0.$$

Their common hypotheses are an actual compact interior selected tail and uniform positive normal coercivity:

$$s(t) \in K_0 \Subset U, \qquad P(s, q) \geq c_0 |q|^p$$

on a neighborhood of K_0. These conditions are assumptions on the actual trajectory. If the original objective has this form only in a local chart, validity of that chart on the tail must also be established.

Theorem 2 proves that $\alpha > 1 + 2/p$ implies automatic rescaled compactness, absolute integrability of the selected force with physical weight t, point convergence, and finite *full physical length*. Its proof uses neither convexity nor definability. A positive completed-square energy gives explicit decay and length-tail estimates. The numerical boundary $1 + 2/p$ already appears in convex flatness theory [2]; our argument identifies the direct normal dissipation and virial mechanism under the stated exact-tail assumptions.

At $\alpha = 1+2/p$, the limiting normal invariant measures constrain the force. Theorem 3 shows that every selected cluster point admits a zero frozen mean force. The conclusion is necessary, not a claim that arbitrary invariant measures can be followed. Theorem 4 proves point convergence in three jointly convex classes: one scalar normal coordinate, a multiplicative normal shape in any dimension, and finitely many separable scalar normal

modes of common degree. Absolute small-action estimates show that every excited mode retains positive limiting action. Bounded correctors then reduce the selected equation to a damped system with a convex fixed-action energy, including the case of nonunique effective minimizers.

The perspective potential (5.24) has a force hull containing zero in its interior at every selected position, yet all its confined critical trajectories converge. Its separate actions prevent the changing force mixtures that the full hull suggests. Conversely, Proposition 5.2 shows that positive residual normal energy entails infinite physical normal length at equality. Point convergence and finite length therefore separate even within the positive classes.

Section 6 treats the zero-energy boundary directly. On every excited confined critical tail, the selected logarithmic velocity satisfies $k = O(H)$. If the limiting energy is zero, this gives a reciprocal-logarithmic lower bound on H, forces infinite selected length and makes the absolute rescaled selected-force integral diverge. A verified force corrector and an exact positive integrating factor instead force every excited tail to retain positive residual energy. The scalar quantity in that criterion is defined by its formula; no canonical action for arbitrary nonintegrable normal dynamics is assumed.

1.2 Effective transport and phase-volume structure

Theorem 7 extends the structural selection result beyond the scalar and fixed-shape classes. A bounded homogeneous force corrector automatically determines its mean field through enclosed phase volume. Joint convexity makes the nth root of unit-energy phase volume concave, where n is the normal configuration dimension, and the corresponding fixed-volume energy convex. The existing corrected-log and corrected-distance estimates then yield positive residual energy and point selection even with nonunique effective minimizers. Example 1 verifies the full hypothesis for a smooth semialgebraic rotating quartic whose frozen shape is not orthogonally separable and whose parameter dependence is not multiplicative in fixed coordinates. The conclusion remains conditional on actual compact selected confinement.

Theorem 8 treats excited tails with zero residual energy without a corrector. The relative bound $k = O(H)$ makes the normalized velocity $v = k/E$ bounded and the energy clock $\mathrm{d}\tau = E\,\mathrm{d}\sigma$ unbounded. Every sequence of energy-time translates has a joint velocity–normal-state measure limit. The normal conditional probabilities are frozen invariant; the velocity marginal satisfies a stationary damped-force equation. The exact limiting logarithmic-energy loss contains the second velocity moment, including a possible variance term. Every smooth frozen first integral has a transported limiting level, which restricts the accessible invariant-force hull. The separable perspective example recovers its fixed-action gradient restriction in this language. These results are necessary outer limits; they do not construct an orbit from a chosen measure path.

Common invariant force means suffice for qualitative single-valued closure in Corollary 8.1, but not for the corrector estimates of Theorem 7. Under joint convexity, every complete compact translated gradient trajectory is constant at an effective minimizer. Corollary 8.2 therefore gives minimizer-set attraction, $v \to 0$, $\sigma E(\sigma) \to \infty$ and $\tau(\sigma)/\log\sigma \to \infty$. It excludes the reciprocal energy scale of the cold drift construction in this common-mean class without proving positive residual energy or general point selection. Theorem 9 adds a geometric restriction: equality in volume-root concavity along any selected segment forces the frozen potential itself to be constant on that segment. Its tangential degeneracy on a flat minimizing set does not, without an additional estimate, make the off-set tangential force integrable.

1.3 Construction and relation to earlier work

The global realization uses the local potential

$$P(s,q) = (1 + Cs^2)V(q) + \eta s\big(T(q) - cV(q)\big), \qquad C = K\eta, \tag{1.4}$$

with one selected coordinate and two normal coordinates. A symmetric quartic perturbation of an ellipse has a transverse homoclinic billiard orbit by the result of Delshams, Fedorov and Ramírez-Ros [9]. The steep-potential persistence theorem of Rapoport, Rom-Kedar and Turaev [13] gives a corresponding network for one fixed homogeneous semialgebraic normal potential $V = g^p$. A nonnegative polynomial T separates the means of an axial and a nonaxial periodic branch. A small positive s^2V term retains joint convexity.

At the critical scaling

$$a = 2/p, \qquad \rho = t^a/a, \qquad \sigma = \log t, \qquad q = t^{-a}Z,$$

the normal equation is Hamiltonian and the total rescaled energy satisfies

$$\frac{\mathrm{d}}{\mathrm{d}\sigma}\left(\frac{|Z_\rho|^2}{2} + P(s,Z) + \frac{|s_\sigma|^2}{2}\right) = -a|s_\sigma|^2.$$

The classical action formulas and branch-switching framework in Brännström–Gelfreich [6] and Brännström–De Simone–Gelfreich [7] are the natural precedent for this transport mechanism. In particular, geometric shadowing of continuous curves of arbitrary length is already part of that theory. Here the selected momentum is dissipative, the small parameter decreases with time, the tracking error must tend to zero, and the orbit must arise from zero-velocity data at the singular endpoint. Those requirements are proved directly.

The energy transport results of [10, 12] provide further adiabatic precedents. In [10], infinite-horizon transport explicitly requires uniform cross-form and return-time estimates. The zero-energy argument here verifies such uniformity after normalizing the total rescaled energy, retaining its logarithm and both the physical and target clocks. Its

shrinking tracking tube yields exact energy exhaustion through repeated motion. These comparisons concern the stated mechanisms and hypotheses, not a priority claim.

The coding argument keeps the actual logarithmic clock when a future word changes. Its local differentiability uses two exponential weights. A nonautonomous inclination estimate transports the one-parameter transverse cold curve across every required future leaf, with the stable boundary retained in the half-line memory estimates. A finite inward maneuver reaches a region of strict opposite forces. Feedback then follows arbitrary sufficiently slow targets at finite squared logarithmic speed. An exact all-time cold normal bound allows one convex semialgebraic extension to preserve the entire preparation and every retained target orbit. Independent copies follow the coordinates of one vector target, and finite differences between their entry clocks disappear because that target's speed tends to zero.

The invariant-measure limit framework is classical [4, 17]; its use here requires the shrinking-energy normalization and the distinct generator cancellations for conditional invariance and selected-velocity stationarity. Standard differential-inclusion limit-set theory [18] does not provide converse realization or selection of one point in a continuum. Likewise, phase-volume adiabatic invariance [19, 20] and convex-volume inequalities, including the marginal theorem of Prékopa [21], are established ingredients. The present phase-volume selection theorem is an exact infinite-horizon critical-flow specialization with an explicit differentiable-corrector assumption; it is not inferred from an ergodic finite-interval approximation.

The broader convergence background includes constant-damping convex dynamics [3], asymptotically vanishing damping and flat minimizing intervals [8], general-convex trajectory convergence above $\alpha = 3$ [11], and subcritical objective estimates [1]. Cabot, Engler and Gadat [8] already discuss billiard-like difficulties in higher-dimensional flat regions; the scalar full-state result in that paper is different from a scalar normal mode coupled to selected coordinates. The present results give a unified comparison within the exact homogeneous setting rather than a classification of all convex semialgebraic flows.

1.4 Organization and conventions

Section 2 establishes the exact equations for arbitrary damping. Sections 3–5 prove the finite-length and critical action results. Section 6 gives the relative-energy and integrating-factor results. Section 7 identifies the phase-volume geometry and proves corrector selection, including the rotating example. Section 8 proves the joint-measure energy-time law and its transport and singleton consequences. Section 9 establishes flat-volume rigidity. Section 10 constructs continuum targets with arbitrarily small dissipation. Sections 11–16 construct the accessible convex normal network, the moving-clock code, and its regular cold target orbits. Sections 17–18 globalize once and prove Theorem 1. Section 19 proves the zero-residual-energy cold interval construction.

Unindexed constants in estimates may change from line to line. The coefficient $C = K\eta$ in (1.4) is fixed. The positive-energy coding uses compact parameter rectangles. In Section 19, the actual energy may vanish; its normalized normal-energy band remains compact and positive, and all required chart bounds are proved uniformly along the negative logarithmic-energy half-line. A claim about a compact selected tail never implies entry into that tail from arbitrary initial data.

2 Exact normal scaling and the damping surplus

Let $U \subset \mathbb{R}^d$ be open and let $P \in C^2(U \times \mathbb{R}^n)$ satisfy

$$P(s, \lambda q) = \lambda^p P(s, q), \qquad \lambda \geq 0, \quad p > 2. \tag{2.1}$$

We consider a classical solution $X = (s, q)$ on $[t_0, \infty)$, $t_0 > 0$, of

$$s_{tt} + \frac{\alpha}{t} s_t + P_s(s, q) = 0, \qquad q_{tt} + \frac{\alpha}{t} q_t + P_q(s, q) = 0. \tag{2.2}$$

The standing *confined-tail assumptions* in Sections 3–9 are that there is a compact set $K_0 \Subset U$, a neighborhood N of K_0, and $c_0 > 0$ such that

$$s(t) \in K_0 \quad (t \geq t_0), \qquad P(s, q) \geq c_0 |q|^p \quad (s \in N,\ q \in \mathbb{R}^n). \tag{2.3}$$

For a potential agreeing with this model only in a local chart, the actual trajectory must stay in that chart on the specified tail. We do not deduce capture from (2.3). Compactness and homogeneity give uniform constants with

$$c_0 |Z|^p \leq P(s, Z) \leq C_0 |Z|^p, \qquad |P_s(s, Z)| \leq C_1 |Z|^p \leq C_f P(s, Z) \quad (s \in K_0). \tag{2.4}$$

Indeed, the restrictions of P and P_s to the compact unit normal sphere are bounded, and both have degree p.

Set

$$\begin{aligned} &a = \frac{2}{p}, \quad b = \alpha - 1, \quad \Delta = b - a, \quad \rho = \frac{t^a}{a}, \quad \sigma = \log t, \quad \varepsilon = t^{-a} = \frac{1}{a\rho}, \\ &q = \varepsilon Z, \quad W = Z_\rho, \quad k = s_\sigma, \qquad H = \frac{|W|^2}{2} + P(s, Z), \quad E = H + \frac{|k|^2}{2}. \end{aligned} \tag{2.5}$$

The parameter Δ is the damping surplus. The word *critical* below refers to $\Delta = 0$ relative to the normal homogeneity degree; it does not refer to the usual general-convex threshold $\alpha = 3$.

Proposition 2.1 (Exact transformed equations)**.** For constant α, system (2.2) is equivalent to

$$\begin{aligned} &Z_\rho = W, \quad W_\rho = -P_Z - \Delta\varepsilon W + a\Delta\varepsilon^2 Z, \\ &s_\sigma = k, \qquad k_\sigma = -bk - P_s. \end{aligned} \tag{2.6}$$

Here and below unlabelled arguments of P and its derivatives are (s, Z). The energy identities are

$$H_\sigma = k \cdot P_s - \Delta|W|^2 + a\Delta\varepsilon Z \cdot W, \qquad E_\sigma = -b|k|^2 - \Delta|W|^2 + a\Delta\varepsilon Z \cdot W. \tag{2.7}$$

If $B = \varepsilon Z \cdot W$ and $C_2 = \varepsilon^2|Z|^2$, then

$$B_\sigma = |W|^2 - pP - bB + a\Delta C_2, \qquad (C_2)_\sigma = -2aC_2 + 2B. \tag{2.8}$$

For $b \neq 0$, the compensated position satisfies

$$z = s + \frac{k}{b}, \qquad z_\sigma = -\frac{P_s}{b}. \tag{2.9}$$

Proof. Since $\rho_t = t^{a-1}$, direct differentiation gives

$$\begin{aligned} s_t &= t^{-1}k, & s_{tt} &= t^{-2}(k_\sigma - k), \\ q_t &= t^{-1}(W - a\varepsilon Z), & q_{tt} &= t^{-2}\big[\varepsilon^{-1}W_\rho - (1+a)W + a(1+a)\varepsilon Z\big]. \end{aligned}$$

Homogeneity and $ap = 2$ give $P_s(s, q) = t^{-2}P_s(s, Z)$ and $P_q(s, q) = t^{-2}\varepsilon^{-1}P_Z(s, Z)$. Substitution proves (2.6). Since $\rho_\sigma = \varepsilon^{-1}$, differentiating H cancels the Hamiltonian work and gives the first identity in (2.7); adding the selected kinetic derivative gives the second. Finally $Z \cdot P_Z = pP$ and $\varepsilon_\sigma = -a\varepsilon$ give (2.8). The selected equation proves (2.9) directly. ∎

The physical energy is related to these variables by

$$E_{\text{phys}} := \frac{|X_t|^2}{2} + P(s, q) = t^{-2}\left[\frac{|k|^2}{2} + \frac{|W - a\varepsilon Z|^2}{2} + P(s, Z)\right]. \tag{2.10}$$

In particular, decay of a potential gap alone cannot be substituted for a full physical-energy or velocity estimate.

3 Strict excess damping and finite physical length

Theorem 2 (Finite length above the homogeneous critical line)**.** *Assume* (2.1)–(2.3) *and* $\alpha > 1 + 2/p$. *Then the rescaled state* (s, Z, W, k) *is bounded,* $E \to 0$, *and*

$$\int_{t_0}^{\infty} tP(s(t), q(t))\,\mathrm{d}t < \infty, \qquad \int_{t_0}^{\infty} t|P_s(s(t), q(t))|\,\mathrm{d}t < \infty. \tag{3.1}$$

The complete physical tail has finite length and converges to $(s_\infty, 0)$ *for some* $s_\infty \in K_0$. *Put*

$$R = \min\left\{\frac{2p(\alpha - 1 - 2/p)}{p+2}, \frac{4}{p-2}\right\}. \tag{3.2}$$

For every $0 < r < R$,

$$E = O(t^{-r}), \qquad E_{\text{phys}} = O(t^{-2-r}), \qquad |X_t| = O(t^{-1-r/2}), \tag{3.3}$$

$$\int_t^\infty |X_\tau|\,\mathrm{d}\tau = O(t^{-r/2}), \qquad |q| = O(t^{-2/p-r/p}),$$
$$|s - s_\infty| + \int_t^\infty |s_\tau|\,\mathrm{d}\tau = O(t^{-r}), \qquad |s_t| = O(t^{-1-r}).$$

Neither convexity nor semialgebraicity is required.

3.1 Rescaled compactness and absolute force integrability

Proof of boundedness and (3.1)**.** Write $\nu = 1/2+1/p < 1$. Coercivity implies $|Z\cdot W| \le \sqrt{2}c_0^{-1/p}E^\nu$. Since $b, \Delta > 0$, (2.7) gives

$$E_\sigma \le A\varepsilon E^\nu, \qquad A = a\Delta\sqrt{2}c_0^{-1/p}.$$

Integration yields

$$E(\sigma)^{1-\nu} \le E(\sigma_0)^{1-\nu} + \frac{A(1-\nu)}{a}\varepsilon(\sigma_0). \tag{3.4}$$

At a zero of E, this follows by applying the calculation to $E+\eta$ and letting $\eta \downarrow 0$. Alternatively, $E = 0$ forces $Z = W = k = 0$, and uniqueness of the original equation makes the solution stationary. Thus E, Z, W, k are bounded; selected confinement was already assumed.

It follows that $B = O(\varepsilon)$ and $C_2 = O(\varepsilon^2)$ in (2.8). Both are integrable in σ. Define

$$\mathcal{R}(\sigma) = a\Delta\int_\sigma^\infty B(u)\,\mathrm{d}u, \qquad \mathcal{L} = E + \mathcal{R}.$$

Then $\mathcal{R} \to 0$, $\mathcal{L}$ is bounded below, and

$$\mathcal{L}_\sigma = -b|k|^2 - \Delta|W|^2.$$

Consequently E has a nonnegative limit and $k, W \in L^2(\,\mathrm{d}\sigma)$. Integrating the virial equation gives

$$\begin{aligned} p\int_\sigma^S P\,\mathrm{d}u = &\int_\sigma^S |W|^2\,\mathrm{d}u + B(\sigma) - B(S) \\ &- b\int_\sigma^S B\,\mathrm{d}u + a\Delta\int_\sigma^S C_2\,\mathrm{d}u. \end{aligned} \tag{3.5}$$

All terms on the right have finite limits as $S \to \infty$. Since $P \ge 0$, this proves $P \in L^1(\,\mathrm{d}\sigma)$. Equation (2.4) proves $P_s \in L^1(\,\mathrm{d}\sigma)$ absolutely. The change of variables $\mathrm{d}\sigma = \mathrm{d}t/t$ and homogeneity give (3.1).

Now $E = |k|^2/2 + |W|^2/2 + P$ is integrable and has a limit, so that limit is zero. In particular $Z, W, k \to 0$. No uniform continuity of the rapidly oscillating W is needed. Integrating the energy and virial identities to infinity also gives

$$\int_\sigma^\infty (b|k|^2 + \Delta|W|^2)\,\mathrm{d}u + \int_\sigma^\infty P\,\mathrm{d}u + \int_\sigma^\infty |P_s|\,\mathrm{d}u \le C\big(E(\sigma) + \varepsilon(\sigma)\big). \tag{3.6}$$

The constants absorb the positive fixed coefficients. ∎

The selected equation alone now proves finite selected length. Write $f = P_s(s, Z)$ along the rescaled trajectory. Variation of constants in $k_\sigma + bk = -f$ yields

$$\int_\sigma^\infty |k(u)|\,\mathrm{d}u \le \frac{1}{b}\left(|k(\sigma)| + \int_\sigma^\infty |f(u)|\,\mathrm{d}u\right). \tag{3.7}$$

Thus $s \to s_\infty \in K_0$ and (2.9) gives the absolutely convergent formula

$$s_\infty = s(\sigma_0) + \frac{k(\sigma_0)}{b} - \frac{1}{b}\int_{\sigma_0}^\infty P_s\,\mathrm{d}\sigma. \tag{3.8}$$

Normal convergence follows from $q = \varepsilon Z$. The next estimate controls the full normal length as well.

3.2 A positive completed-square energy

Proof of the rates and full length in Theorem 2. Put

$$\theta = \frac{2\Delta}{p+2}, \qquad \beta = p\theta = 2(\Delta - \theta), \qquad Y = W + \theta\varepsilon Z,$$

and introduce the nonnegative energy

$$\mathcal{F} = \frac{|k|^2}{2} + P(s, Z) + \frac{|Y|^2}{2} = E + \theta B + \frac{\theta^2}{2}C_2. \tag{3.9}$$

The square is essential for positivity. The transformed equations give

$$Z_\sigma = \varepsilon^{-1}Y - \theta Z, \qquad Y_\sigma = -\varepsilon^{-1}P_Z - (\Delta - \theta)Y + (\Delta - \theta)(a + \theta)\varepsilon Z.$$

The selected and normal work terms cancel upon differentiating (3.9). Since $p\theta = 2(\Delta - \theta)$,

$$\mathcal{F}_\sigma = -\beta\mathcal{F} - \frac{2b - \beta}{2}|k|^2 + (\Delta - \theta)(a + \theta)\varepsilon Z \cdot Y. \tag{3.10}$$

Here $2b - \beta = 2(a + \theta) > 0$. Coercivity therefore gives

$$\mathcal{F}_\sigma \le -\beta\mathcal{F} + C\varepsilon\mathcal{F}^\nu. \tag{3.11}$$

Let

$$\eta_0 = 1 - \nu = \frac{p-2}{2p}, \qquad \kappa = \eta_0\beta, \qquad \delta_0 = \frac{a}{\eta_0} = \frac{4}{p-2}.$$

For a nonstationary solution $\mathcal{F} > 0$ at every finite time: a zero would force $k = Z = Y = 0$, hence the stationary state by uniqueness. Multiplication of (3.11) by $\eta_0\mathcal{F}^{\eta_0 - 1}$ and integration give, with $\tau = \sigma - \sigma_0$,

$$\mathcal{F}(\sigma)^{\eta_0} \le e^{-\kappa\tau}\mathcal{F}(\sigma_0)^{\eta_0} + C\eta_0\varepsilon(\sigma_0)\int_0^\tau e^{-\kappa(\tau - v)}e^{-av}\,\mathrm{d}v. \tag{3.12}$$

The last integral is $(e^{-a\tau} - e^{-\kappa\tau})/(\kappa - a)$ if $\kappa \neq a$, and $\tau e^{-a\tau}$ otherwise. Thus

$$\mathcal{F} = \begin{cases} O(t^{-\min\{\beta,\delta_0\}}), & \beta \neq \delta_0, \\ O\big(t^{-\delta_0}(\log t)^{1/\eta_0}\big), & \beta = \delta_0. \end{cases} \tag{3.13}$$

Moreover

$$E \leq 2\mathcal{F} + C\varepsilon^2 \mathcal{F}^{2/p}. \tag{3.14}$$

For $R = \min\{\beta, \delta_0\}$, $2a + 2R/p \geq R$, so E satisfies the same endpoint bounds. In particular $E = O(t^{-r})$ for each $0 < r < R$.

The exact identity $Y = tq_t + (a + \theta)q$, or (2.10), gives

$$|X_t| \leq Ct^{-1}\mathcal{F}^{1/2} + Ct^{-1-a}\mathcal{F}^{1/p}.$$

Because $r < \delta_0$ implies $a + r/p > r/2$, this proves $|X_t| = O(t^{-1-r/2})$, $E_{\text{phys}} = O(t^{-2-r})$, and the full length-tail bound in (3.3). Coercivity gives $|q| = O(t^{-a-r/p})$.

Finally $f = P_s = O(t^{-r})$. The identity

$$\beta - b = \frac{(p-2)b - 4}{p+2}$$

shows $R \leq b$: if $b \leq \delta_0$, then $\beta \leq b$; if $b \geq \delta_0$, then $\delta_0 \leq b$. Hence $r < b$, and variation of constants gives $k = O(t^{-r})$. Integration yields the stronger selected rates in (3.3). ∎

The equality $\beta = \delta_0$ occurs at $\alpha = (p+2)/(p-2)$, a rate crossover distinct from $1 + 2/p$. The logarithmic factor in (3.13) is an elementary upper bound; its optimality is not asserted. All length statements concern the specified positive-time tail. A regular cold solution has finite length on every earlier bounded interval.

3.3 Relation to convex flatness estimates

For $x_* = (s_*, 0)$ in the uniformly coercive selected neighborhood,

$$\langle \nabla P(s,q), (s - s_*, q)\rangle = pP + (s - s_*) \cdot P_s \geq \big(p - C_f|s - s_*|\big)P. \tag{3.15}$$

Thus every $\gamma < p$ satisfies the local flatness inequality $P \leq \langle \nabla P, x - x_*\rangle/\gamma$ sufficiently near x_*. The threshold $\alpha > 1 + 2/\gamma$ and its convex trajectory-convergence context are already recorded by Aujol, Dossal and Rondepierre [2, Definition 2.1 and Section 3]. Whenever the complete hypotheses of the applicable convex theorem hold, one may choose $\gamma < p$ to recover the strict numerical boundary $\alpha > 1 + 2/p$. Theorem 2 supplies a direct force-integrability and full-length proof under exact homogeneity and actual selected confinement, and does not require convexity.

Su, Boyd and Candès [16, Theorem 7] obtain objective estimates under a powered-objective convexity hypothesis, which for $|x|^p$ also reflects this boundary. May [11] proves trajectory convergence and weighted energy integrability for general convex potentials

when $\alpha > 3$. Under additional geometric assumptions, flat-minimum rate and finite-length results in [2, 14] involve a stronger damping range. These comparisons concern their stated hypotheses; a local inequality such as (3.15) neither proves capture into a chart nor replaces those hypotheses.

4 Critical mean forces and invariant measures

Throughout this section $\alpha = 1 + a$, so $\Delta = 0$. Equations (2.6)–(2.7) reduce to

$$Z_\rho = W, \quad W_\rho = -P_Z, \quad s_\sigma = k, \quad k_\sigma = -ak - P_s, \quad H_\sigma = k \cdot P_s, \quad E_\sigma = -a|k|^2. \tag{4.1}$$

Thus the normal system is Hamiltonian at fixed s, while the selected motion dissipates the total rescaled energy.

Proposition 4.1 (Compact-tail consequences and the exact force criterion)**.** Under (2.3), the rescaled trajectory is bounded and

$$E \to E_\infty \geq 0, \qquad \int_{\sigma_0}^{\infty} |k|^2 \, \mathrm{d}\sigma < \infty, \qquad k \to 0, \quad H \to E_\infty, \quad q \to 0. \tag{4.2}$$

Selected point convergence is equivalent to convergence of the vector improper integral

$$\int_{\sigma_0}^{\infty} P_s(s(\sigma), Z(\sigma)) \, \mathrm{d}\sigma = \int_{t_0}^{\infty} t P_s(s(t), q(t)) \, \mathrm{d}t. \tag{4.3}$$

Absolute convergence of this integral implies finite selected length.

Proof. The nonnegative energy in (4.1) is nonincreasing. Coercivity bounds Z, W, k, and its dissipation identity gives $k \in L^2$. Since $k_\sigma = -ak - P_s$ is bounded, k is uniformly continuous. A uniformly continuous square-integrable function tends to zero: otherwise a sequence of disjoint intervals of fixed positive length would contribute a fixed positive amount to its square integral. Hence $H = E - |k|^2/2 \to E_\infty$, and $q = \varepsilon Z \to 0$.

The exact compensated-position identity is

$$\left(s + \frac{k}{a}\right)_\sigma = -\frac{P_s}{a}, \qquad s(S) - s(\sigma_0) = \frac{k(\sigma_0) - k(S)}{a} - \frac{1}{a}\int_{\sigma_0}^{S} P_s \, \mathrm{d}u. \tag{4.4}$$

Since $k(S) \to 0$, this proves the equivalence. The physical expression follows by homogeneity. If $P_s \in L^1$, variation of constants as in (3.7) gives $k \in L^1$. ∎

The integral in (4.3) may converge conditionally. A bounded primitive need not converge, and $k \in L^2$ need not imply $k \in L^1$. Also, Proposition 4.1 does not assert $k_\sigma \to 0$.

4.1 The frozen force set

For s in the coercive neighborhood N and $H \geq 0$, define

$$\begin{aligned} \Sigma_{s,H} &= \{(Z,W) : |W|^2/2 + P(s,Z) = H\}, \\ \mathcal{M}(s,H) &= \{\text{invariant Borel probability measures on } \Sigma_{s,H}\}, \\ \mathcal{G}(s,H) &= \left\{ \int P_s(s,Z)\,\mathrm{d}\mu : \mu \in \mathcal{M}(s,H) \right\}. \end{aligned} \tag{4.5}$$

The generator of the frozen Hamiltonian vector field is

$$\mathcal{A}_s\phi = W \cdot \phi_Z - P_Z \cdot \phi_W.$$

Coercivity makes the energy shells compact and the flow on them complete. Weak limits of orbital averages give invariant measures. A measure is invariant if and only if $\int \mathcal{A}_s\phi\,\mathrm{d}\mu = 0$ for every C^1 test function. For the converse, apply the generator identity to ϕ composed with the frozen flow and differentiate its integral.

Proposition 4.2. The sets $\mathcal{M}(s,H)$ and $\mathcal{G}(s,H)$ are nonempty, compact and convex. Their graphs are closed on compact subsets of $N \times [0,\infty)$. For $H > 0$,

$$\mathcal{G}(s,H) = H\mathcal{G}(s,1), \tag{4.6}$$

and every $\mu \in \mathcal{M}(s,H)$ satisfies

$$\int P\,\mathrm{d}\mu = \frac{2H}{p+2}, \qquad \int |W|^2\,\mathrm{d}\mu = \frac{2pH}{p+2}. \tag{4.7}$$

At $H = 0$, $\mathcal{M}(s,0) = \{\delta_{(0,0)}\}$ and $\mathcal{G}(s,0) = \{0\}$.

Proof. Weak compactness, the energy-support constraint, and the generator identities prove compactness and closed graphs; convexity is immediate. The dilation $(Z,W) \mapsto (H^{1/p}Z, H^{1/2}W)$ conjugates the unit-energy flow to the energy-H flow after a constant time change. It bijects invariant measures, and P_s scales by H, proving (4.6). Finally

$$\mathcal{A}_s(Z \cdot W) = |W|^2 - pP = 2H - (p+2)P.$$

Its invariant mean is zero, which proves (4.7). ∎

Homogeneity therefore fixes the mean potential energy, but does not generally fix the mean selected force. On a fixed compact energy shell, all invariant measures have the same force mean $\bar{f}$ if and only if the frozen orbit averages of P_s converge to $\bar{f}$ uniformly over initial phase points. Indeed, failed uniform convergence would yield a sequence of long occupation measures whose invariant weak limit has a different mean; conversely, integrate the uniform convergence against any invariant measure. This is a statement about the force observable and does not require unique ergodicity. It supplies no rate and, by itself, no infinite-time integrable averaging error.

4.2 Actual occupation measures

Proposition 4.3 (An outer slow limit)**.** Consider solutions on a fixed slow interval $[0, T]$ with a common compact rescaled range and with fast equations

$$X_j' = \varepsilon_j^{-1} e^{a\tau}(W_j, -P_Z(s_j, Z_j)), \qquad X_j = (Z_j, W_j), \quad \varepsilon_j \downarrow 0,$$

while $s_j' = k_j$, $k_j' = -ak_j - P_s(s_j, Z_j)$ and $H_j' = k_j \cdot P_s(s_j, Z_j)$. A subsequence of (s_j, k_j, H_j) converges uniformly to (s, k, H). For almost every τ there is $\mu_\tau \in \mathcal{M}(s(\tau), H(\tau))$ such that, with $\bar{f}(\tau) = \int P_s \, \mathrm{d}\mu_\tau$,

$$s' = k, \qquad k' = -ak - \bar{f}, \qquad H' = k \cdot \bar{f}, \qquad \bar{f} \in \mathcal{G}(s, H). \tag{4.8}$$

The limiting total energy still satisfies $(H + |k|^2/2)' = -a|k|^2$.

Proof. The slow variables are uniformly Lipschitz, so Arzelà–Ascoli gives their uniform convergence. Weak compactness and disintegration of $\mathrm{d}\tau \, \delta_{X_j(\tau)}$ give $\mathrm{d}\tau \, \mu_\tau$. The convergent slow energy imposes the energy-support constraint. For a C^1 phase test ϕ and a smooth compactly supported scalar test ψ,

$$\int \psi \mathcal{A}_{s_j} \phi(X_j) \, \mathrm{d}\tau = -\varepsilon_j \int (\psi' - a\psi) e^{-a\tau} \phi(X_j) \, \mathrm{d}\tau \longrightarrow 0.$$

Uniform convergence of s_j permits passage to the frozen generator. A countable dense family of tests proves invariance almost everywhere, simultaneously for all tests. Passing to the weak integral forms of the slow equations proves (4.8); its energy identity follows by cancellation. ∎

Invariant-measure descriptions of nonuniquely averaged slow dynamics have a substantial history; see Artstein and Vigodner [4]. Proposition 4.3 is only the compact finite-window statement proved above. It neither realizes arbitrary selections from $\mathcal{G}$ nor supplies cold accessibility.

Theorem 3 (Necessary zero mean force)**.** *Every selected cluster point s_* of a confined critical tail satisfies*

$$0 \in \mathcal{G}(s_*, E_\infty). \tag{4.9}$$

If $E_\infty > 0$ and the set $\{s \in K_0 : 0 \in \mathcal{G}(s, 1)\}$ is totally disconnected, the trajectory converges to a point.

Proof. Choose $\sigma_j \to \infty$ with $s(\sigma_j) \to s_*$ and fix $h > 0$. Since $k \to 0$, $s \to s_*$ uniformly on these translated windows, and $H \to E_\infty$ there. Let

$$\mu_j = \frac{1}{h} \int_{\sigma_j}^{\sigma_j + h} \delta_{(Z(u), W(u))} \, \mathrm{d}u.$$

A subsequence converges weakly. Its mean force tends to zero because

$$\frac{1}{h}\int_{\sigma_j}^{\sigma_j+h} P_s\,\mathrm{d}u = -\frac{k(\sigma_j+h)-k(\sigma_j)}{h} - \frac{a[s(\sigma_j+h)-s(\sigma_j)]}{h} \longrightarrow 0.$$

For every phase test ϕ,

$$\int_{\sigma_j}^{\sigma_j+h} \mathcal{A}_{s(u)}\phi(X(u))\,\mathrm{d}u = [\varepsilon\phi(X)]_{\sigma_j}^{\sigma_j+h} + a\int_{\sigma_j}^{\sigma_j+h} \varepsilon\phi(X)\,\mathrm{d}u = O(\varepsilon(\sigma_j)).$$

Replacing $s(u)$ by s_* costs $o(1)$. Each weak limit is therefore invariant at (s_*, E_∞) and has zero mean force, proving (4.9).

A bounded continuous trajectory has a nonempty compact connected omega-limit set: it is the nested intersection of the compact connected closures of its tail images. If $E_\infty > 0$, (4.6) puts this set in the stated zero-force locus. A connected subset of a totally disconnected set is a singleton. Since $q \to 0$, full positional convergence follows. ∎

At zero residual energy the condition (4.9) is automatic. Even at positive energy, an entire interval of zero-force barycenters does not imply that the actual normal motion can switch between the corresponding invariant measures. Section 5.5 gives a convex homogeneous example of this obstruction.

5 Action constraints and critical point selection

We retain the exact critical equations and confined-tail hypotheses. The force set describes all frozen invariant means. A trajectory may preserve additional quantities that prevent it from exploring that entire set. We first prove a bounded-corrector criterion and then verify it, including arbitrarily small actions, in three convex classes.

5.1 Periodic action identities and a bounded force corrector

For a differentiable parameter family of periodic frozen orbits $L_c(s,H)$, with period T_c, the unnormalized action is

$$J_c(s,H) = \oint_{L_c(s,H)} W\cdot\,\mathrm{d}Z.$$

Its classical derivative formulas are

$$(J_c)_H = T_c, \qquad (J_c)_s = -\int_0^{T_c} P_s(s,Z_c(r))\,\mathrm{d}r, \qquad \bar{f}_c = -\frac{(J_c)_s}{(J_c)_H}. \tag{5.1}$$

To see this, parametrize the orbit on a fixed circle. Variation of $\oint W\,\mathrm{d}Z$ and integration by parts give the integral of $\delta W\cdot Z_\rho - W_\rho\cdot\delta Z$. Hamilton's equations and variation of the energy constraint identify this with $\delta H - P_s\cdot\delta s$, integrated over one period. Periodicity cancels the boundary terms. Homogeneity gives

$$\nu = \frac{1}{2}+\frac{1}{p}, \qquad J_c(s,H) = H^\nu A_c(s), \qquad \bar{f}_c(s,H) = -\frac{H}{\nu}\nabla\log A_c(s). \tag{5.2}$$

The branch index matters. A multidimensional energy shell need not be a single periodic orbit, and its branches can have different action coefficients. These formulas and periodic-branch drift are classical; see [6, 7].

Proposition 5.1 (Infinite-time corrected reduction)**.** Suppose $E_\infty > 0$. On a neighborhood of the resulting compact positive-energy region, let $I(s, H) > 0$ be C^2 with $I_H > 0$, and let $h(s, I)$ be its inverse in energy, defined on a product neighborhood of the selected set and the relevant action values. Set

$$\bar{f}(s, H) = -I_s/I_H = \nabla_s h(s, I)\big|_{I=I(s,H)}.$$

Assume there is a vector-valued C^1 corrector $A(s, X)$, $X = (Z, W)$, such that A and its selected derivative A_s at fixed X are bounded and

$$\mathcal{A}_s A = P_s - \bar{f}(s, H(s, X)). \tag{5.3}$$

Then $I \to I_\infty > 0$ with $I - I_\infty = O(\varepsilon)$. Writing $h_\infty(s) = h(s, I_\infty)$, there is a corrected velocity $v \to 0$ such that

$$s_\sigma = v + r_s, \qquad v_\sigma = -av - \nabla h_\infty(s) + r_v, \qquad |r_s| + |r_v| \le C\varepsilon. \tag{5.4}$$

Moreover $h_\infty(s(\sigma)) \to E_\infty$.

Proof. Write $f = P_s$, $B = A_s k$ and $w = I_H k$. The full chain rule and the energy equation give

$$\begin{aligned} f - \bar{f} &= \varepsilon A_\sigma - \varepsilon B, & I_\sigma &= w \cdot (f - \bar{f}), \\ (I - \varepsilon w \cdot A)_\sigma &= \varepsilon\big[(aw - w_\sigma) \cdot A - w \cdot B\big]. \end{aligned} \tag{5.5}$$

The states, k_σ, H_σ, and all needed derivatives of I are bounded on the positive band. Hence w, w_σ, B are bounded. Integration of the last equation, using $\int_\sigma^\infty \varepsilon \, \mathrm{d}u = \varepsilon(\sigma)/a$, proves existence of I_∞ and the $O(\varepsilon)$ error. The positive energy band bounds I below by a positive constant.

Set $v = k + \varepsilon A$. Direct differentiation gives

$$s_\sigma = v - \varepsilon A, \qquad v_\sigma = -av - \bar{f} + \varepsilon B.$$

Smoothness and $I - I_\infty = O(\varepsilon)$ imply $\bar{f} - \nabla h_\infty = O(\varepsilon)$, proving (5.4). Also $v \to 0$ by Proposition 4.1, and $h_\infty(s) = H + O(\varepsilon) \to E_\infty$.

For clarity, the actual force error itself need not be absolutely integrable or tend to zero. Its improper primitive nevertheless has an $O(\varepsilon)$ tail, since

$$\int_u^v (f - \bar{f}) \, \mathrm{d}\sigma = [\varepsilon A]_u^v + \int_u^v \varepsilon(aA - B) \, \mathrm{d}\sigma. \tag{5.6}$$

∎

Lemma 1 (Convex corrected-distance argument)**.** *In Proposition 5.1, suppose U is convex and h_∞ is convex on U. Then s converges to a minimizer of h_∞ in U, even when that minimizer is not unique.*

Proof. Let $g = \nabla h_\infty$. Every selected cluster point u satisfies $g(u) = 0$. Indeed, integrate the second equation in (5.4) over a fixed late interval starting at a subsequence with $s \to u$. The first equation, $v \to 0$ and $r_s \to 0$ make s uniformly close to u on that interval. All other terms in the integrated second equation tend to zero, leaving the interval length times $g(u)$. Convexity makes u a global minimizer in U.

For any such u, put

$$y = s + v/a, \qquad Q_u = \tfrac{1}{2}|y-u|^2 + h_\infty(s)/a^2.$$

The cancellation of the $v \cdot g$ terms gives exactly

$$(Q_u)_\sigma = -\frac{(s-u)\cdot g(s)}{a} + (y-u)\cdot(r_s + r_v/a) + \frac{g(s)\cdot r_s}{a^2}. \tag{5.7}$$

The first term is nonpositive; the remaining terms are absolutely integrable because all states are bounded and the errors are $O(\varepsilon)$. Since Q_u is bounded below, it has a limit. The convergence of $h_\infty(s)$ implies existence of a limit for $|y-u|$, hence for $|s-u|$. Choosing u to be a cluster point makes that limit zero. ∎

This is a finite-dimensional integrable-error form of the classical convex heavy-ball convergence mechanism; see Alvarez [3]. It proves point convergence and does not require, or assert, finite selected length in a flat minimizing set.

5.2 Scalar wells and common normal shapes

For one scalar normal variable, homogeneity necessarily gives

$$P(s,Z) = c_+(s)(Z_+)^p + c_-(s)((-Z)_+)^p, \qquad c_\pm > 0, \tag{5.8}$$

where $r_+ = \max\{r, 0\}$. The coefficients are C^2. Define

$$C_p = 2\sqrt{2}\int_0^1 \sqrt{1-r^p}\,\mathrm{d}r, \qquad D(s) = C_p\big(c_+(s)^{-1/p} + c_-(s)^{-1/p}\big). \tag{5.9}$$

Each positive-energy shell is one periodic orbit and has a unique orbital probability measure. Its action and fixed-action energy are

$$I(s,H) = D(s)H^\nu, \qquad \bar{f} = -\frac{H}{\nu}\nabla\log D, \qquad h(s,I) = \left(\frac{I}{D(s)}\right)^{1/\nu}. \tag{5.10}$$

A corrector for $P_s - \bar{f}$ is obtained by integrating that mean-zero observable from the section $Z = 0, W > 0$. Its period integral vanishes by (5.1), so the primitive is single valued. Continuous differentiability of the flow, section and return time gives a C^1 corrector on

each compact positive-energy band. At turning points the nonzero acceleration provides a flow coordinate, while at crossings the velocity is nonzero; no division by a vanishing turning-point velocity is involved.

If P is jointly convex on a convex selected domain, the normal Hessian Schur complement on either half-line gives

$$D^2c - \frac{p}{p-1}c^{-1}\nabla c \otimes \nabla c \geq 0. \tag{5.11}$$

This is equivalent to concavity of $c^{-1/(p-1)}$. Raising this positive concave function to the power $(p-1)/p \in (0,1)$ proves concavity of $c^{-1/p}$. Therefore D is concave. The function $x \mapsto x^{-1/\nu}$ is convex and decreasing, so (5.10) is convex in s at fixed $I > 0$. Proposition 5.1 and Lemma 1 apply on a positive limiting-energy tail.

The same conclusion holds for a multiplicative normal shape in any dimension,

$$P(s, Z) = c(s)V_0(Z), \qquad c > 0, \tag{5.12}$$

where V_0 is positive coercive and homogeneous of degree p. The invariant virial gives

$$\bar{f}(s, H) = \frac{2H}{p+2}\nabla \log c(s).$$

Here one may take

$$I = c^{-1/p}H^{\nu}, \qquad h(s, I) = I^{1/\nu}c^{2/(p+2)}, \qquad A = -\frac{\nabla \log c}{p+2}(Z \cdot W). \tag{5.13}$$

Indeed, $\mathcal{A}_s(Z \cdot W) = 2H - (p+2)P$ proves the corrector identity. This scalar action-like quantity need not be the action of an individual multidimensional orbit; the displayed identities are what is required. Convexity restricted to any nonzero normal ray gives (5.11), and hence convexity of $h(\cdot, I)$. No assumption on integrability or chaotic behavior of the frozen normal flow is needed.

5.3 Absolute action estimates near zero energy

The positive-band corrector does not alone justify passage to zero normal energy. We give the absolute estimate that closes this issue and also controls individual modes of a separable system.

In this subsection unlabelled P and P_s are evaluated at (s, q), and the physical frozen generator is $\mathcal{A}_s = u \cdot \partial_q - P_q \cdot \partial_u$. Use physical velocities $v = s_t$ and $u = q_t$, and define

$$E_N = \frac{|u|^2}{2} + P(s, q), \qquad L = \frac{|v|^2}{2} + \frac{|u + aq/t|^2}{2} + P(s, q). \tag{5.14}$$

At critical damping $t^2L = E$ exactly. Thus, without any action assumption,

$$L = O(t^{-2}), \quad q = O(t^{-2/p}), \quad u, v = O(t^{-1}), \quad E_{\text{phys}} = O(t^{-2}). \tag{5.15}$$

The velocity bound follows from the square in L and the coercive bound on q.

Lemma 2 (Absolute physical action correction)**.** *For a nonzero scalar well* (5.8) *or multiplicative normal state* (5.12)*, let $I(s, E_N)$ be the corresponding action in* (5.10) *or* (5.13)*, and put $K = t^\alpha I$. There is a corrected quantity $\widehat{K}$ such that*

$$\begin{aligned} |\widehat{K} - K| &\le C t^\alpha E_N^{2/p}(t^{-1} + |v|), \\ |\widehat{K}'| &\le C t^\alpha E_N^{2/p}\big[t^{-2} + |v|/t + |v|^2 + E_{\mathrm{phys}}\big]. \end{aligned} \tag{5.16}$$

For a scalar mode in a finite separable system, the same estimate holds with its individual normal energy and the total physical energy on the right.

Proof. A nonzero normal state remains nonzero at every finite time by uniqueness, so $E_N > 0$.

For $P = c(s)V_0(q)$ write

$$\begin{aligned} \eta_0 &= \frac{1}{2} - \frac{1}{p}, \qquad D_0 = |u|^2 - pP, \qquad Q = \frac{q \cdot u}{E_N}, \\ C(t) &= \frac{2\alpha/t + \nabla \log c \cdot v}{2p}, \qquad \widehat{K} = K(1 + CQ). \end{aligned}$$

Euler homogeneity and $E_N' = -(\alpha/t)|u|^2 + v \cdot P_s$ give

$$\begin{aligned} (\log I)' + \alpha/t &= -C D_0/E_N, \\ Q' &= D_0/E_N - (E_N'/E_N + \alpha/t)Q, \\ \widehat{K}' &= K\big[C' - C(E_N'/E_N + \alpha/t) - C^2 D_0/E_N\big]Q. \end{aligned} \tag{5.17}$$

Coercivity implies $|Q| \le C E_N^{-\eta_0}$. The ratios D_0/E_N and E_N'/E_N are bounded respectively by C and $C(t^{-1} + |v|)$. Bounded selected coefficients and $v' = -(\alpha/t)v - P_s$ give

$$|C'| \le C\big[t^{-2} + |v|/t + |v|^2 + E_{\mathrm{phys}}\big].$$

Since $\nu - \eta_0 = 2/p$, the absolute product $|KQ|$ is bounded by $C t^\alpha E_N^{2/p}$. Substitution proves (5.16). In particular no smallness of CQ or relative approximation $\widehat{K}/K \approx 1$ is used.

For an asymmetric scalar well, put $I = D(s)E_N^\nu$. In the physical frozen phase plane choose primitives

$$\mathcal{A}_s A = P_s - \bar{f}(s, E_N), \qquad \mathcal{A}_s B = u^2 - I/I_E, \tag{5.18}$$

with zero value at the crossing $q = 0, u > 0$. Both sources have zero orbital mean: the second uses $\oint u \, \mathrm{d}q = I$ and period I_E. Hence the primitives are single valued. Uniformly for compact s and small positive energy,

$$|A| + |B| + |A_s| + |B_s| + |uA_u| + |uB_u| \le C E_N^\nu. \tag{5.19}$$

Here selected derivatives are taken at fixed (q, u). To verify the bounds, normalize $q = E_N^{1/p} Q_0$, $u = E_N^{1/2} U_0$. The unit-energy orbits form a continuously differentiable compact

family with a transverse crossing. The sources have size E_N and the period scales as $E_N^{\nu-1}$, so each primitive is E_N^{ν} times a continuously differentiable normalized primitive. Differentiation at fixed phase state uses $E_{N,s} = P_s = O(E_N)$ and $uE_{N,u} = u^2 = O(E_N)$; the derivatives of the normalized variables are bounded. This proves all terms in (5.19) on the entire shell, including the turning points. Also

$$I_E = O(E_N^{\nu-1}), \quad I_{EE} = O(E_N^{\nu-2}), \quad I_{Es} = O(E_N^{\nu-1}).$$

Let $\delta = \alpha/t$, $R_0 = v \cdot A - \delta B$, and define $\widehat{K} = t^\alpha[I - I_E R_0]$. The uncorrected identity is

$$(t^\alpha I)' = t^\alpha I_E[v \cdot \mathcal{A}_s A - \delta \mathcal{A}_s B].$$

Taking the full damped derivative of R_0 therefore gives

$$\begin{aligned} \widehat{K}' = -t^\alpha \big\{ &[(\alpha/t) I_E + (I_E)'] R_0 \\ &+ I_E[\, v' \cdot A - \delta' B + v \cdot (A_s v) - \delta u\, v \cdot A_u \\ &\qquad - \delta B_s \cdot v + \delta^2 u B_u \,] \big\}. \end{aligned} \tag{5.20}$$

The terms involving the momentum derivatives of both primitives are included here. The energy equation implies $|(I_E)'| \le C E_N^{\nu-1}(t^{-1} + |v|)$. Equations (5.19)–(5.20), the bound on v', and $2\nu - 1 = 2/p$ prove (5.16).

For a scalar mode j of a separable system, its energy equation remains $E_j' = -(\alpha/t)u_j^2 + v \cdot (P_j)_s$. The same computation applies. Only the selected acceleration changes to $v' = -(\alpha/t)v - \sum_l (P_l)_s$, whose force is bounded by the total physical energy. This proves the last assertion. ∎

Corollary 5.1 (Every excited action survives)**.** *In the scalar, multiplicative, and finite separable scalar classes, every non-identically-zero normal mode has a strictly positive limiting scaled physical action K. A zero action limit occurs exactly for an identically zero mode.*

Proof. Using (5.15) and $\alpha = 1 + 2/p$ in (5.16) gives

$$|\widehat{K} - K| = O(t^{-a}), \qquad |\widehat{K}'| = O(t^{-1-a}). \tag{5.21}$$

Thus $K \to \ell \ge 0$ and $K - \ell = O(t^{-a})$. Suppose $\ell = 0$. Since its action coefficient stays above and below positive constants,

$$E_N = O(t^{-n_*}), \qquad n_* = (\alpha + a)/\nu = 2 + \frac{4}{p+2}. \tag{5.22}$$

For a single separable mode, the same estimate follows using its own energy; other mode actions need not vanish.

Let $H_j = |W_j|^2/2 + P_j(s, Z_j)$ denote the associated rescaled energy, or the whole normal energy in the multiplicative case. The physical relation gives

$$H_j \le C(t^2 E_j + |q_j|^2) \le C\big(t^{2-n_*} + t^{-2n_*/p}\big) = O(t^{-\delta_*}), \qquad \delta_* = \frac{4}{p+2} > 0. \tag{5.23}$$

But for a nonzero mode $H_j > 0$ at finite time and

$$(H_j)_\sigma = k \cdot (P_j)_s, \qquad |(\log H_j)_\sigma| \le C|k| \longrightarrow 0.$$

Choose σ_1 so late that the last upper bound is at most $\delta_*/2$. Then

$$H_j(\sigma) \ge H_j(\sigma_1) e^{-\delta_*(\sigma-\sigma_1)/2},$$

contradicting (5.23). Thus $\ell > 0$. An identically zero mode has zero action. This argument uses an absolute small-action estimate, not a relative expansion near a vanishing energy. ∎

5.4 The critical convergence theorem

Theorem 4 (Point selection in action-constrained convex valleys)**.** *Assume* (2.1)–(2.3)*, let U be convex, let P be jointly convex, and take $\alpha = 1 + 2/p$. All scalar coefficients below are positive and C^2 on U, and V_0 is positive coercive and C^2. Suppose one of the following holds:*

1. *there is one scalar normal variable, with the possibly asymmetric well* (5.8)*;*
2. *$P(s,q) = c(s)V_0(q)$ is a multiplicative normal shape in any normal dimension;*
3. *$P(s,q) = \sum_{j=1}^n P_j(s,q_j)$ is a sum of degree-p scalar wells of the form* (5.8)*.*

Then $X(t)$ converges to $(s_\infty, 0)$. Every nonzero mode has positive limiting action. If the normal state is excited, $E_\infty > 0$; if it is identically zero, the full physical tail has finite length.

Proof. For the first two cases, Corollary 5.1 gives a positive action limit whenever the normal state is nonzero. Indeed,

$$K = D(s)[t^2 E_N]^\nu, \qquad t^2 E_N = \frac{|W - aq|^2}{2} + P(s,Z), \qquad |t^2 E_N - H| = O(\varepsilon),$$

with $D = c^{-1/p}$ in the multiplicative case. Since D is uniformly positive and bounded, the normal energy stays in a compact positive band. Proposition 4.1 then gives $E_\infty > 0$. The correctors and convex fixed-action energies established above prove selected convergence by Lemma 1.

For the separable case, discard the identically zero modes. Each remaining mode has $K_j \to \ell_j > 0$ by Corollary 5.1, hence has a compact positive rescaled-energy band. Apply the identities (5.5) to each mode with H_j in place of H. They use only its exact equation $(H_j)_\sigma = k \cdot (P_j)_s$ and bounded k_σ, so forces from the other modes do not invalidate the argument. They show that the rescaled actions satisfy $I_j - \ell_j = O(\varepsilon)$; the physical-rescaled comparison just displayed identifies their limits.

Put

$$h_*(s) = \sum_j h_j(s, \ell_j), \qquad v_* = k + \varepsilon \sum_j A_j(s, Z_j, W_j).$$

Summing the corrected equations gives

$$s_\sigma = v_* + r_s, \qquad (v_*)_\sigma = -a v_* - \nabla h_*(s) + r_v, \qquad |r_s| + |r_v| \le C\varepsilon.$$

Joint convexity of P restricts to convexity of each summand when all other normal coordinates vanish. Consequently each $h_j(\cdot, \ell_j)$, and hence h_*, is convex. Also $v_* \to 0$ and $h_*(s) = H + O(\varepsilon) \to E_\infty$. The corrected-distance argument proves convergence with no uniqueness requirement on the effective minimizers.

If every normal mode is identically zero, the selected equation is $s_{tt} + \alpha s_t/t = 0$, so $s_t = Ct^{-\alpha}$. Since $\alpha > 1$, the full physical tail has finite length and converges. In all cases $q \to 0$ by Proposition 4.1. ∎

5.5 A force-hull interior without accessible drift

For even $p > 2$, consider on $|s| < 1$ the potential

$$P(s, Z) = \frac{|Z_1|^p}{(1+s)^{p-1}} + \frac{|Z_2|^p}{(1-s)^{p-1}}. \tag{5.24}$$

Each summand is the perspective of $|x|^p$ composed with a positive affine denominator. Thus P is jointly convex, semialgebraic, smooth on the strip, and uniformly normally coercive on compact selected substrips. Theorem 4 applies to every actual confined critical tail.

Nevertheless its frozen force set at any $H > 0$ is exactly

$$\mathcal{G}(s, H) = \left[-\frac{2(p-1)H}{(p+2)(1+s)}, \frac{2(p-1)H}{(p+2)(1-s)} \right]. \tag{5.25}$$

Indeed, the two axial orbital measures give the endpoints. For any invariant measure, the separate energies H_1, H_2 are nonnegative and sum to H. Applying the virial to each mode gives the mean force

$$\frac{2(p-1)}{p+2} \left[-\frac{\int H_1 \, \mathrm{d}\mu}{1+s} + \frac{\int H_2 \, \mathrm{d}\mu}{1-s} \right],$$

which lies in the displayed interval; convex combinations of the endpoint measures fill it. Zero lies strictly inside this force hull everywhere in the strip, yet every confined critical trajectory selects a point. Separate asymptotic actions prevent arbitrary changing energy allocations between the modes. The force hull is therefore an outer constraint, not a switching theorem.

5.6 Critical point convergence can have infinite length

Proposition 5.2 (Positive residual normal energy)**.** For any confined critical tail with $E_\infty > 0$, the physical normal length is infinite:

$$\int_{t_0}^{\infty} |q_t(t)|\,\mathrm{d}t = \infty.$$

In particular this holds for every excited trajectory in Theorem 4, even though that trajectory converges to a point.

Proof. At $\Delta = 0$, (2.8) reads $B_\sigma = |W|^2 - pP - aB$. Compactness gives $B = O(\varepsilon)$, hence

$$\int_{\sigma_0}^{S} (|W|^2 - pP)\,\mathrm{d}\sigma = O(1).$$

Since $P = H - |W|^2/2$ and $H \to E_\infty$,

$$\int_{\sigma_0}^{S} |W|^2\,\mathrm{d}\sigma = \frac{2p}{p+2}\int_{\sigma_0}^{S} H\,\mathrm{d}\sigma + O(1) = \frac{2pE_\infty}{p+2}(S - \sigma_0) + o(S).$$

Choose $M_0 > 0$ bounding $|W|$. Then $|W| \geq |W|^2/M_0$, so its absolute logarithmic-time integral diverges. The exact physical velocity gives

$$\int_{t_0}^{T} |q_t|\,\mathrm{d}t \geq \int_{\sigma_0}^{\log T} |W|\,\mathrm{d}\sigma - a\int_{\sigma_0}^{\log T} \varepsilon|Z|\,\mathrm{d}\sigma.$$

The second integral stays finite, proving the assertion. ∎

For a general nonseparable critical family, positivity of E_∞ remains a hypothesis. Corollary 5.1 does not extend that positivity conclusion to arbitrary homogeneous normal dynamics.

6 The zero-residual-energy boundary

We retain the confined-tail assumptions (2.3) and the critical equations (4.1). An *excited* normal tail means a solution not on the invariant set $Z = W = 0$. Coercivity and uniqueness imply $H > 0$ at every finite time on such a tail. The distinction between zero residual energy and an identically zero normal state is essential. The estimates in this section do not require convexity or semialgebraicity.

6.1 Relative energy and absolute force integrability

Proposition 6.1 (The selected velocity has normal-energy order)**.** Every excited confined critical tail satisfies $|k| \leq CH$ for all sufficiently large σ. If $E_\infty = 0$, then

$$E = H + O(H^2), \qquad |H_\sigma| \leq CH^2, \qquad H(\sigma) \geq \frac{1}{H(S)^{-1} + C(\sigma - S)} \quad (\sigma \geq S). \tag{6.1}$$

In particular $E/H \to 1$, $\int_S^\infty H \,\mathrm{d}\sigma = \infty$, and

$$E(\sigma) = a \int_\sigma^\infty |k(u)|^2 \,\mathrm{d}u. \tag{6.2}$$

Proof. The relative force bound (2.4) gives $|P_s| \le CH$. Hence the exact work equation and Proposition 4.1 imply

$$|(\log H)_\sigma| \le C|k| \longrightarrow 0. \tag{6.3}$$

For every $d > 0$, there is S_d such that, for $S_d \le u \le \sigma$,

$$H(u) \le H(\sigma) e^{d(\sigma-u)}, \qquad H(\sigma) \ge H(S_d) e^{-d(\sigma-S_d)}. \tag{6.4}$$

Fix $0 < d < a$ and put $S = S_d$. Variation of constants in the selected equation yields

$$|k(\sigma)| \le e^{-a(\sigma-S)}|k(S)| + C\int_S^\sigma e^{-a(\sigma-u)} H(u) \,\mathrm{d}u \le e^{-a(\sigma-S)}|k(S)| + \frac{C}{a-d} H(\sigma).$$

By (6.4), the first term divided by $H(\sigma)$ tends to zero. This proves $k = O(H)$ without assuming a positive limiting energy. Now $E = H + |k|^2/2$ and $|H_\sigma| \le C|k|H$ prove the first two estimates in (6.1). Integrating $|(H^{-1})_\sigma| \le C$ proves the lower bound and the divergent integral. Integration of $E_\sigma = -a|k|^2$, using $E_\infty = 0$, proves (6.2). ∎

Writing $\gamma = (p-2)/(2p) > 0$, the energy-normalized fast rate obeys

$$\varepsilon^{-1} H^\gamma \ge \frac{c e^{a\sigma}}{(1+\sigma)^\gamma} \longrightarrow \infty$$

after shifting the origin of σ. Vanishing normal amplitude therefore does not by itself destroy fast–slow separation on an excited zero-energy tail. This bound alone is not an averaging or realization theorem.

Theorem 5 (Absolute integrability at zero residual energy)**.** *Suppose $E_\infty = 0$. The following properties are equivalent:*

1. $\int_S^\infty |P_s(s(\sigma), Z(\sigma))| \,\mathrm{d}\sigma < \infty$;
2. $\int_S^\infty |k(\sigma)| \,\mathrm{d}\sigma < \infty$;
3. $Z = W = 0$ *identically on the tail.*

On this branch the full physical trajectory has finite length, and $s_\infty = s(S) + k(S)/a$. Every excited zero-energy tail instead has infinite selected length and a divergent absolute rescaled selected-force integral.

Proof. Duhamel's formula and Tonelli's theorem give

$$\int_S^\infty |k| \,\mathrm{d}\sigma \le \frac{|k(S)|}{a} + \frac{1}{a}\int_S^\infty |P_s| \,\mathrm{d}\sigma. \tag{6.5}$$

Thus the first property implies the second. If the second held on an excited tail, (6.3) would make $\log H$ converge to a finite real number. This would give $\lim H > 0$, contrary to $E_\infty = 0$. Hence the second implies the third. On the zero-normal branch, $P_s = 0$ and $k(\sigma) = e^{-a(\sigma - S)} k(S)$. This proves both absolute integrability statements and the displayed limit. The selected length is $\int |k| \, \mathrm{d}\sigma$ in either physical or logarithmic time, and the normal coordinate is identically zero. ∎

Theorem 5 concerns absolute integrals. Point convergence is equivalent only to convergence of the signed vector integral in (4.3). An excited zero-energy tail may not be ruled out by replacing that signed condition with an unproved absolute-integrability assertion. The construction in Section 19 will show that such tails do exist and may fail to select a point.

6.2 An exact integrating factor and a force corrector

Let $\nu = (p+2)/(2p) = 1 - \gamma$, and retain the frozen generator $\mathcal{A}_s = W \cdot \partial_Z - P_Z \cdot \partial_W$.

Theorem 6 (Positive residual energy from an integrating factor)**.** *Suppose there is a positive C^1 function $D(s)$, bounded above and below away from zero on a neighborhood of K_0, and a vector-valued C^1 function $A(s, Z, W)$ on its positive-energy normal states such that*

$$b(s) = -\frac{1}{\nu} \nabla \log D(s), \qquad P_s(s, Z) - H b(s) = \mathcal{A}_s A(s, Z, W), \qquad |A| + |A_s| \le C H^\nu. \tag{6.6}$$

Here A_s is differentiated at fixed (Z, W). Then every excited confined critical tail has $E_\infty > 0$. More precisely, the explicitly defined quantity

$$I = D(s) H^\nu$$

has a limit $I_\infty > 0$, with $\log I - \log I_\infty = O(\varepsilon)$ and $I - I_\infty = O(\varepsilon)$.

Proof. On an excited tail put

$$B = A/H, \qquad \mathcal{J} = \log I - \nu \varepsilon k \cdot B.$$

Since $\mathcal{A}_s H = 0$, the corrector equation gives $\mathcal{A}_s B = P_s/H - b$. The full chain rule, with the indicated fixed-normal-state selected derivative, is

$$B_\sigma = \varepsilon^{-1} \mathcal{A}_s B + B_s k.$$

The integrating-factor identity cancels the averaged work:

$$(\log I)_\sigma = \nabla \log D \cdot k + \nu \frac{k \cdot P_s}{H} = \nu k \cdot \mathcal{A}_s B.$$

Using $\varepsilon_\sigma = -a\varepsilon$ and $k_\sigma = -ak - P_s$ now gives the exact identity

$$\mathcal{J}_\sigma = \nu \varepsilon \big[(2ak + P_s) \cdot B - k \cdot (B_s k) \big]. \tag{6.7}$$

The relative force estimate implies

$$B_s = \frac{A_s}{H} - \frac{A \otimes P_s}{H^2}, \qquad |B| + |B_s| \le C H^{\nu-1}.$$

By Proposition 6.1,

$$|\varepsilon k \cdot B| \le C\varepsilon H^{\nu}, \qquad |\mathcal{J}_\sigma| \le C\varepsilon(H^{\nu} + H^{\nu+1}). \tag{6.8}$$

The energy is bounded, so the right sides are $O(\varepsilon)$. Since $\int_\sigma^\infty \varepsilon(u)\,\mathrm{d}u = \varepsilon(\sigma)/a$, $\mathcal{J}$ has a finite real limit with an $O(\varepsilon)$ tail error. Its correction tends to zero at the same rate. Thus $\log I$ has a finite real limit, and exponentiation gives $I_\infty > 0$ and both claimed rates. The upper bound for D then bounds H away from zero. ∎

The quantity I in this theorem is an exact integrating-factor expression for the averaged work equation. Without additional structure it is not a canonical action of the normal orbit. The homological-equation and corrected-invariant method is classical; see, for example, Avendaño-Camacho, Vallejo and Vorobiev [5]. The particular identity (6.7) incorporates critical damping and the exponentially changing clock.

The hypothesis on A_s must be checked after homogeneous extension at fixed (Z, W). A uniformly C^1 unit-energy corrector with a weighted-homogeneous extension

$$A(s, \lambda^{1/p} Z, \lambda^{1/2} W) = \lambda^{\nu} A(s, Z, W), \qquad \lambda > 0,$$

is sufficient when that extension has the stated selected-derivative bound. A tangential derivative on a moving energy shell is not a substitute. In the scalar and multiplicative classes of Section 5, the already constructed correctors satisfy these bounds. For a sum of independent modes, the separate mode quantities should still be retained.

Corollary 6.1 (Correctors with finite-power growth)**.** *The same positive-residual-energy conclusion holds if the corrector assumption in* (6.6) *is replaced by*

$$\mathcal{A}_s B = P_s/H - b(s), \qquad |B| + |B_s| \le C(1 + H^{-M})$$

for some finite $M \ge 0$, with the same D and b. The $O(\varepsilon)$ convergence of I follows as well.

Proof. Identity (6.7) is unchanged. If $M > 0$, choose $d > 0$ with $Md < a$ in (6.4). The boundedness of k, P_s shows that both $\varepsilon k \cdot B$ and the right side of (6.7) decay exponentially. For $M = 0$ this follows directly. Hence $\log I$ has a finite real limit and H stays bounded away from zero. On that positive band the bounds improve to $O(\varepsilon)$, including the tail integral of (6.7). ∎

Neither equality of all frozen invariant-force means nor joint convexity alone supplies the differentiable corrector in Theorem 6. If two periodic branches have distinct means, its single field b cannot satisfy the corrector equation on both: integration against their orbital measures would give two different values of Hb. This is exactly the obstruction exploited below.

7 Phase volume and critical point selection

The integrating factor in Theorem 6 need not be guessed when one force corrector applies to the whole normal energy shell. It is determined by enclosed phase volume. Joint convexity then supplies a convex fixed-volume energy in arbitrary normal dimension. This combines the zero-energy estimate with the corrected-distance argument, and extends the multiplicative class to verified changes of normal orientation.

Throughout this section, $U \subset \mathbb{R}^d$ is open and convex, $P \in C^2(U \times \mathbb{R}^n)$ is jointly convex and homogeneous of degree $p > 2$ in its normal variable, and the actual critical trajectory satisfies (2.3). Write $\nu = 1/2 + 1/p$. The dimension n below is the normal configuration dimension, so normal phase space has dimension $2n$.

7.1 Volume geometry and the invariant force mean

Lemma 3 (Propagation of normal coercivity)**.** *Coercivity at one interior selected parameter implies locally uniform normal coercivity throughout U.*

Proof. Convexity and normal homogeneity give $P(s,0) = 0$, $P_Z(s,0) = 0$ and $P(s,Z) \geq 0$. Suppose $P(s_0, z) = 0$ for some $z \neq 0$. For any $s \in U$, choose $0 < \theta < 1$ small enough that $s_1 = (s - \theta s_0)/(1-\theta) \in U$. Joint convexity gives

$$P(s,z) \leq \theta P(s_0, z/\theta) + (1-\theta)P(s_1, 0) = 0.$$

Thus that direction vanishes at every selected parameter, contradicting coercivity at the given parameter. Positivity on each unit normal sphere and compactness give a uniform positive lower bound on every compact parameter subset of U. ∎

Define the configuration body and enclosed phase volume by

$$\begin{aligned} B_s &= \{Z : P(s,Z) \leq 1\}, \\ \mathcal{V}(s,h) &= \mathrm{vol}_{2n}\{(Z,W) : |W|^2/2 + P(s,Z) \leq h\}, \\ \mathcal{V}_1(s) &= \mathcal{V}(s,1), \qquad D_{\mathrm{vol}}(s) = \mathcal{V}_1(s)^{1/n}. \end{aligned} \tag{7.1}$$

These bodies are compact and have nonempty interiors by Lemma 3. Weighted scaling gives

$$\mathcal{V}(s,h) = \mathcal{V}_1(s) h^{n\nu}, \qquad I_{\mathrm{vol}}(s,h) = D_{\mathrm{vol}}(s) h^{\nu} = \mathcal{V}(s,h)^{1/n}. \tag{7.2}$$

Thus I_{vol} is the nth root of phase volume, not the raw volume or the $2n$th root. It need not be an action of an individual normal orbit.

Proposition 7.1 (Phase-volume force and convex fixed-volume energy)**.** The functions $\mathcal{V}_1$ and D_{vol} are positive and C^2 on U, and D_{vol} is concave. At every $h > 0$, the normalized microcanonical probability $\mu^{\mathrm{mc}}_{s,h}$ is frozen-flow invariant and satisfies

$$\int P_s \, \mathrm{d}\mu^{\mathrm{mc}}_{s,h} = h b_{\mathrm{vol}}(s), \qquad b_{\mathrm{vol}} = -\frac{1}{n\nu} \nabla \log \mathcal{V}_1 = -\frac{1}{\nu} \nabla \log D_{\mathrm{vol}}. \tag{7.3}$$

For every fixed phase-volume root $\iota > 0$, the energy

$$U_\iota(s) = \left(\frac{\iota}{D_{\mathrm{vol}}(s)}\right)^{1/\nu} \tag{7.4}$$

is positive, C^2 and convex on U, with $\nabla U_\iota = U_\iota b_{\mathrm{vol}}$.

Proof. Put $Q(s) = \mathrm{vol}_n(B_s)$. Polar coordinates and integration over the momentum ball give

$$\begin{aligned} Q(s) &= \frac{1}{n}\int_{\mathbb{S}^{n-1}} P(s,\omega)^{-n/p}\,\mathrm{d}\omega, \\ \mathcal{V}_1(s) &= C_{n,p}Q(s), \qquad C_{n,p} = \frac{(2\pi)^{n/2}\Gamma(1+n/p)}{\Gamma(1+n\nu)}. \end{aligned} \tag{7.5}$$

For the second identity, the momentum-ball integral equals

$$\kappa_n 2^{n/2} Q(s)\frac{n}{p}\int_0^1 u^{n/p-1}(1-u)^{n/2}\,\mathrm{d}u, \qquad \kappa_n = \frac{\pi^{n/2}}{\Gamma(1+n/2)}.$$

The beta-integral formula yields the constant above. Local uniform coercivity permits two selected derivatives under the spherical integral, proving the regularity.

Write $H_s(Z,W) = |W|^2/2 + P(s,Z)$, where this subscript indicates the frozen parameter. Every positive shell is regular: a vanishing phase gradient would give $W = 0$ and $P_Z = 0$, and Euler's identity would force $P = 0$. Coarea and differentiation of the compact regular sublevel give

$$\partial_h \mathcal{V} = \int_{H_s=h} \frac{\mathrm{d}\Sigma}{|\nabla_{Z,W} H_s|}, \qquad \nabla_s \mathcal{V} = -\int_{H_s=h} \frac{P_s\,\mathrm{d}\Sigma}{|\nabla_{Z,W} H_s|}. \tag{7.6}$$

The probability measure

$$\mathrm{d}\mu^{\mathrm{mc}}_{s,h} = \frac{1}{\partial_h \mathcal{V}(s,h)}\frac{\mathrm{d}\Sigma}{|\nabla_{Z,W} H_s|}$$

is invariant, since Hamiltonian flow preserves volume and energy and hence these regular-shell disintegrations. It is not generally normalized Euclidean surface area. Dividing the identities in (7.6) and using (7.2) proves (7.3); no ergodicity assumption enters.

For $s_\theta = (1-\theta)s_0 + \theta s_1$, joint convexity gives

$$(1-\theta)B_{s_0} + \theta B_{s_1} \subset B_{s_\theta}.$$

The Brunn–Minkowski inequality in configuration dimension n makes $Q^{1/n}$ concave. Equation (7.5) therefore makes D_{vol} concave. Composition with the positive, convex, decreasing function $r \mapsto \iota^{1/\nu} r^{-1/\nu}$ proves convexity of (7.4). Differentiation proves its gradient identity. ∎

The dimension in this argument matters. Brunn–Minkowski directly in phase space gives concavity of $\mathcal{V}^{1/(2n)}$ in (s,h); the stronger concavity of $\mathcal{V}_1^{1/n}$ uses (7.5). Fixed-volume

energy convexity also follows from Prékopa's marginal theorem [21, Theorem 6], applied to the convex phase sublevel family. These are classical convex-geometric consequences. No semialgebraicity of the integrated volume is asserted.

If $\mathcal{G}(s,1)$ is a singleton, its value must be $b_{\mathrm{vol}}(s)$, because the microcanonical measure is one of the invariant probabilities in (4.5). More generally, $b_{\mathrm{vol}}(s) \in \mathcal{G}(s,1)$ even when other means exist. Thus phase volume identifies one distinguished mean, but does not choose the mean of a particular normal trajectory.

7.2 A force corrector yields point selection

Theorem 7 (Phase-volume corrector selection)**.** *Under the assumptions of this section, suppose that above a neighborhood of the actual selected compact tail there are a field $b(s)$ and a vector-valued C^1 function $A(s,Z,W)$ on the positive-energy normal states such that*

$$\mathcal{A}_s A = P_s - Hb(s), \qquad |A| + |A_s| \le CH^{\nu}. \tag{7.7}$$

The selected derivative A_s is taken at fixed (Z,W). Then $b = b_{\mathrm{vol}}$ wherever (7.7) holds, and every excited confined critical tail satisfies

$$\begin{gathered} E_\infty > 0, \qquad I_{\mathrm{vol}}(s,H) \longrightarrow \iota_\infty > 0, \qquad I_{\mathrm{vol}}(s,H) - \iota_\infty = O(\varepsilon), \\ s(\sigma) \longrightarrow s_\infty \in \operatorname*{argmin}_U U_{\iota_\infty}. \end{gathered} \tag{7.8}$$

In particular the physical trajectory converges to $(s_\infty, 0)$, whether or not the effective minimizer is unique. The zero-normal branch also converges and has finite physical length.

Proof. Integrate (7.7) against the invariant microcanonical probability at energy $h > 0$. The generator integral vanishes, so (7.3) implies $b = b_{\mathrm{vol}}$. Integration against any invariant probability shows, in addition, that $\mathcal{G}(s,h) = \{hb_{\mathrm{vol}}(s)\}$. The identification gives exactly the integrating factor required by Theorem 6, with $D = D_{\mathrm{vol}}$. Its exact corrected-log proof therefore yields $\iota_\infty > 0$, $E_\infty > 0$ and the stated error, without presupposing a positive energy band.

Now set $U_\infty = U_{\iota_\infty}$ and $\widetilde{k} = k + \varepsilon A$. The full chain rule gives the exact equations

$$s_\sigma = \widetilde{k} - \varepsilon A, \qquad \widetilde{k}_\sigma = -a\widetilde{k} - Hb_{\mathrm{vol}} + \varepsilon A_s k. \tag{7.9}$$

The positive band and (7.8) imply $H = U_\infty(s) + O(\varepsilon)$. Proposition 7.1 gives $\nabla U_\infty = U_\infty b_{\mathrm{vol}}$, so (7.9) is the actual integrably perturbed system (5.4), with $\widetilde{k} \to 0$ and $U_\infty(s) \to E_\infty$. Its potential is convex on all of U, even though the corrector was assumed only near the tail. Lemma 1 proves point convergence to a minimizer. Finally, on $Z = W = 0$, one has $k_\sigma = -ak$ and hence integrable selected speed; the normal coordinate vanishes identically. ∎

Theorem 7 adds no separate exactness assumption on b and no separate convexity assumption on the effective energy. Both follow from volume geometry. Its dynamical requirement remains a verified differentiable corrector with the fixed-state parameter bound.

Common invariant means alone do not supply such a solution of the cohomological equation or an integrable averaging error. Corollary 6.1 also permits a finite-power-growth normalized corrector for positivity; once the positive energy band is established, the same reduction proves selection.

Phase-volume adiabatic invariance is classical, originating with Hertz [19] and developed, with explicit finite-parameter and almost-everywhere quantifiers, by Kasuga [20]. Those results are not invoked as orbitwise infinite-time estimates here. The new conclusion in Theorem 7 follows from the exact critical identities and the stronger stated corrector hypothesis, including the potentially vanishing-energy boundary.

7.3 Changing the orientation of a nonseparable well

Let

$$P(s,Z) = c(s)V_0(R(s)Z), \tag{7.10}$$

where $c > 0$, the matrix $R(s)$ is orthogonal and C^2, and V_0 is positive coercive, C^2 and homogeneous of degree p. Assume joint convexity on the selected domain. For each selected coordinate define

$$\lambda_j = \partial_{s_j} \log c, \qquad \Omega_j = R^T \partial_{s_j} R.$$

The matrix Ω_j is skew-symmetric, and the frozen identities are

$$\begin{aligned} P_{s_j} &= \lambda_j P + P_Z \cdot \Omega_j Z, \\ \mathcal{A}_s(Z \cdot W) &= 2H - (p+2)P, \\ \mathcal{A}_s(W \cdot \Omega_j Z) &= -P_Z \cdot \Omega_j Z. \end{aligned} \tag{7.11}$$

Indeed $\partial_{s_j} R = R\Omega_j$, and $W \cdot \Omega_j W = 0$. Thus

$$A_j = -\frac{\lambda_j}{p+2} Z \cdot W - W \cdot \Omega_j Z, \qquad b_j = \frac{2\lambda_j}{p+2} \tag{7.12}$$

satisfies (7.7). On compact parameter sets its coefficients and selected derivatives are bounded, while coercivity gives $|Z||W| \le CH^\nu$. This proves both required bounds at fixed normal state. Orthogonal changes preserve normal volume, so

$$\mathcal{V}_1(s) = \mathcal{V}_{0,1} c(s)^{-n/p}, \qquad U_\iota(s) = C_\iota c(s)^{2/(p+2)}. \tag{7.13}$$

No integrability or ergodicity assumption on V_0 is needed. A parameter-dependent rotation is not a fixed change of coordinates in the full equations; its angular-momentum term is precisely the second term of (7.12).

Example 1 (An explicit convex rotating quartic)**.** Take $p = 4$, one selected coordinate, two normal coordinates, and $K = 1000$. On $U = (-1/K, 1/K)$ put

$$\begin{aligned} c(s) = 1 + Ks^2, \qquad R(s) &= \frac{1}{1+s^2} \begin{pmatrix} 1-s^2 & -2s \\ 2s & 1-s^2 \end{pmatrix}, \\ u(s) = \text{first row of } R(s), \qquad P(s,Z) &= c(s)\big[|Z|^4 + (u(s)\cdot Z)^4\big]. \end{aligned} \tag{7.14}$$

This smooth semialgebraic family is jointly convex, uniformly normally coercive, and non-multiplicative in fixed normal coordinates. Its frozen shape is not orthogonally separable. Every excited actual compact confined critical tail selects $s_\infty = 0$ and has $E_\infty > 0$.

Proof. Write $r = |Z|$, $\ell = u \cdot Z$, and $V = r^4 + \ell^4$. On U,

$$|u| = 1, \quad |u'| \le 2, \quad |u''| \le 8, \quad 1 \le c \le 2, \quad |c'| \le 2, \quad c'' = 2K.$$

Direct differentiation gives

$$\begin{gathered} V_{ZZ} \ge 4r^2 \,\mathrm{Id}, \qquad |V_Z| \le 8r^3, \qquad |V_s| \le 8r^4, \qquad |V_{sZ}| \le 32r^3, \\ V_{ss} = 12\ell^2\ell_s^2 + 4\ell^3\ell_{ss} \ge -32r^4. \end{gathered} \tag{7.15}$$

Consequently $P_{ZZ} \ge 4r^2 \,\mathrm{Id}$, $|P_{sZ}| \le 80r^3$ and $P_{ss} \ge (2K - 96)r^4$. For $r > 0$ the selected Schur complement is at least

$$P_{ss} - P_{sZ}P_{ZZ}^{-1}P_{Zs} \ge (2K - 96 - 80^2/4)r^4 = 304r^4 > 0.$$

At $Z = 0$ all Hessian blocks vanish continuously. This proves joint convexity on the convex product domain. Coercivity is immediate from $P \ge |Z|^4$.

At $s = 0$ the frozen shape is $V_0 = |Z|^4 + Z_1^4$, with angular expansion

$$V_0/r^4 = \frac{11}{8} + \frac{1}{2}\cos(2\theta) + \frac{1}{8}\cos(4\theta).$$

The ratio of fourth-harmonic amplitude to constant term is $1/11$, invariant under orthogonal coordinate changes. For a positive separated quartic $Ax^4 + By^4$ that ratio is $1/3$, so no such separation exists. Central symmetry would force any separated scalar half-wells to be even, and does not provide another representation. This is orthogonal, kinetic-energy-preserving nonseparability; no statement about arbitrary nonlinear changes of variables is needed. Also $c'(0) = 0$ but $P_s(0, Z) = -8Z_1^3Z_2$, whose ratio to V_0 varies with direction. The family is therefore not multiplicative in fixed coordinates.

Equation (7.13) gives

$$U_\iota(s) = C_\iota(1 + 1000s^2)^{1/3}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}s^2}(1 + 1000s^2)^{1/3} = \frac{2000(3 - 1000s^2)}{9(1 + 1000s^2)^{5/3}} > 0$$

on U. Its unique minimizer is zero, so Theorem 7 proves the stated conclusion. The class has genuine excited confined solutions: take $s = k = 0$ and a nonzero normal oscillation along either coordinate axis. Each axis is invariant and $P_s(0, Z) = 0$ there. This observation does not establish capture of any larger set of initial data. ∎

8 Effective transport in the energy clock

At zero residual energy, the unnormalized shell in (4.5) collapses and its zero-force condition becomes automatic. Energy normalization retains a nondegenerate normal shell

and an infinite slow clock. The resulting limit law keeps both the distribution of selected velocities and the normal invariant measures conditional on those velocities. Smooth frozen first integrals retain further restrictions that the full force hull discards.

In this section the actual tail satisfies (2.3), is excited, and has $E_\infty = 0$. Unless explicitly added, neither joint convexity nor a corrector is assumed. All assertions are necessary laws for actual trajectories, not converse realization statements.

8.1 Exact normalization and the two small scales

Put $\gamma = (p-2)/(2p) = 1-\nu$ and define

$$\frac{\mathrm{d}\tau}{\mathrm{d}\sigma} = E, \qquad v = \frac{k}{E}, \qquad z = E^{-1/p}Z, \qquad w = E^{-1/2}W, \qquad x = (z,w), \qquad \delta = \varepsilon E^{-\gamma}. \tag{8.1}$$

Here v is the normalized selected velocity, not the corrected velocity of (7.9). Proposition 6.1 gives $k = O(H)$, $E/H \to 1$ and $E \geq c/(1+\sigma)$ on a sufficiently late tail. Hence $\tau \to \infty$ and (s,v,x) remains in a fixed compact set. Write $|v| \leq V_*$ there. Define

$$\begin{gathered} J_s(x) = |w|^2/2 + P(s,z), \qquad f_s(x) = P_s(s,z), \qquad X_s(x) = (w, -P_z(s,z)), \\ \mathcal{L}_s\phi = w\cdot\phi_z - P_z\cdot\phi_w, \qquad \mathcal{D}\phi = \frac{z}{p}\cdot\phi_z + \frac{w}{2}\cdot\phi_w. \end{gathered} \tag{8.2}$$

Direct differentiation of the exact critical equations gives

$$\begin{aligned} J_s(x) &= 1 - E|v|^2/2, & s_\sigma &= Ev, \\ v_\sigma &= -av - f_s(x) + aE|v|^2 v, & x_\sigma &= \delta^{-1}X_s(x) + aE|v|^2(z/p, w/2), \\ E_\sigma &= -aE^2|v|^2, & \delta_\sigma &= -a\delta(1-\gamma E|v|^2). \end{aligned} \tag{8.3}$$

Equivalently, in energy time,

$$\begin{aligned} s_\tau &= v, & v_\tau &= E^{-1}[-av - f_s(x)] + a|v|^2 v, \\ x_\tau &= (E\delta)^{-1}X_s(x) + a|v|^2(z/p, w/2), & E_\tau &= -aE|v|^2. \end{aligned} \tag{8.4}$$

The reciprocal lower bound implies $\delta/E^N \to 0$ for every fixed $N \geq 0$. All tests below are defined on an open neighborhood of a fixed compact near-unit-energy region containing the late tail, so their derivatives are unambiguous. The general clock comparisons are only

$$c\log(1+\sigma) \leq \tau(\sigma) + C, \qquad \tau(\sigma) = o(\sigma). \tag{8.5}$$

The second follows by averaging $E \to 0$. Thus $\log\log t$ is a lower comparison scale, not a universal exact energy-time law.

Use the already defined invariant measures at unit energy and put

$$\mathcal{Q}(s) = -\frac{1}{a}\mathcal{G}(s,1). \tag{8.6}$$

By Proposition 4.2, this is an upper semicontinuous set-valued map with uniformly bounded nonempty compact convex values and closed graph on the selected compact neighborhood. Lower semicontinuity is not assumed.

8.2 Instantaneous attraction to the force graph

Proposition 8.1 (Damped-memory graph attraction)**.** Every tail under consideration satisfies

$$\operatorname{dist}\big((s(\sigma), v(\sigma)), \operatorname{graph} \mathcal{Q}\big) \longrightarrow 0. \tag{8.7}$$

In particular, if $(s(\sigma_j), v(\sigma_j)) \to (s_*, v_*)$, then $v_* \in \mathcal{Q}(s_*)$.

Proof. Fix a late starting time S and introduce the probability measure

$$\mu_\sigma = a \int_S^\sigma e^{-a(\sigma-u)} \delta_{x(u)} \, \mathrm{d}u + e^{-a(\sigma-S)} \delta_{x(S)}. \tag{8.8}$$

Variation of constants in (8.3) gives

$$v(\sigma) + \frac{1}{a} \int f_{s(\sigma)}(x) \, \mathrm{d}\mu_\sigma(x) \longrightarrow 0. \tag{8.9}$$

To verify the freezing and remainder errors, split the integral into the recent window $[\sigma - L, \sigma]$ and the remote part. The latter has mass at most e^{-aL}, apart from the initial atom with the same bound. On the recent window,

$$\sup |s(u) - s(\sigma)| \le V_* L E(\sigma - L) \longrightarrow 0, \qquad \sup E(u) \longrightarrow 0.$$

These bounds handle the frozen force and the cubic remainder, respectively. First let $\sigma \to \infty$ for fixed L, then $L \to \infty$. The same splitting proves

$$\int |J_{s(\sigma)}(x) - 1| \, \mathrm{d}\mu_\sigma(x) \longrightarrow 0. \tag{8.10}$$

For a C^1 normal test use the exact identity

$$\mathcal{L}_{s(u)} \phi(x(u)) = \delta(u) \frac{\mathrm{d}}{\mathrm{d}u} \phi(x(u)) - a\delta(u) E(u) |v(u)|^2 \mathcal{D}\phi(x(u)).$$

Integration by parts on the recent window against the exponential weight gives vanishing boundary and integral terms, since

$$a\delta + \delta_\sigma = a\gamma E |v|^2 \delta.$$

The remote part is bounded by Ce^{-aL}. Freezing s as above consequently gives

$$\int \mathcal{L}_{s(\sigma)} \phi \, \mathrm{d}\mu_\sigma \longrightarrow 0. \tag{8.11}$$

At any joint cluster point (s_*, v_*), extract a weakly convergent subsequence of μ_σ. Equations (8.10)–(8.11) put its limit in $\mathcal{M}(s_*, 1)$; equation (8.9) gives its force mean $-av_*$. This proves the cluster assertion and, by compactness, (8.7). ∎

The conclusion concerns distance to the graph. Closed-graph compactness alone does not yield the stronger same-fiber assertion $\operatorname{dist}(v(\sigma), \mathcal{Q}(s(\sigma))) \to 0$. That assertion does follow if $\mathcal{Q}$ is Hausdorff-continuous on the relevant compact set, in particular in the singleton case below. No failure of same-fiber attraction for the present Hamiltonian class is claimed; it is simply not supplied by this argument.

8.3 A joint-measure limit law

For any sequence $\tau_j \to \infty$, write on eventually defined translated windows

$$S_j(u) = s(\tau_j + u), \quad V_j(u) = v(\tau_j + u), \quad X_j(u) = x(\tau_j + u), \quad e_j(u) = E(\tau_j + u),$$
$$R_j(u) = \log \frac{e_j(u)}{e_j(0)}. \tag{8.12}$$

The exact variables here are viewed as functions of τ through its increasing inverse. The exponent ν continues to denote $1/2 + 1/p$; we use Λ_u for the joint limiting probability measure.

Theorem 8 (Energy-time transport with conditional invariance)**.** *After extraction, there are complete locally Lipschitz functions $S : \mathbb{R} \to K_0$ and $R : \mathbb{R} \to \mathbb{R}$, with $R(0) = 0$, and a measurable family of probability measures $\Lambda_u(\,\mathrm{d}v,\,\mathrm{d}x)$ on a fixed compact state set, with the following properties.*

1. *$S_j \to S$ and $R_j \to R$ locally uniformly. For every continuous state test G and compactly supported continuous time test χ,*

$$\int \chi(u) G(V_j(u), X_j(u))\,\mathrm{d}u \longrightarrow \int \chi(u) \int G(v,x)\,\mathrm{d}\Lambda_u\,\mathrm{d}u. \tag{8.13}$$

   Tests continuous also in u and the converging parameter $S_j(u)$ are permitted.

2. *For almost every u, Λ_u is supported on $J_{S(u)}(x) = 1$. In its disintegration*

$$\Lambda_u(\,\mathrm{d}v,\,\mathrm{d}x) = \eta_u(\,\mathrm{d}v)\mu_{u,v}(\,\mathrm{d}x), \tag{8.14}$$

   one has $\mu_{u,v} \in \mathcal{M}(S(u), 1)$ for η_u-almost every v.

3. *For almost every u, simultaneously for every C^1 velocity test ψ,*

$$\int \nabla\psi(v) \cdot [-av - f_{S(u)}(x)]\,\mathrm{d}\Lambda_u = 0. \tag{8.15}$$

4. *Define $m(u) = \int v\,\mathrm{d}\Lambda_u$ and $q_2(u) = \int |v|^2\,\mathrm{d}\Lambda_u$. Then almost everywhere*

$$S' = m = -\frac{1}{a}\int f_{S(u)}\,\mathrm{d}\Lambda_u, \qquad R' = -aq_2, \qquad \eta_u(\mathcal{Q}(S(u))) = 1. \tag{8.16}$$

   In particular,

$$S' \in \mathcal{Q}(S), \qquad R' = -a\left(|S'|^2 + \int |v - m|^2\,\mathrm{d}\Lambda_u\right) \le -a|S'|^2. \tag{8.17}$$

5. *For any C^1 frozen first integral $\Phi(s,x)$ on a neighborhood of the near-unit shells, satisfying $\mathcal{L}_s\Phi(s,\cdot)=0$, there is a locally Lipschitz function c_Φ such that*

$$\begin{aligned}\Phi(S_j(u),X_j(u)) &\longrightarrow c_\Phi(u) \quad \textit{locally uniformly},\\ \Lambda_u\{(v,x):\Phi(S(u),x)=c_\Phi(u)\} &= 1 \quad \textit{for almost every } u,\end{aligned} \tag{8.18}$$

and

$$c_\Phi' = \int \big[v\cdot\Phi_s(S,x)+a|v|^2\mathcal{D}\Phi(S,x)\big]\,\mathrm{d}\Lambda_u. \tag{8.19}$$

For a finite or countable family these statements hold together, with a common full-measure set in the countable case.

6. *The relative energy and original logarithmic clock satisfy, locally uniformly,*

$$\frac{e_j(u)}{e_j(0)} \longrightarrow e^{R(u)}, \qquad e_j(0)[\sigma(\tau_j+u)-\sigma(\tau_j)] \longrightarrow \int_0^u e^{-R(r)}\,\mathrm{d}r. \tag{8.20}$$

Proof. *Compactness and the moment identities.* The bounds $|S_j'|\le V_*$, $R_j'=-a|V_j|^2$ and $R_j(0)=0$ give local uniform compactness on every bounded interval. Diagonal extraction gives complete limits. Weak compactness of $\mathrm{d}u\,\delta_{(V_j(u),X_j(u))}$ and disintegration give (8.13). Uniform convergence of S_j extends it to the stated parameter-dependent tests. The shell identity in (8.3) and local uniform $e_j\to 0$ prove unit-shell support. Passing to distributional derivatives in $S_j'=V_j$ and $R_j'=-a|V_j|^2$ gives $S'=m$, $R'=-aq_2$. The identity $q_2=|m|^2+\int|v-m|^2\,\mathrm{d}\Lambda_u$ proves the variance formula once the force law is established. Finally, integrate $e_j(0)/e_j(u)=e^{-R_j(u)}$ to obtain (8.20), also for negative u with the oriented integral.

Conditional frozen invariance. Take compactly supported C^1 time test χ, C^1 velocity test h, and C^1 normal test ϕ. Put $\delta_j(u)=\delta(\tau_j+u)$. Multiply the derivative of $h(V_j)\phi(X_j)$ by $e_j\delta_j$. Equation (8.4) gives the principal term $h(V_j)\mathcal{L}_{S_j}\phi(X_j)$; the velocity derivative contributes terms bounded by $C\delta_j$, and the dilation contributes terms bounded by $Ce_j\delta_j$. Integration by parts against χ leaves the additional differentiated weight

$$(e_j\delta_j)' = -a\delta_j + a(\gamma-1)e_j\delta_j|V_j|^2. \tag{8.21}$$

Every remainder tends to zero locally uniformly. Thus

$$\int\chi(u)\int h(v)\mathcal{L}_{S(u)}\phi(x)\,\mathrm{d}\Lambda_u\,\mathrm{d}u = 0.$$

Countable dense families of velocity and normal tests give one full-measure set of times on which these identities hold for every test. Disintegration in v then gives the generator criterion for $\mu_{u,v}$, proving conditional invariance. The argument does not assume that v is frozen on normal windows.

Velocity stationarity and support. The selected equation gives exactly

$$e_j\frac{\mathrm{d}}{\mathrm{d}u}\psi(V_j) = \nabla\psi(V_j)\cdot[-aV_j - f_{S_j}(X_j)] + ae_j|V_j|^2\nabla\psi(V_j)\cdot V_j.$$

Integration by parts against χ makes the left side vanish in the limit because e_j and $e_j' = -ae_j|V_j|^2$ tend to zero locally uniformly. The last term does too. A countable family dense in the C^1 norm on the compact velocity ball gives one full-measure set on which (8.15) holds for all tests. Coordinate tests prove the force identity in (8.16); the quadratic test proves the useful work balance

$$\int v \cdot f_{S(u)}(x)\,\mathrm{d}\Lambda_u = -aq_2(u). \tag{8.22}$$

For a fixed admissible u, conditional invariance gives

$$\bar f(v) = \int f_{S(u)}\,\mathrm{d}\mu_{u,v}, \qquad \bar v(v) = -\bar f(v)/a \in \mathcal{Q}(S(u)).$$

Let $Q = \mathcal{Q}(S(u))$ and use the C^1 test $\psi(v) = \operatorname{dist}(v,Q)^2/2$. Its gradient is $v - \Pi_Q(v)$, where Π_Q is metric projection onto the closed convex set Q. The projection inequality gives

$$[v - \Pi_Q(v)] \cdot [v - \bar v(v)] \geq \operatorname{dist}(v,Q)^2.$$

Stationarity consequently yields

$$0 = -a\int [v - \Pi_Q(v)] \cdot [v - \bar v(v)]\,\mathrm{d}\eta_u \leq -a\int \operatorname{dist}(v,Q)^2\,\mathrm{d}\eta_u.$$

This proves support of the entire velocity marginal in Q, not only membership of its mean.

Transport and concentration of first integrals. The exact first-integral chain rule is

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\Phi(s,x) = v \cdot \Phi_s + a|v|^2\mathcal{D}\Phi. \tag{8.23}$$

Its right side is bounded on the compact state region, so the translated traces are equi-bounded and equi-Lipschitz. Every local uniform sublimit equals $\int \Phi(S(u),x)\,\mathrm{d}\Lambda_u$ almost everywhere by (8.13). Continuous representatives with this property agree everywhere; hence the whole already extracted subsequence converges locally uniformly to one c_Φ. Apply (8.13) also to the square of the trace. Its first and second moments are c_Φ and c_Φ^2, proving concentration. Distributional passage in (8.23) gives (8.19). This works for each fixed Φ without another extraction; intersecting countably many full-measure sets handles a countable family. As a check, $\Phi = J_s$ gives $c_\Phi = 1$, $\mathcal{D}\Phi = J_s$ and $\Phi_s = f_s$, so its transport identity is exactly (8.22). ∎

Invariant-measure limits for singularly perturbed systems are classical; Artstein–Vigodner [4] and the accessible invariant-measure formulation in Artstein [17, Theorem 3.1 and Remark 3.2] are direct precedents. Theorem 8 verifies the additional normalization, scale cancellation and exact dissipation for the shrinking-energy critical tail. It makes no ergodicity, mixing-rate or bounded-corrector assumption. The variance term is a possible defect of weak velocity convergence; the theorem does not itself construct a tail on which that defect is positive.

When $d = 1$, stationarity has a useful stronger form. With $\bar{f}(v)$ as above, the finite signed measure $[-av - \bar{f}(v)]\eta_u(\,\mathrm{d}v)$ has distributional derivative zero on $\mathbb{R}$. It is therefore a constant multiple of Lebesgue measure; compact support forces that constant to vanish. Hence $\bar{f}(v) = -av$ for η_u-almost every v. In higher dimension stationarity instead gives a divergence-free vector measure, which need not vanish. The projection argument is valid in every dimension.

8.4 Transported levels restrict the available force hull

For finitely many frozen first integrals $\Phi_1, \dots, \Phi_\ell$, define on their feasible parameter-level set

$$\begin{aligned} \mathcal{M}(s,c) &= \{\mu \in \mathcal{M}(s,1) : \mu\{\Phi_i(s,x) = c_i \ (1 \le i \le \ell)\} = 1\}, \\ \mathcal{Q}(s,c) &= -\frac{1}{a}\left\{ \int f_s \,\mathrm{d}\mu : \mu \in \mathcal{M}(s,c) \right\}. \end{aligned} \tag{8.24}$$

For the limiting levels $c_i = c_{\Phi_i}$, conditional invariance and concentration put almost every $\mu_{u,v}$ in $\mathcal{M}(S(u), c(u))$. These fibers are consequently nonempty; repeating the convex-projection proof yields

$$\eta_u(\mathcal{Q}(S(u), c(u))) = 1, \qquad S' \in \mathcal{Q}(S, c). \tag{8.25}$$

The levels evolve by (8.19), rather than being freely reset when a different invariant measure is chosen. Proposition 8.1 also applies to the augmented state

$$(s, \Phi_1(s,x), \dots, \Phi_\ell(s,x), v).$$

Indeed, equation (8.23) shows that each level changes by $O(E)$ on a fixed logarithmic-time memory window, so the occupation proof concentrates on its limiting level. This is again graph attraction. An arbitrary finite-tail same-level fiber can be empty; for $\Phi = J_s$ the actual level is $1 - E|v|^2/2$, whereas the frozen probabilities in (8.24) lie on $J_s = 1$.

If Φ has weighted homogeneous degree κ, meaning

$$\Phi(s, \lambda^{1/p} z, \lambda^{1/2} w) = \lambda^\kappa \Phi(s, z, w),$$

then $\mathcal{D}\Phi = \kappa\Phi$. Concentration and $R' = -aq_2$ imply

$$(e^{\kappa R} c_\Phi)' = e^{\kappa R} \int v \cdot \Phi_s(S, x) \,\mathrm{d}\Lambda_u. \tag{8.26}$$

If Φ_s has zero mean under every relevant frozen invariant measure, conditional invariance makes the right side zero. This conserves the energy-restored quantity in each translated limit. It does not assert a finite nonzero asymptotic action of the original trajectory.

For example, for independent scalar modes put

$$J_i = |w_i|^2/2 + g_i(s)|z_i|^p, \qquad \Phi_i = g_i(s)^{-1/p} J_i^\nu, \qquad g_i > 0. \tag{8.27}$$

These are C^1 frozen first integrals even at $J_i = 0$: their z_i, w_i and selected derivatives are bounded respectively by $CJ_i^{1/2}$, $CJ_i^{1/p}$ and CJ_i^{ν}, and extend by zero. On a fixed modal level the virial identity gives

$$\int g_i |z_i|^p \, \mathrm{d}\mu = \frac{J_i}{p\nu}.$$

Differentiation of (8.27) at fixed state then shows that every such invariant mean of $(\Phi_i)_s$ vanishes. Equation (8.26) gives constants $C_i = e^{\nu R} c_i \geq 0$. At least one is positive, since $\sum J_i = 1$. Define

$$U_C(s) = \sum_i C_i^{1/\nu} g_i(s)^{1/(p\nu)}. \tag{8.28}$$

Solving for each J_i and summing yields

$$e^{-R} U_C(S) = 1, \qquad e^R = U_C(S). \tag{8.29}$$

The same virial identity gives the restricted mean force $e^{-R} \nabla U_C(S) = \nabla \log U_C(S)$. Thus the restricted hull is a singleton and

$$S' = -\frac{1}{a} \nabla \log U_C(S), \qquad q_2 = |S'|^2, \qquad R' = -\frac{1}{a} |\nabla \log U_C(S)|^2. \tag{8.30}$$

For the perspective example (5.24), where $p = 4$ and $g_1 = (1+s)^{-3}$, $g_2 = (1-s)^{-3}$, this becomes $U_C = A/(1+s) + B/(1-s)$ with $A, B \geq 0$, $A + B > 0$. A recurrent nonconstant scalar oscillation cannot obey this gradient equation. The formula identifies explicitly the transported constraint missing from the unrestricted force hull; it is not a new existence assertion or a replacement for the original positive-action estimates.

8.5 Singleton force means and the convex rate obstruction

Corollary 8.1 (Qualitative closure without a corrector)**.** *Suppose $\mathcal{G}(s, 1) = \{b(s)\}$ on the relevant compact selected set. Then b is continuous there,*

$$v(\sigma) + b(s(\sigma))/a \longrightarrow 0, \tag{8.31}$$

and every translated limit in Theorem 8 satisfies

$$S' = -b(S)/a, \qquad V_j \longrightarrow -b(S)/a \quad \textit{locally uniformly}, \qquad R' = -|b(S)|^2/a. \tag{8.32}$$

In particular S is C^1 and the variance term vanishes.

Proof. Closed graph, uniform boundedness and singleton values imply continuity of b. Proposition 8.1 then gives (8.31). Its combination with local uniform convergence of S_j gives the strong velocity convergence, and the exact first- and second-moment identities give (8.32). ∎

This is uniqueness of the invariant force mean, not unique ergodicity. It supplies no integrable bound on the error in (8.31). If P is defined on an open coercive parameter neighborhood, the microcanonical calculation (7.3), which did not use convexity, identifies

$$b = -\frac{1}{n\nu}\nabla \log \mathcal{V}_1, \qquad S' = \frac{1}{an\nu}\nabla \log \mathcal{V}_1(S), \qquad R + \frac{1}{n\nu}\log \mathcal{V}_1(S) = \text{constant}. \quad (8.33)$$

Without singleton means this formula still identifies the microcanonical mean, but not an orbit's selected force.

Corollary 8.2 (Convex singleton means exclude reciprocal energy decay)**.** *Assume additionally that P is jointly convex on $U \times \mathbb{R}^n$, where U is open and convex, and that the actual compact tail lies in the interior of U. Suppose $\mathcal{G}(s,1)$ is a singleton on a neighborhood of K_0. Define*

$$U_{\text{eff}}(s) = \mathcal{V}_1(s)^{-1/(n\nu)}.$$

Then U_{eff} is positive, C^2 and convex on U, and its common force near the tail is $\nabla \log U_{\text{eff}}$. Every excited zero-residual-energy tail satisfying these hypotheses has

$$M := \operatorname*{argmin}_U U_{\text{eff}} \neq \varnothing, \qquad \operatorname{dist}(s(\sigma), M) \longrightarrow 0, \qquad v(\sigma) \longrightarrow 0, \quad (8.34)$$

and

$$\sigma E(\sigma) \longrightarrow \infty, \qquad \frac{\tau(\sigma)}{\log \sigma} \longrightarrow \infty. \quad (8.35)$$

Thus $E(\sigma) \asymp \sigma^{-1}$ and $\tau(\sigma) \asymp \log \sigma$ are excluded in this class, without a corrector.

Proof. Lemma 3 and Proposition 7.1 give the global volume definition and convexity. Every translated limit is a complete trajectory in K_0 of

$$S' = -\frac{\nabla U_{\text{eff}}(S)}{aU_{\text{eff}}(S)}, \qquad \frac{\mathrm{d}}{\mathrm{d}u}U_{\text{eff}}(S) = -\frac{|\nabla U_{\text{eff}}(S)|^2}{aU_{\text{eff}}(S)}.$$

The positive potential is bounded above and below on K_0. Its monotonicity therefore makes the backward integral of $|\nabla U_{\text{eff}}(S)|^2$ finite. That integrand is uniformly continuous, since S is Lipschitz and the gradient is uniformly continuous on the compact selected region. It tends to zero as $u \to -\infty$. A backward cluster point $s_- \in K_0$ thus satisfies $\nabla U_{\text{eff}}(s_-) = 0$, and convexity makes it a global minimizer on U. At every finite u, monotonicity yields

$$U_{\text{eff}}(S(u)) \leq \lim_{r \to -\infty} U_{\text{eff}}(S(r)) = U_{\text{eff}}(s_-) = \min_U U_{\text{eff}}.$$

The reverse inequality holds automatically. Every complete compact translated trajectory is consequently constant at a minimizer.

Each cluster point of the actual selected tail is the value at zero of one such translated limit. Compactness gives attraction to M, and continuity makes the common force tend to zero. Corollary 8.1 gives $v \to 0$. Finally the exact equation says

$$(1/E)_\sigma = a|v|^2 \longrightarrow 0.$$

Averaging gives $(1/E(\sigma))/\sigma \to 0$, hence $\sigma E(\sigma) \to \infty$. For every $L > 0$, eventually $E(\sigma) \geq L/\sigma$. Integration and division by $\log \sigma$ give $\liminf \tau/\log \sigma \geq L$; letting L increase proves the second limit. ∎

Combining (8.5) and (8.35) gives $\log \sigma = o(\tau)$ and $\tau = o(\sigma)$ under this corollary. The conclusion is minimizer-set attraction and a rate obstruction. It neither proves point selection when M has many points nor excludes all excited zero-residual-energy tails. If M is a singleton, compactness of course converts the proved attraction into point convergence.

8.6 Outer limits and accessible motion

Theorem 8 does not realize arbitrary paths of invariant measures. Frozen invariance, velocity stationarity and smooth first-integral transport can still leave transition constraints between invariant normal components. A general nonintegrable flow need not have enough smooth first integrals to describe all such constraints. Set-valued asymptotic-pseudotrajectory results, such as [18], likewise do not turn an outer inclusion into realization or point selection. We do not require an additional global localization of that theory here.

The genuine cold trajectory of Theorem 11 shows why a universal single-valued velocity law cannot replace these restrictions. Its nonconstant smooth periodic scalar target $S_0(\tau)$ satisfies the strong tracking limits (19.46). Translation by its period gives S_0, up to phase, as an actual energy-time limit. At every interior value of its oscillation interval, the two traversals have opposite nonzero velocities. They cannot both satisfy an autonomous single-valued law $S' = G(S)$. This obstruction comes from the actual constructed orbit, not from a freely chosen force-hull path. In that construction velocity tracking is strong and the variance term of (8.17) is zero.

9 Rigidity of flat phase-volume minima

The convex volume energy can have many minimizers. Their frozen normal geometry is nevertheless rigid: equal maximal phase volume along a selected segment forces the entire frozen potential to be constant there. This conclusion is geometric. It does not supply the quantitative averaging or force integrability missing from Corollary 8.2.

For the geometric result alone, let $U \subset \mathbb{R}^d$ be open and convex and let P be continuous, jointly convex, positive for $Z \neq 0$, and positively homogeneous of degree $p > 1$ in Z, with locally uniform normal coercivity. Put

$$B_s = \{Z : P(s, Z) \leq 1\}, \qquad d_B(s) = \mathrm{vol}_n(B_s)^{1/n}.$$

The bodies need not be centrally symmetric. Fiber inclusion and Brunn–Minkowski make d_B concave. In the critical-flow setting, $D_{\mathrm{vol}} = C_{n,p}^{1/n} d_B$ by (7.5).

Theorem 9 (Equality in volume concavity forces a constant family)**.** *If for some distinct* $s_0, s_1 \in U$ *and* $0 < \theta < 1$,

$$d_B((1-\theta)s_0 + \theta s_1) = (1-\theta)d_B(s_0) + \theta d_B(s_1), \tag{9.1}$$

then $B_s = B_{s_0}$ *and* $P(s, Z) = P(s_0, Z)$ *for every* $s \in [s_0, s_1]$ *and every* Z*. In particular* $d_B(s_0) = d_B(s_1)$*. Conversely, a constant family gives equality. Thus volume-root concavity is strict on every segment whose endpoint potentials differ.*

Proof. For a fixed dual normal vector w define the partial conjugate

$$P^*(s, w) = \sup_Z \{w \cdot Z - P(s, Z)\}.$$

Superlinear normal growth gives a finite attained supremum. If Z_0, Z_1 are endpoint maximizers, joint convexity at their interpolant gives

$$P^*(s_\theta, w) \geq (1-\theta)P^*(s_0, w) + \theta P^*(s_1, w), \qquad s_\theta = (1-\theta)s_0 + \theta s_1. \tag{9.2}$$

This is concavity in the selected parameter; the conjugate remains convex in its dual normal variable.

Let $h_s(w) = \max_{Z \in B_s} w \cdot Z$. The origin is interior, so $h_s(w) > 0$ for $w \neq 0$, without any symmetry assumption. Homogeneity identifies $P(s, \cdot)$ with the pth power of the gauge of B_s. Maximizing first on the body and then radially gives

$$P^*(s, w) = \max_{r \geq 0} \{r h_s(w) - r^p\} = (p-1)p^{-p'} h_s(w)^{p'}, \qquad p' = \frac{p}{p-1} > 1. \tag{9.3}$$

The maximizing radius is $(h_s(w)/p)^{1/(p-1)}$; at $w = 0$ both sides vanish. Hence (9.2) implies

$$h_{s_\theta}(w)^{p'} \geq (1-\theta)h_{s_0}(w)^{p'} + \theta h_{s_1}(w)^{p'}. \tag{9.4}$$

Set $B_i = B_{s_i}$ and $B_\theta^{\mathrm{mix}} = (1-\theta)B_0 + \theta B_1$. Joint convexity and Brunn–Minkowski give

$$d_B(s_\theta) \geq \mathrm{vol}_n(B_\theta^{\mathrm{mix}})^{1/n} \geq (1-\theta)d_B(s_0) + \theta d_B(s_1).$$

Equality (9.1) forces equal volumes in the inclusion $B_\theta^{\mathrm{mix}} \subset B_{s_\theta}$. Nested full-dimensional compact convex bodies of equal volume are equal: a proper inclusion would give, by separation and an interior ball, a positive-volume region in the larger body outside the smaller one. Consequently

$$h_{s_\theta}(w) = (1-\theta)h_{s_0}(w) + \theta h_{s_1}(w).$$

Insert this into (9.4). Strict convexity of the positive scalar power $r^{p'}$ gives the reverse inequality, with equality only if $h_{s_0}(w) = h_{s_1}(w)$. Thus the endpoint supports coincide for every w, so $B_0 = B_1$ and their volumes agree.

A concave function agreeing with its affine chord at one interior point agrees with that chord on the whole segment. Indeed, subtract the chord; a positive value at any other point could be combined with the opposite endpoint to force a positive value at the given interior zero. Applying this to d_B shows it is constant on $[s_0, s_1]$. For any subsegment the same equality argument now makes its endpoint bodies identical. All bodies on the original segment therefore coincide; so do their gauges and their pth powers $P(s, \cdot)$. ∎

The proof uses no classification of equality cases in Brunn–Minkowski and in particular does not assume centered or symmetric bodies. Merely translating equal-volume bodies would violate the concavity of their positive support powers unless the translation vanished.

9.1 Tangential degeneracy of the minimizing set

Return to the C^2, $p > 2$ critical setting of Section 7. Recall that $\nu = 1/2 + 1/p$ and that $\iota > 0$ fixes the phase-volume root, so $U_\iota = (\iota/D_{\mathrm{vol}})^{1/\nu}$. Whenever the effective minimizer set is nonempty, it is independent of ι:

$$M = \operatorname*{argmin}_U U_\iota = \operatorname*{argmax}_U D_{\mathrm{vol}}. \tag{9.5}$$

It is convex by concavity of D_{vol}.

Corollary 9.1 (Frozen and tangential rigidity on flat minima)**.** *For every $u, v \in M$ and every Z,*

$$P(u, Z) = P(v, Z). \tag{9.6}$$

Let $T = \operatorname{span}(M - M)$ and let Π_T be orthogonal projection onto this selected subspace. At every $u \in M$, including relative boundary points,

$$\Pi_T P_s(u, Z) = 0, \qquad P_{ss}(u, Z)v = 0, \qquad P_{Zs}(u, Z)v = 0 \quad (v \in T). \tag{9.7}$$

On any compact selected neighborhood where nearest points in M exist and remain in a compact subset of U, there is a modulus $\omega_0(r) \to 0$ as $r \downarrow 0$ such that, for $r = \operatorname{dist}(s, M)$ sufficiently small,

$$|\Pi_T P_s(s, Z)| \le |Z|^p r \omega_0(r). \tag{9.8}$$

On a uniformly coercive actual tail this is bounded by $CHr\omega_0(r)$. If the selected Hessian is locally Lipschitz, the bound improves to CHr^2.

Proof. Constancy of D_{vol} on each segment in M and Theorem 9 prove (9.6). Fix $u \in M$. The vectors $y - u$, $y \in M$, span T, because $y - z = (y - u) - (z - u)$. On each segment $[u, y]$, the selected restriction of $P(\cdot, Z)$ is constant. Its one-sided derivative at u gives $P_s(u, Z) \cdot (y - u) = 0$. Its second derivative vanishes on the open segment and also at its endpoints by C^2 continuity. Thus

$$(y - u)^T P_{ss}(u, Z)(y - u) = 0.$$

The full joint Hessian is positive semidefinite. A vector with zero quadratic form for such a matrix lies in its kernel; apply this to $(y-u,0)$. Both Hessian identities in (9.7) follow for these spanning directions and hence for all of T.

For the local estimate let $u \in M$ be a nearest point to s and $r = |s-u|$. All connecting segments in the stated localization lie in a common compact subset of U. Symmetry of P_{ss} and (9.7) give $\Pi_T P_{ss}(u,\omega) = 0$ for every unit normal vector ω. Therefore

$$\Pi_T P_s(s,\omega) = \int_0^1 \Pi_T[P_{ss}(u+t(s-u),\omega) - P_{ss}(u,\omega)](s-u)\,\mathrm{d}t.$$

Uniform continuity on the compact parameter region times the unit sphere bounds this by $r\omega_0(r)$. Normal homogeneity yields (9.8), including $Z = 0$ directly. Coercivity gives $|Z|^p \le CH$ along the actual tail. A locally Lipschitz Hessian permits $\omega_0(r) \le Cr$, proving the last assertion. ∎

The nearest-point localization is substantive because M is closed relative to U, not necessarily closed in the ambient space. Even within this localization, (9.8) does not prove integrability of the tangential force along the actual orbit: no rate or summability of $Hr\omega_0(r)$ has been obtained. Thus flat-volume rigidity does not replace the corrector in Theorem 7, produce a positive limiting phase volume from common means alone, or settle point selection for a nontrivial minimizing set in Corollary 8.2.

10 Continuum targets and dissipative force control

The connectedness of a bounded trajectory's omega-limit set is a general topological restriction. Under an available force margin, the finite-dissipation identity places no further restriction on the shape of a positional target continuum. The next construction makes this statement quantitative.

Lemma 4 (A slow smooth traversal of any continuum)**.** *Let $U_0 \subset \mathbb{R}^m$ be open and connected, let $s_e \in U_0$, and let $K \subset U_0$ be nonempty, compact and connected. There is a C^∞ curve $Q : [0,\infty) \to U_0$ with relatively compact image such that*

$$Q(0) = s_e, \qquad Q^{(j)}(0) = 0 \ (j \ge 1), \qquad \omega(Q) = K, \qquad \sup_{v\ge 0} |Q^{(j)}(v)| < \infty \ (j \ge 1). \tag{10.1}$$

For $M > 0$, the curve

$$S_M(u) = Q\left(\log \frac{M+u}{M}\right) \tag{10.2}$$

has the same omega-limit set and satisfies, with constants independent of M,

$$|S_M^{(j)}(u)| \le \frac{C_j}{(M+u)^j} \quad (1 \le j \le 3), \qquad \int_0^\infty |S_M'(u)|^2\,\mathrm{d}u \le \frac{C_1^2}{M}. \tag{10.3}$$

Proof. For each $\delta > 0$, any two points of K are connected by a finite δ-chain in K. Indeed, the points chain-connected to a fixed point form a relatively open and closed subset of K: every relative δ-ball lies in a single chain component. Connectedness gives one component.

Choose $\delta_n \downarrow 0$ with $\delta_1 < \frac{1}{2}\operatorname{dist}(K, U_0^c)$. At stage n, visit a finite δ_n-net of K, joining successive net points and the endpoint of the preceding stage by finite δ_n-chains in K. Repeat an endpoint if necessary so that each stage contributes a step. The resulting sequence (x_j) lies in K and satisfies

$$|x_{j+1} - x_j| \longrightarrow 0, \qquad \omega(x_j) = K.$$

Join s_e to x_0 by a finite polygonal path in U_0, which exists because an open connected Euclidean domain is polygonally connected. The initial path and all later edges have compact closure in U_0; the later edges stay in the fixed δ_1-neighborhood of K.

Choose a nondecreasing $\vartheta \in C^\infty([0,1])$ from zero to one whose positive-order derivatives vanish at both endpoints. On successive unit intervals interpolate every edge by

$$Q(j+\tau) = x_j + \vartheta(\tau)(x_{j+1} - x_j), \qquad 0 \le \tau \le 1,$$

using the same rule on the finite initial path. The concatenation is C^∞, is flat at zero, and has every fixed-order derivative bounded by a constant times the maximal edge length. The distance of any late interpolating segment from K is at most its edge length and tends to zero. Visits to the nets supply every point of K as a cluster point, so $\omega(Q) = K$.

The time change in (10.2) is increasing and unbounded. With $v = \log((M+u)/M)$,

$$S_M' = \frac{Q'(v)}{M+u}, \quad S_M'' = \frac{Q''(v) - Q'(v)}{(M+u)^2}, \quad S_M''' = \frac{Q'''(v) - 3Q''(v) + 2Q'(v)}{(M+u)^3}.$$

These identities and integration of $(M+u)^{-2}$ prove (10.3). ∎

The curve approaches K and need not lie in K. Requiring a path within K would incorrectly exclude connected sets that are not path connected. No definability assumption is imposed on the target or on the initial datum that will encode it.

10.1 What a force hull proves at the relaxed level

Let $b_i(s,H)$, $1 \le i \le r$, be continuous frozen force branches. The relaxed critical equations are

$$s' = k, \qquad k' = -ak - b, \qquad H' = k \cdot b, \qquad b \in \mathcal{B}(s,H) := \operatorname{co}\{b_1(s,H), \dots, b_r(s,H)\}. \tag{10.4}$$

Every solution satisfies $(H + |k|^2/2)' = -a|k|^2$. This is positional control subject to an energy budget. At $k = 0$ the branch vector fields are $(0, -b_i, 0)$, so their convex hull has no interior in the full (s,k,H) space. No nonconstant controlled loop can return to its initial full state.

If zero is interior to $\mathcal{B}(s,H)$ throughout a compact parameter set D, there is a uniform $r_* > 0$ with

$$\overline{B}(0,r_*) \subset \mathcal{B}(s,H) \qquad ((s,H) \in D). \tag{10.5}$$

To prove this, the continuous support margin

$$\gamma(s,H) = \min_{|v|=1} \max_i v \cdot b_i(s,H)$$

is positive and has a positive minimum on D. The support-function characterization of convex set inclusion gives (10.5).

If the branches are C^j, $j \geq 1$, one can also choose C^j mixing weights, strictly positive uniformly, for all sufficiently small forces. Here are the details. At one parameter value, choose a small $d > 0$ such that

$$-\frac{d\sum_i b_i}{1-rd} \in \mathcal{B}.$$

A convex representation of this vector, multiplied by $1-rd$ and augmented by d in every coordinate, represents zero with positive coefficients. The matrix with columns $(b_i, 1)$ has full row rank, so

$$R = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}$$

is a smooth right inverse. It corrects that positive representation at nearby parameters. A finite partition of unity glues these local representations into $\lambda^0(s,H)$ while preserving the equations and positivity. Then

$$\lambda(s,H,b) = \lambda^0(s,H) + R(s,H)(b,0)$$

has sum one, represents b, and remains uniformly positive for $|b|$ sufficiently small.

Proposition 10.1 (Relaxed realization with arbitrarily small dissipation)**.** Suppose $U_0 \subset \mathbb{R}^m$ is open and connected, $0 < H_- < H_+$, and zero is interior to the branch force hull throughout $U_0 \times (H_-, H_+)$. Fix (s_e, H_e) there. For every continuum $K \subset U_0$ and every $\delta_E > 0$, equations (10.4) have a solution from $(s_e, 0, H_e)$ with

$$\omega(s) = K, \qquad k \to 0, \qquad H_- < H \leq H_e < H_+, \qquad a\int_0^\infty |k|^2 \,\mathrm{d}u < \delta_E.$$

For differentiable branches the mixing weights can be chosen differentiably and strictly positive.

Proof. Take $S = S_M$ from Lemma 4 and set

$$k = S', \quad E(u) = H_e - a\int_0^u |S'(v)|^2 \,\mathrm{d}v, \quad H = E - \tfrac{1}{2}|S'|^2, \quad b = -aS' - S''. \tag{10.6}$$

A compact neighborhood of the target image and a compact energy interval around H_e have a uniform force ball. For large M, (10.3) puts b inside it and makes

$$a\int_0^\infty |S'|^2 \,\mathrm{d}u + \tfrac{1}{2}\sup|S'|^2 \leq aC_1^2/M + C_1^2/(2M^2)$$

smaller than the available energy margin, while the first term is less than δ_E. The first two equations in (10.4) hold by construction, and $H' = -a|S'|^2 - S' \cdot S'' = k \cdot b$ proves the third. The mixing construction above supplies the last assertion. ∎

Geometric shadowing of arbitrary slow curves through periodic-action networks is established in [7], following the finite-concatenation framework of [6]. Proposition 10.1 does not replace a shadowing theorem. For an actual mechanical trajectory, the required additional facts are a realizable transition network, vanishing rather than fixed tracking error, control of the decreasing energy, and access from the singular cold manifold. We prove these for one scalar selected coordinate in the next sections and then synchronize independent copies.

11 A convex homogeneous valley with a hyperbolic normal network

We construct a local objective with one selected coordinate $s \in \mathbb{R}$ and two normal coordinates $q = (x, z) \in \mathbb{R}^2$. The degree of homogeneity will be a sufficiently large, but fixed and finite, even integer p. The dynamical scaling used later is $a = 2/p$ and $\alpha = 1 + a$.

11.1 An algebraic table and its gauge

Fix $A > B > 0$ and a sufficiently small number $\delta_b > 0$. Consider the convex body

$$\mathcal{C}_{\delta_b} = \left\{ (x, z) : \frac{x^2}{A^2} + \frac{z^2}{B^2} + \delta_b \frac{z^4}{B^4} \le 1 \right\}. \tag{11.1}$$

Its defining function has Hessian

$$\begin{pmatrix} 2/A^2 & 0 \\ 0 & 2/B^2 + 12\delta_b z^2/B^4 \end{pmatrix} > 0.$$

Consequently $\mathcal{C}_{\delta_b}$ is compact and strictly convex, and its boundary is smooth with strictly positive curvature. It contains the origin in its interior and is invariant under both coordinate reflections.

Let g be its Minkowski gauge. Put

$$L(q) = \frac{x^2}{A^2} + \frac{z^2}{B^2}, \qquad D(q) = \frac{z^4}{B^4}.$$

The equation $q/g(q) \in \partial \mathcal{C}_{\delta_b}$ gives

$$g(q)^2 = \frac{L(q) + \sqrt{L(q)^2 + 4\delta_b D(q)}}{2}, \qquad g(0) = 0. \tag{11.2}$$

In particular g is semialgebraic, convex, positively homogeneous of degree one, positive away from zero, and smooth on $\mathbb{R}^2 \setminus \{0\}$.

Lemma 5. *For every integer $p > 2$, the function $V_p = g^p$ is convex and homogeneous of degree p. It is semialgebraic and belongs to $C^r(\mathbb{R}^2)$ for every integer $r < p$. Its derivatives of order $j < p$ vanish at zero and satisfy*

$$\|D^j V_p(q)\| \le C_j |q|^{p-j} \qquad (q \neq 0). \tag{11.3}$$

Moreover, there are $v_0, \lambda_0 > 0$, depending on p, such that

$$V_p(q) \ge v_0 |q|^p, \qquad D^2 V_p(q) \ge \lambda_0 |q|^{p-2} I \qquad (q \neq 0). \tag{11.4}$$

Proof. Convexity follows by composing the convex nonnegative gauge with the convex increasing function $r \mapsto r^p$. Smoothness on the unit circle and homogeneity give (11.3). Extending each derivative of order less than p by zero at the origin, induction on its order proves the claimed differentiability there. Positivity and compactness of the unit circle give the first inequality in (11.4).

For the second inequality, positive curvature of the level sets of g implies that $D^2 g$ is positive on nonzero vectors tangent to those level sets. Since g is convex, $D^2 g$ is positive semidefinite, and

$$D^2(g^p) = p g^{p-1} D^2 g + p(p-1) g^{p-2} \nabla g \otimes \nabla g.$$

The second term is positive on every vector not tangent to a level set; the first is positive on every nonzero tangent vector. Thus $D^2 V_p$ is positive definite away from zero. Compactness on the unit circle, followed by homogeneity, gives the uniform lower bound. ∎

11.2 Persistence of a transverse homoclinic

The boundary in (11.1) is the symmetric quartic perturbation

$$\frac{x^2}{A^2} + \frac{z^2}{B^2} = 1 + \varepsilon_b \frac{z^4}{B^4}, \qquad \varepsilon_b = -\delta_b.$$

By Delshams–Fedorov–Ramírez-Ros [9, Theorem 4.4, p. 1162], for sufficiently small nonzero ε_b the major-axis billiard orbit has transverse primary bi-asymptotic trajectories. In the billiard flow these are homoclinic to the same major-axis periodic orbit. We fix one such δ_b once and for all. No limit in δ_b will subsequently be taken.

These homoclinics are regular. Their tails approach the normally incident major-axis impacts; their finite middle portions have no grazing impacts. The local return neighborhoods and the finitely many connecting orbit segments used below may therefore be chosen at positive distance from the billiard singularity set.

To pass to a smooth mechanical Hamiltonian, introduce the auxiliary family

$$V(q;\varepsilon) = g(q)^{1/\varepsilon}, \qquad \varepsilon \downarrow 0. \tag{11.5}$$

The final potential will be a single member with $\varepsilon = 1/p$ and p even. Fix a finite differentiability order $r \ge 4$; the degree is chosen large enough that the frozen periodic

families and section maps used below are C^3 in their parameters and section variables. For $1/\varepsilon > r+2$, all differentiability requirements at $q=0$ hold. The following calculations verify the steep-potential Conditions I–IV of Rapoport–Rom-Kedar–Turaev [13, Section 3.1].

Interior convergence. If $\mathcal{K} \Subset \operatorname{int} \mathcal{C}_{\delta_b}$, then $g \le r_0 < 1$ on $\mathcal{K}$. For $0 \le j \le r+1$, the chain rule and the homogeneous derivative bounds for g give

$$\sup_{\mathcal{K}} |D^j V(\cdot;\varepsilon)| \le C_j (1+\varepsilon^{-1})^j r_0^{\varepsilon^{-1}-j} \longrightarrow 0.$$

The same estimate includes the origin by continuous extension.

Boundary representation and monotonicity. On a fixed collar of the boundary take

$$Q(q) = g(q) - 1, \qquad W(Q;\varepsilon) = (1+Q)^{1/\varepsilon}.$$

The pattern Q is smooth, is zero on the boundary, and has nonzero gradient there: Euler's identity gives $q \cdot \nabla g(q) = g(q) = 1$. The representation $V = W(Q;\varepsilon)$ is exact on this collar, and

$$\partial_Q W = \varepsilon^{-1}(1+Q)^{1/\varepsilon - 1} > 0.$$

If a pattern on a neighborhood of the whole closed table is desired, extend Q smoothly through a smaller interior region. This auxiliary extension does not change either V or its collar representation.

Barrier limit and inverse width. For each fixed collar coordinate $Q < 0$, $W(Q;\varepsilon) \to 0$; for each fixed $Q > 0$, it tends to $+\infty$. The inverse offset is

$$\mathcal{Q}(W;\varepsilon) = W^{\varepsilon} - 1.$$

On every interval $0 < W_1 \le W \le W_2 < \infty$ its value is $O(\varepsilon)$, and, for $1 \le j \le r+1$,

$$\partial_W^j \mathcal{Q} = \varepsilon(\varepsilon-1)\cdots(\varepsilon-j+1) W^{\varepsilon-j} = O(\varepsilon) \tag{11.6}$$

uniformly. This verifies the inverse-width condition to the required order.

The persistence theorem of Rapoport–Rom-Kedar–Turaev [13, Theorem 5, Section 3.4] therefore applies: the hyperbolic regular periodic orbit and its transverse regular homoclinic persist for every sufficiently small ε. Here the theorem number refers also to the publicly available arXiv version of that paper. Choosing a sufficiently large finite even integer p gives the following result.

Proposition 11.1. For a sufficiently large finite even p, the Hamiltonian

$$\mathcal{H}(q,v) = \frac{1}{2}|v|^2 + V(q), \qquad V = g^p, \tag{11.7}$$

has at energy $1/2$ a hyperbolic axial periodic orbit L_- with a transverse homoclinic. It also has a nonaxial hyperbolic periodic orbit L_+, transversely connected to L_- in both directions. The two periodic orbits, finite connecting pieces, and local hyperbolic return neighborhoods can be chosen as a compact transition network. Their continuations to every compact positive-energy interval have uniform hyperbolicity, transversality, C^3 section maps, and return-time bounds.

Proof. The persistence argument proves the first assertion. The invariant plane $z = v_z = 0$ contains a scalar oscillator whose periodic orbit approaches the major-axis billiard orbit. Uniqueness of its local continuation identifies this oscillator with L_-.

For clarity, the hyperbolic-set consequence is the transverse-homoclinic construction of Smale [15]; its local product argument is recalled here. Take a small section through L_- and straighten its stable and unstable curves. A finite segment of the transverse homoclinic maps an outgoing unstable strip across an incoming stable strip. Composing this transition with sufficiently many local iterates before and after it produces a hyperbolic return rectangle with one branch staying near the periodic orbit and a second branch making the homoclinic excursion. Nested intersections of the forward and backward strips give its symbolic product structure. A periodic itinerary containing an excursion yields a hyperbolic periodic orbit L_+ not contained in the axis. Itineraries asymptotic to the axial word in one time direction and to the chosen excursion word in the other give the two transverse connections. All middle connecting pieces are finite; their asymptotic tails lie in the local hyperbolic boxes. Thus no unbounded transition time is required.

Finally, homogeneity transports this construction between energies. If $(q(\tau), v(\tau))$ is a solution at energy $1/2$, then, with $\rho_H = (2H)^{1/p}$,

$$q_H(\tau) = \rho_H q(\rho_H^{(p-2)/2}\tau), \qquad v_H(\tau) = \rho_H^{p/2} v(\rho_H^{(p-2)/2}\tau) \tag{11.8}$$

is a solution at energy H. On a compact interval $0 < H_- \le H \le H_+$ these coordinate and time changes and their inverses have bounded derivatives. The asserted uniform bounds follow. ∎

11.3 Force separation and a fixed convex strip

Define the homogeneous polynomial

$$T(x, z) = z^2(x^2 + z^2)^{(p-2)/2}. \tag{11.9}$$

It is nonnegative, vanishes on the axis, and has strictly positive time average on L_+. For a periodic orbit of a degree-p homogeneous mechanical potential, averaging

$$\frac{\mathrm{d}}{\mathrm{d}\tau}(q \cdot v) = |v|^2 - q \cdot \nabla V = |v|^2 - pV$$

gives the virial identity

$$\langle V\rangle = \frac{2H}{p+2}. \tag{11.10}$$

Choose

$$0 < c < \frac{\langle T\rangle_{L_+}}{\langle V\rangle_{L_+}} \quad \text{at energy } H = 1/2. \tag{11.11}$$

Then $T - cV$ has negative mean on L_- and positive mean on L_+. The same signs hold at every positive energy, since both averages scale linearly with H. The axes, δ_b, and c may be chosen rational inside the relevant strict inequalities.

Fix a strip width $S > 0$ and choose $K > 0$ so large that $s_a = c/(2K) < S/8$. With a sufficiently small $\eta > 0$, set

$$\begin{aligned} C &= K\eta, \qquad s_a = \frac{c}{2K}, \\ P(s,q) &= (1+Cs^2)V(q) + \eta s\,[T(q) - cV(q)] \\ &= A_0(s)V(q) + \eta s T(q), \qquad A_0(s) = 1 + Cs^2 - \eta cs. \end{aligned} \tag{11.12}$$

The symbol A_0 denotes this coefficient and is distinct from the table axis A.

Proposition 11.2 (The convex valley and its two normal branches)**.** The parameters in (11.12) can be chosen so that the following properties hold on $|s| \le S$.

1. P is semialgebraic, jointly convex, and has any prescribed finite differentiability below p. It is homogeneous of degree p in q, is even in x and z, and vanishes exactly when $q = 0$. Uniformly on the strip,

$$P(s,q) \ge \tfrac{1}{2}v_0|q|^p, \qquad P_{qq}(s,q) \ge \tfrac{1}{2}\lambda_0|q|^{p-2}I \quad (q \ne 0).$$

2. The frozen Hamiltonian $H_s(q,v) = |v|^2/2 + P(s,q)$ has C^3 families of hyperbolic periodic orbits $L_-(s,H)$ and $L_+(s,H)$, with transverse connections in both directions. On every fixed compact positive-energy band these form a uniform finite transition network with one stable and one unstable transverse direction. Fixed sufficiently long local iterates give cross-form maps with arbitrarily strong stable and inverse-unstable contractions, uniformly in (s,H).

3. The axial branch L_- lies in $z = v_z = 0$, and

$$P_s(s_a, x, 0) = 0 \qquad \text{for every } x. \tag{11.13}$$

   Thus $s = s_a$ is an exact selected equilibrium whenever the normal motion is axial.

4. Write $b_\pm(s,H) = \langle P_s\rangle_{L_\pm(s,H)}$. Then

$$b_-(s,H) = 2C(s-s_a)\langle V\rangle_{L_-(s,H)} = \frac{4CH(s-s_a)}{(p+2)A_0(s)}, \tag{11.14}$$

   whereas $b_+(s_a,H) = \eta\langle T\rangle_{L_+(s_a,H)} > 0$. There is $d_0 > 0$ such that $b_- < 0 < b_+$ for $s_a - d_0 < s < s_a$. On any compact subinterval of this interval and any compact positive-energy band, both signs have a uniform strict margin.

Proof. All estimates below are first made on $|q| = 1$ and then extended by homogeneity. For fixed S and K, the difference $P - V$ and its normal derivatives are $O(\eta)$ uniformly on $|s| \le S$. Therefore, after decreasing η, the positivity and normal Hessian bounds in item 1 hold. Let M_1 bound

$$|(2Ks - c)\nabla V(q) + \nabla T(q)| \quad \text{on } |s| \le S,\ |q| = 1.$$

The mixed and selected Hessian entries satisfy

$$|P_{sq}(s,q)| \le \eta M_1 |q|^{p-1}, \qquad P_{ss}(s,q) = 2K\eta V(q) \ge 2K\eta v_0 |q|^p.$$

For $q \neq 0$, the Schur-complement cost is bounded by

$$P_{sq} P_{qq}^{-1} P_{qs} \le \frac{2\eta^2 M_1^2}{\lambda_0} |q|^p.$$

Taking η small enough that $2\eta M_1^2/\lambda_0 \le K v_0$ makes the Schur complement nonnegative. At $q = 0$ every second derivative is zero, since $p > 2$. This proves joint convexity on the whole strip. Regularity, semialgebraicity, and symmetry follow from the explicit formula.

At energy $1/2$, the frozen family is a uniformly small smooth perturbation of the network of Proposition 11.1. The periodic families persist by the implicit function theorem for their section maps, because the nontrivial Floquet multipliers are off the unit circle. Their local stable and unstable curves depend smoothly on the parameters. On a fixed section through each connecting segment, transverse intersections persist by the implicit function theorem. Compactness of $[-S, S]$ gives uniform bounds after a single sufficiently small choice of η; the exact homogeneous scaling (11.8) gives the bounds on every compact positive-energy band.

Here is the local-map content of the last assertion in item 2. In small stable–unstable section coordinates, a local periodic iterate contracts the stable direction and its inverse contracts the unstable direction. The fixed maps along the finite connecting segments have bounded derivatives, and transversality makes the required unstable projections invertible. Composing each connecting map with m local iterates at both ends and solving for the incoming unstable variable in terms of the outgoing one gives the cross form. Its stable and inverse-unstable derivatives are bounded by a fixed transition constant times the corresponding mth powers of the local contraction factors. Choose m first, then shrink the boxes, to make the cross-form contraction as small as desired. Smooth parameter dependence and compactness preserve the choice uniformly. There are finitely many such maps and all their return times are bounded above and below by positive constants. Infinite residence near a periodic orbit is represented by repeated local edges, rather than by an edge of infinite duration.

Both normal coordinate reflections leave the vector field invariant, so the axis remains invariant. Since $T(x, 0) = 0$ and $A_0'(s) = 2C(s - s_a)$, (11.13) follows pointwise. On the

axial orbit the potential is $A_0(s)V$, and the virial identity gives $A_0(s)\langle V\rangle = 2H/(p+2)$, proving (11.14). At $s = s_a$ one has $P_s = \eta T$ on the entire normal plane. The continued excursion orbit is still nonaxial, hence its average of T is positive. Continuity gives positivity of b_+ on a neighborhood of s_a; the axial formula gives negativity of b_- immediately to its left. Homogeneity and compactness give all stated uniform sign margins. ∎

Remark. The table perturbation is fixed before the steepness is chosen. Only one sufficiently large finite even p is then retained, and all later coupling and extension parameters are fixed real numbers. In particular the objective is semialgebraic with finite complexity. The conclusion needed for Theorem 1 is existence at one exponent $\alpha = 1+2/p \in (1, 3/2)$, not a construction at every damping exponent.

12 The homogeneous construction and regular cold data

We now turn from the general selection criteria to the construction of accessible drift. Throughout this part, $P = P(s,q)$ is the homogeneous valley supplied by Proposition 11.2, with $q \in \mathbb{R}^2$. Thus

$$P(s,q) = (1+Cs^2)V(q) + \eta s\big(T(q) - cV(q)\big), \qquad C = K\eta, \qquad s_a = \frac{\eta c}{2C} = \frac{c}{2K}. \tag{12.1}$$

The normal degree is a sufficiently large even integer p. We put

$$a = \frac{2}{p}, \qquad \alpha = 1 + a. \tag{12.2}$$

The normal velocity denoted by W in Section 2 will here be denoted by V_1, to distinguish it from the homogeneous potential $V = g^p$. The constants defining P are fixed. All constants below may depend on these choices and on a fixed compact positive-energy interval. They do not depend on the length of a symbolic word or on the time at which its construction begins. We initially work in the selected-coordinate strip on which P is convex and nonnegative.

Lemma 6 (Exact critical scaling)**.** *For $t > 0$, set*

$$\rho = \frac{t^a}{a}, \quad \sigma = \log t, \quad q(t) = t^{-a}Z(\rho), \quad V_1 = Z_\rho, \quad k = \frac{ds}{d\sigma}, \quad \varepsilon = t^{-a} = \frac{1}{a\rho}. \tag{12.3}$$

Then the homogeneous equation

$$s_{tt} + \frac{\alpha}{t}s_t + P_s(s,q) = 0, \qquad q_{tt} + \frac{\alpha}{t}q_t + P_q(s,q) = 0 \tag{12.4}$$

is equivalent to

$$\begin{aligned} Z_\rho &= V_1, & (V_1)_\rho &= -P_Z(s,Z), \\ s_\rho &= \varepsilon k, & k_\rho &= \varepsilon(-ak - P_s(s,Z)), \qquad H = \frac{1}{2}|V_1|^2 + P(s,Z). \\ H_\rho &= \varepsilon k P_s(s,Z), & \varepsilon_\rho &= -a\varepsilon^2, \end{aligned} \tag{12.5}$$

In particular, with $E = H + k^2/2$,

$$E_\rho = -\frac{k^2}{\rho}, \qquad E_\sigma = -ak^2. \tag{12.6}$$

Proof. Since $d\rho/dt = t^{a-1}$, direct differentiation gives

$$q_t = t^{-1} Z_\rho - a t^{-a-1} Z$$

and

$$q_{tt} = t^{a-2} Z_{\rho\rho} - (a+1) t^{-2} Z_\rho + a(a+1) t^{-a-2} Z.$$

The last two terms cancel against $(\alpha/t) q_t$ because $\alpha = a+1$. Homogeneity and $ap = 2$ give $P_q(s, t^{-a} Z) = t^{a-2} P_Z(s, Z)$, proving the normal equation. Also $s_t = k/t$, so the selected equation becomes $k_\sigma = -ak - P_s(s, Z)$. Since $d\sigma/d\rho = \varepsilon$, this is the asserted equation for k. Differentiating H cancels the normal Hamiltonian terms and leaves $H_\rho = P_s s_\rho$. Finally, $E_\rho = \varepsilon k P_s + \varepsilon k(-ak - P_s) = -a\varepsilon k^2 = -k^2/\rho$. ∎

Lemma 7 (The cold endpoint)**.** *For cold initial positions* (s_0, q_0)*, the regular solution of* (12.4) *satisfies, in the variables of* (12.3)*,*

$$\begin{aligned} Z(\rho) &= a q_0 \rho - \int_0^\rho (\rho - r) P_Z(s(r), Z(r))\, dr, \\ k(\rho) &= -\frac{1}{a\rho} \int_0^\rho P_s(s(r), Z(r))\, dr, \\ s(\rho) &= s_0 - \frac{1}{a^2} \int_0^\rho u^{-2} \int_0^u P_s(s(r), Z(r))\, dr\, du. \end{aligned} \tag{12.7}$$

Conversely, the regular solution of these integral equations gives the cold solution of (12.4)*. In particular,*

$$Z = a q_0 \rho + O(\rho^{p+1}), \qquad k = O(\rho^p), \qquad s = s_0 + O(\rho^p). \tag{12.8}$$

Its dependence on the cold positions is continuously differentiable, to any fixed finite order allowed by the regularity of P*.*

Proof. Local cold existence can be seen directly in the physical variables. Writing $X = (s, q)$, the integral equation is

$$X(t) = X_0 - \int_0^t r^{-\alpha} \int_0^r u^\alpha \nabla P(X(u))\, du\, dr. \tag{12.9}$$

On a fixed neighborhood of X_0 with gradient Lipschitz constant L, its contraction constant on $[0, T]$ is at most $LT^2/[2(\alpha+1)]$. Taking T small gives the unique local solution. The differentiated equation shows that $X_t(0) = 0$ and $X_{tt}(0) = -\nabla P(X_0)/(\alpha+1)$. Differentiating the integral operator with respect to X_0 gives smooth finite-dimensional

dependence; the corresponding derivative equations are Volterra equations with the same small-interval bounds.

The normal equation in (12.5), integrated twice with $Z(0) = 0$ and $Z_\rho(0) = aq_0$, gives the first line of (12.7). The equation for k reads $(\rho k)_\rho = -P_s/a$. Regularity fixes its integration constant to zero and gives the second line. Integrating $s_\rho = k/(a\rho)$ gives the third. Since $Z = O(\rho)$, homogeneity gives $P_Z(s, Z) = O(\rho^{p-1})$ and $P_s(s, Z) = O(\rho^p)$; substitution proves (12.8). Conversely these bounds justify all the integrations and yield $q = q_0 + O(\rho^p)$ and $s = s_0 + O(\rho^p)$. The identity $\rho^p = t^2/a^p$ then gives the physical cold regularity. Uniqueness in (12.9) identifies the two solutions. ∎

Lemma 8 (A bound on the whole cold trajectory)**.** *On every interval where the regular cold solution stays in the strip of nonnegativity of P,*

$$E(0) = \frac{a^2|q_0|^2}{2}, \qquad |Z_\rho(\rho)| \le a|q_0|, \qquad |q(t)| \le |q_0|. \tag{12.10}$$

Proof. The expansion (12.8) makes (12.6) integrable at $\rho = 0$ and gives the displayed value of $E(0)$. Since $P \ge 0$ and E is nonincreasing, $|Z_\rho|^2/2 \le E(0)$. Integrating from $Z(0) = 0$ yields $|Z(\rho)| \le a|q_0|\rho$. Now $q = Z/(a\rho)$. ∎

Lemma 9 (An exact axial reference and transverse cold access)**.** *Fix $q_* > 0$ and use cold positions $(s_0, q_0) = (s_a, (q_*, 0))$. The rescaled solution is*

$$s = s_a, \qquad k = 0, \qquad Z(\rho) = (X(\rho), 0), \qquad X(0) = 0, \quad X_\rho(0) = aq_*.$$

It is the axial periodic normal orbit at energy $H_* = a^2q_*^2/2$. The cold curve obtained by replacing q_0 with (q_*, ζ) has a nonzero transverse unstable component at $\zeta = 0$. Its first selected, axial, energy, and section-time variations vanish there.

Proof. The formula for s_a and $T(x, 0) = 0$ give $P_s(s_a, x, 0) = 0$ pointwise. Reflection symmetry gives $P_{q_2}(s_a, x, 0) = 0$, so the asserted reference is exact. The remaining scalar normal equation has a positive homogeneous restoring potential; its positive-energy orbit is periodic.

Reflection in q_2 makes the selected and axial coordinates even in ζ and the transverse coordinate odd. The nonzero first variation therefore solves

$$w_{\rho\rho} + A(\rho)w = 0, \qquad w(0) = 0, \quad w_\rho(0) = a, \qquad A(\rho) = P_{q_2q_2}(s_a, X(\rho), 0). \tag{12.11}$$

Since $X(-\rho) = -X(\rho)$ and P is even in the first normal coordinate, A is even and periodic. If M is its full-period monodromy and $J = \mathrm{diag}(1, -1)$, reversibility gives $M^{-1} = JMJ$. Together with $\det M = 1$, this implies that the two diagonal entries of M are equal. Writing them as m, hyperbolicity gives $m_{12}m_{21} = m^2 - 1 > 0$. Thus the vertical line spanned by $(0, 1)$ is neither eigendirection and in particular is not the stable direction. This proves the unstable component assertion.

The selected and axial first variations vanish by the same reflection, and the energy first variation vanishes as well. On a section through an axial origin crossing, the crossing-time derivative is zero because the axial first variation is zero and the reference crossing velocity is nonzero. Advancing to a fixed later section preserves these statements. ∎

13 Coding with a variable slow parameter

We use the two periodic families L_- and L_+ from Proposition 11.2. The former is axial. Fix a compact rectangle of (s, H)-values, with H bounded away from zero, and work on a slightly larger open rectangle where both families and their transverse connections persist. A bounded k-interval is included in the center coordinates. The use of cross forms is related to the coding descriptions in [6, 7]. The estimates below are proved for the moving parameter ε and the dissipative center equations (12.5).

Lemma 10 (Uniform cross forms)**.** *There are stable and unstable section coordinates (x, y), finitely many return branches indexed by $c, d \in \{-, +\}$, and constants $0 < \lambda < 1$ and $0 < T_{\min} < T_{\max} < \infty$ such that the return maps of (12.5) have cross form*

$$\begin{aligned} x_{i+1} &= f_{cd}(x_i, y_{i+1}, w_i, \varepsilon_i), \\ y_i &= g_{cd}(x_i, y_{i+1}, w_i, \varepsilon_i), \\ w_{i+1} &= w_i + \varepsilon_i \Phi_{cd}(x_i, y_{i+1}, w_i, \varepsilon_i), \\ \varepsilon_{i+1} &= \frac{\varepsilon_i}{1 + a\varepsilon_i T_{cd}(x_i, y_{i+1}, w_i, \varepsilon_i)}, \qquad w = (s, k, H). \end{aligned} \tag{13.1}$$

The joint derivative of (f_{cd}, g_{cd}) in the fast input coordinates has norm at most λ in a product maximum norm. The images have a fixed positive margin inside the fast boxes. All required derivatives through order two are bounded uniformly, and $T_{\min} \le T_{cd} \le T_{\max}$. For a constant-symbol branch $c \to c$, the frozen reference follows a fixed number of periods of L_c.

Proof. Near each hyperbolic periodic orbit choose local stable and unstable coordinates. A fixed long local iterate contracts the stable direction and its inverse contracts the unstable direction. A transverse connecting segment has a nonsingular unstable-to-unstable block in the corresponding boundary-value description. Solving for the incoming unstable coordinate in terms of the outgoing one therefore gives a cross form. Adding m local iterates before and after the connecting segment makes its stable and inverse-unstable derivatives at most $C\nu^m$, where $\nu < 1$ is a uniform local contraction rate. Choose m once so these bounds are strictly below one, and then shrink the boxes to obtain the image margin. There are only finitely many branches. Compactness of the parameter rectangle makes the choices uniform and gives positive lower and finite upper flight-time bounds.

For $\varepsilon = 0$, the center is frozen. Over each of the fixed finite return flights, the full vector field differs in the center variables by $O(\varepsilon)$, with the same bounds for the derivatives

needed here. Smooth dependence and section transversality preserve the cross form for small ε. Integration of the w-equations gives the third line of (13.1). The last line is exact, because $\varepsilon^{-1} = a\rho$. Taking smaller boxes and a smaller maximal ε preserves the contraction and image margins. ∎

For the coding construction, extend the maps smoothly outside a larger center rectangle. The fast coefficients can be evaluated at a smooth clamp to that larger rectangle, and the center drift can be cut off away from the working region. Set

$$\varepsilon = \delta\vartheta, \tag{13.2}$$

where $\delta > 0$ is fixed and small. The center step in (w, ϑ) is then C^2-close to the identity by $O(\delta)$. Extend the ϑ displacement outside $[0, 1]$, keeping zero fixed and making the displacement vanish near a larger positive endpoint. The extension may be chosen so that positive ϑ stays bounded under both forward and backward steps and

$$\varepsilon_j \le C\varepsilon_i e^{C\delta|j-i|}. \tag{13.3}$$

The near-identity center maps are global diffeomorphisms. These extensions are used only to define coded leaves. A separate first-exit argument will show that every retained physical prefix and the limiting orbit stay in the unmodified region.

Lemma 11 (Coded center leaves and local regularity)**.** *For every two-sided code $\omega \in \{-,+\}^{\mathbb{Z}}$ and every center value at a fixed index, the extended maps have a unique orbit in the fast boxes with that code and center value. At each index the fast coordinates are a locally C^1 function of the center, with bounds uniform in the code and the index.*

Proof. Anchor the center $c_0 = (w_0, \vartheta_0)$ at index zero. For a proposed bounded fast sequence $U = (x_i, y_i)_{i\in\mathbb{Z}}$, the center maps in (13.1) determine a unique forward and backward sequence $c_i(U)$. For $r > 1$, use the distance

$$\|U - \widetilde{U}\|_r = \sup_{i\in\mathbb{Z}} r^{-|i|}|U_i - \widetilde{U}_i|.$$

The product of the closed fast boxes is complete for this distance. The center difference recurrence, anchored at zero, gives

$$\|c(U) - c(\widetilde{U})\|_r \le \frac{C\delta}{r - 1 - C\delta}\|U - \widetilde{U}\|_r. \tag{13.4}$$

Indeed forward differences satisfy $D_{i+1} \le (1 + C\delta)D_i + C\delta|U_i - \widetilde{U}_i|$ with $D_0 = 0$. The weighted geometric sum gives (13.4), and the inverse center maps give the same estimate backwards.

Substitute this center sequence into the first two lines of (13.1). The resulting operator on fast sequences has Lipschitz constant at most

$$\kappa_r = \lambda r + \frac{C\delta}{r - 1 - C\delta}. \tag{13.5}$$

Choose $1 < r < 1/\lambda$, then decrease δ so that $\kappa_r < 1$. The operator maps the product of the fast boxes into itself and has a unique fixed point. Recentring the argument at any index gives the asserted graph representation.

For completeness, local differentiability is obtained using two weights; it does not follow by asserting that the nonlinear operator is Fréchet differentiable in the same weighted product norm. Choose

$$1 < r_1, \qquad r_1^2 < r_2 < 1/\lambda, \tag{13.6}$$

and take δ small enough that both contraction constants in (13.5) are below one and $e^{C\delta} < r_1$. If the anchored center changes by h, the fixed-point and center estimates give

$$|\Delta U_i| + |\Delta c_i| \le C|h| r_1^{|i|}. \tag{13.7}$$

Solve the pointwise linearized fast and center recurrences, with anchored center variation h, by the same linear contraction in the r_1 norm. This defines a Jacobi sequence $(J_i h, K_i h)$ satisfying the same bound.

The finite-dimensional coefficients in (13.1) have uniform second derivatives. Taylor's theorem therefore bounds the remainders obtained after subtracting the Jacobi equations by $C|h|^2 r_1^{2|i|}$. Center-map remainders carry an additional $O(\delta)$ factor. By (13.6), these forcing sequences have r_2 norm at most $C|h|^2$. The linearized coupled equations have a uniform contraction inverse in that norm. Consequently

$$\|\Delta U - Jh\|_{r_2} + \|\Delta c - Kh\|_{r_2} \le C|h|^2. \tag{13.8}$$

Evaluation at the anchor is continuous in the r_2 norm, proving differentiability of the finite-dimensional local graph.

To prove continuity of its derivative, compare the Jacobi equations at two nearby anchor values. Their coefficient differences are bounded by $C|h| r_1^{|i|}$, and unit-direction Jacobi sequences are bounded by $C r_1^{|i|}$. The difference forcing thus has r_2 norm $O(|h|)$. The same inverse estimate proves continuity, with a uniform local Lipschitz bound, for the derivative evaluated at the anchor. This proves the claimed local C^1 bounds without a same-norm differentiability assertion for the nonlinear product operator. ∎

Lemma 12 (Local parameter and symbolic estimates)**.** *At a fixed center (w_i, ε_i), the coded fast graph differs from its frozen graph with the same code by at most $C\varepsilon_i$. Its derivative with respect to $\ell = \log \varepsilon$ is $O(\varepsilon_i)$. If two codes agree on a window and i has distance L from its nearest end, their fast graphs at the same center differ by at most $C\theta^L$, where $0 < \theta < 1$ is fixed.*

Proof. Recenter the contraction at i. In addition to (13.3), the center recurrence gives

$$|w_j - w_i| \le C\varepsilon_i |j - i| e^{C\delta|j-i|}.$$

Comparing the fixed-point equations with the frozen equations, the weighted forcing is therefore $O(\varepsilon_i)$, because the weight suppresses the displayed polynomial-exponential factor. The contraction gap gives the local frozen comparison.

In the Jacobi equations for variation of ϑ_i, the direct fast coefficient variation has a factor δ, and the induced variation of w_j in a weighted neighborhood has that same factor. The two-weight argument in Lemma 11 gives an $O(\delta)$ local fast derivative in ϑ. Since $\partial_\ell = \vartheta\partial_\vartheta$, the derivative in ℓ is $O(\delta\vartheta) = O(\varepsilon_i)$.

For code locality, use the same proposed fast inputs and anchor the center inside the common window. The two center recurrences agree there. Differences in the fixed-point operators occur only outside the window; their weighted norm is at most Cr^{-L}, including the center response outside it. Dividing by the contraction gap gives the assertion, with $\theta = r^{-1}$ or a slightly larger constant less than one. ∎

In particular, the center dynamics is a full-code dynamics. In general its leading coefficient is not a function of just the current symbol. The actual return times nevertheless imply

$$\frac{1}{\varepsilon_0^{-1} + aT_{\max} i} \le \varepsilon_i \le \frac{1}{\varepsilon_0^{-1} + aT_{\min} i} \qquad (i \ge 0). \tag{13.9}$$

Lemma 13 (A moving-clock prefix estimate)**.** *Suppose two two-sided codes agree at every index $j \le N$, and their coded trajectories have the same initial center. For $0 \le i \le N$,*

$$|w_i - \widetilde{w}_i| + |\log \varepsilon_i - \log \widetilde{\varepsilon}_i| \le C\varepsilon_i \theta_1^{N-i} \tag{13.10}$$

for a fixed $0 < \theta_1 < 1$. The constants are independent of N.

Proof. Use the center coordinates $Y = (w, \ell)$ with $\ell = \log \varepsilon$. Lemmas 11 and 12 show that its return map is identity plus an increment whose value and first derivative are $O(\varepsilon)$. At the same center, the increment difference for the two codes at index i is at most $C\varepsilon_i\theta^{N-i}$. Thus, while the differences are small,

$$D_{i+1} \le (1 + C\varepsilon_i)D_i + C\varepsilon_i\theta^{N-i}, \qquad D_0 = 0.$$

Variation of constants gives

$$D_i \le C \sum_{j<i} \varepsilon_j \theta^{N-j} \exp\left(C \sum_{m=j+1}^{i-1} \varepsilon_m \right). \tag{13.11}$$

The ratio $\varepsilon_j/\varepsilon_i$ and the exponential factor are bounded by $e^{C\delta(i-j)}$, after increasing C. Choose δ so small that the resulting geometric series converges. Then (13.11) is at most $C\varepsilon_i\theta_1^{N-i}$ with a fixed $\theta_1 < 1$. This estimate also closes the small-difference bootstrap. Since $\sigma = -\ell/a$, it controls the actual logarithmic clock. ∎

14 Long periodic blocks and cold inclination

For $c \in \{-,+\}$, let $T_c(s,H)$ be the frozen return period chosen in Lemma 10, and define

$$b_c(s,H) = \frac{1}{T_c(s,H)} \int_0^{T_c(s,H)} P_s(s, Z_c(r;s,H))\, dr, \qquad G_c(s,k,H) = \big(k, -ak - b_c, kb_c\big). \tag{14.1}$$

These have the finite differentiability required below on the fixed parameter rectangle. Replacing a period by a fixed integer multiple does not change b_c.

Lemma 14 (Averaging on a long constant word)**.** *Suppose a coded orbit has symbol c on a block from m through $m+n$. At an interior return i, its increments satisfy*

$$\begin{aligned} w_{i+1} - w_i &= \varepsilon_i T_c(w_i) G_c(w_i) + R_i, \\ \sigma_{i+1} - \sigma_i &= \varepsilon_i T_c(w_i) + r_i, \end{aligned} \tag{14.2}$$

where

$$|R_i| + |r_i| \le C\left[\varepsilon_i^2 + \varepsilon_i\big(\theta^{i-m} + \theta^{m+n-i}\big)\right]. \tag{14.3}$$

Consequently a constant-symbol run over any fixed bounded interval of slow time follows the averaged equation $w_\sigma = G_c(w)$ with error $O(\varepsilon_m)$, in addition to its initial error.

For $0 < h \le h_0$, a word consisting of a $+$ block of intended slow length βh, a $-$ block of intended length $(1-\beta)h$, and the necessary bounded connecting flights has endpoint estimates

$$\begin{aligned} \Delta\sigma &= h + O(h^2 + \varepsilon), \\ w_{\text{end}} &= w + h\big[\beta G_+(w) + (1-\beta) G_-(w)\big] + O(h^2 + \varepsilon). \end{aligned} \tag{14.4}$$

Here ε is its starting parameter. The bounds are uniform when β lies in a fixed compact subinterval of $(0,1)$ and ε/h is sufficiently small.

Proof. For the frozen graph, symbolic locality bounds its distance from the constant-code periodic graph by $C(\theta^{i-m} + \theta^{m+n-i})$. The moving graph adds $O(\varepsilon_i)$ by Lemma 12. Integrating the center vector field along the reference periodic return gives $T_c G_c$. Smooth dependence of the return integral on its fast and center inputs proves the first line of (14.2) and (14.3). The same argument applies to the return time, using

$$\Delta\sigma = \frac{1}{a}\log(1 + a\varepsilon_i T) = \varepsilon_i T + O(\varepsilon_i^2).$$

On a bounded slow-time interval the mesh is $O(\varepsilon_m)$. The sum of its squared mesh sizes is $O(\varepsilon_m)$, and each geometric boundary layer in (14.3) also sums to $O(\varepsilon_m)$. Subtracting the time increment error times G_c from the first line of (14.2) gives an Euler scheme in the actual slow time. Discrete Gronwall, with a fixed finite-time factor, proves the asserted averaged-ODE comparison.

For a single block of intended length $\ell \le h_0$, choose the number of local-period returns to be a nearest integer to

$$\frac{\ell}{\varepsilon T_c(w)}. \tag{14.5}$$

The center varies by $O(\ell)$ during the block and ε changes relatively by $O(\ell)$. The variation of the period and the center field therefore contributes $O(\ell^2)$. Integer rounding costs $O(\varepsilon)$, squared-mesh errors cost $O(\varepsilon\ell)$, and both end layers cost $O(\varepsilon)$. Hence

$$\Delta\sigma = \ell + O(\ell^2 + \varepsilon), \qquad w_{\mathrm{end}} = w + \ell G_c(w) + O(\ell^2 + \varepsilon).$$

Apply this successively to the two blocks. A connecting flight has bounded fast duration and costs $O(\varepsilon)$ in slow time and center displacement. This proves (14.4). The lower bound on both fractions and $\varepsilon \ll h$ ensure that both local-period blocks contain the required number of returns. The use of T_c in (14.5) weights the fractions by slow duration, rather than by the unweighted number of cycles. ∎

We next retain the cold constraint when a future code is selected. The normal dimension two is used here: there is one stable and one unstable transverse section coordinate.

Lemma 15 (A cold inclination estimate)**.** *Let the exact axial cold reference be as in Lemma 9. For every sufficiently late axial return N, the image of a suitably restricted interval of the transverse cold curve is a graph*

$$(x, \mathbf{c}) = \big(x_N(y), \mathbf{c}_N(y)\big), \qquad |y| \le r, \tag{14.6}$$

over a fixed unstable interval. The frozen coordinates straighten the axial unstable manifold to $x = 0$, and $\mathbf{c}$ is the deviation of $(s, k, H, \log\varepsilon)$ from the exact reference at that return. With ε_N^* denoting the reference parameter,

$$\|x_N\|_{C^1} + \|\mathbf{c}_N\|_{C^1} \le C\varepsilon_N^*. \tag{14.7}$$

The whole retained cold preparation stays in a fixed small selected neighborhood of s_a.

Proof. Take a fixed preliminary number J of axial periods so that the reference parameter is small and the cold tangent is in a narrow unstable cone. This is possible because the transverse variational equation is exactly the frozen hyperbolic equation and, by Lemma 9, the cold tangent has a nonzero unstable component. The center first variation is zero. Restricting the original cold interval then makes its image a C^1 unstable graph with small center and stable slopes.

Choose frozen charts depending smoothly on (s, H) that straighten the axial unstable manifold to $x = 0$. Include k and the relative logarithmic clock in $\mathbf{c}$. The exact reference has $(x, y, \mathbf{c}) = (0, 0, 0)$ at every return. By taking a fixed iterate, the frozen stable derivative is at most $\nu < 1$ and the unstable expansion is at least $\mu > 1$. An even iterate

can be used if the unstable multiplier is negative. The actual return map differs in C^1 from the frozen map by $O(\varepsilon_j^*)$. Its center derivatives satisfy

$$D_{\mathbf{c}}\mathbf{c}_{\text{next}} = I + O(\varepsilon_j^*), \qquad D_{(x,y)}\mathbf{c}_{\text{next}} = O(\varepsilon_j^*). \tag{14.8}$$

All bounds remain uniform in a small graph cone. The relative clock coordinate keeps the actual parameter comparable to ε_j^* there.

Write the current graph as $(x_j(y), \mathbf{c}_j(y))$ and set

$$X_j = \sup |x_j'|, \qquad C_j = \sup |\mathbf{c}_j'|.$$

For small slopes and small box radius, the unstable output is strictly expanding as a function of y, with derivative at least a fixed $\mu_0 > 1$. Reparameterize the image by that output and recut to $|y| \le r$. The frozen stable output vanishes at $x = 0$ for every frozen center. Its center derivative is therefore $O(|x|)$, and its unstable derivative has the same vanishing property. The actual map adds $O(\varepsilon_j^*)$ to these bounds. The chain rule, divided by the unstable expansion, yields

$$\begin{aligned} X_{j+1} &\le \kappa_x X_j + CrC_j + C\varepsilon_j^*, \\ C_{j+1} &\le \kappa_c C_j + C\varepsilon_j^* X_j + C\varepsilon_j^*, \end{aligned} \qquad \kappa_x, \kappa_c < 1. \tag{14.9}$$

For example the center factor before division is $1 + O(\varepsilon_j^*)$, whereas the denominator is at least μ_0.

Choose a weighted slope norm $D_j = X_j + W_0 C_j$. Taking the radius and then the maximal parameter sufficiently small makes the nonnegative coefficient matrix in (14.9) contractive in this norm:

$$D_{j+1} \le \kappa D_j + C\varepsilon_j^*, \qquad \kappa < 1. \tag{14.10}$$

Since the return times are uniformly bounded above and below, ε_j^* is comparable to $(j+1)^{-1}$. Thus

$$D_N \le \kappa^{N-J} D_J + C \sum_{j=J}^{N-1} \kappa^{N-1-j} \varepsilon_j^* \le C\varepsilon_N^*. \tag{14.11}$$

To check the last bound, split the sum at $N/2$. In the late half, $\varepsilon_j^* \le C\varepsilon_N^*$; the early half has an exponentially small factor. Each graph passes through the exact reference, so $x_N(0) = \mathbf{c}_N(0) = 0$. The slope bound therefore also bounds its values and proves (14.7).

The fixed-interval covering used in this argument is retained under each recut. Before it is obtained, expansion grows a small interval about zero until it spans the prescribed unstable interval. Thereafter its two endpoint images lie beyond the corresponding endpoints of that interval; restricting the image gives another graph over the entire interval. The preimages form nested shrinking intervals of the original cold curve. Every retained graph is in the invariant small cone, and its values are bounded by its slope times r. On

the fixed early cold prefix, smooth dependence keeps the selected coordinate close to s_a. Between later sections, bounded flight times and the $O(\varepsilon)$ center equations preserve a slightly larger selected bound. This proves the last assertion as well. ∎

Lemma 16 (Half-line coding and cold intersections)**.** *For each future code starting in the axial box, the extended maps have a center-stable graph*

$$y_0 = h_\omega(x_0, Y_0), \qquad Y = (s, k, H, \log \varepsilon), \tag{14.12}$$

with uniform local C^1 bounds. After a fixed number of additional local iterates, its values lie strictly inside the unstable interval of Lemma 15. Each such graph has a unique intersection with the cold graph (14.6) *for N sufficiently large. If two future codes agree through their first L symbols, their cold intersections satisfy*

$$|\Delta y_0| \le C\theta^L, \qquad |\Delta x_0| + |\Delta Y_0| \le C\varepsilon_N^* \theta^L. \tag{14.13}$$

Proof. Use the contraction of Lemma 11 on the half-line $i \ge 0$, prescribing the stable boundary x_0 and the center Y_0. The remaining stable coordinates are generated forward and the unstable coordinates are determined by their infinite future. The weighted estimates and the two-weight regularity proof are unchanged; deleting the negative part of a geometric sum only improves its bound. This gives (14.12). A common prefix of length L makes the operator difference $O(\theta^L)$ and hence

$$\|h_\omega - h_{\widetilde{\omega}}\|_{C^0} \le C\theta^L. \tag{14.14}$$

The strict image margin in the cross forms, or a fixed extra local iterate before a transition, puts all these graphs strictly inside the chosen unstable interval.

The intersection with the transported cold curve is the scalar equation

$$y = h_\omega\big(x_N(y), Y_N^* + \mathbf{c}_N(y)\big).$$

Its right-hand side maps the closed interval strictly inside itself and, by (14.7), has derivative at most $C\varepsilon_N^* < 1$. It therefore has a unique fixed point. Comparing two fixed-point equations and using (14.14) gives the first estimate in (14.13). Composing with the C^1-small cold graph gives the second. Each point is an image of a genuine regular cold datum by Lemma 15; no center value is independently prescribed in this intersection argument. ∎

Lemma 17 (Stable-boundary memory and cold prefix locality)**.** *Consider two half-line coded trajectories whose codes agree through index L. Let $U_i = (x_i, y_i)$ and let Y_i be the center including the logarithmic clock. Then, for $0 \le i \le L$,*

$$|\Delta U_i| \le C|\Delta Y_i| + C\theta^i |\Delta x_0| + C\theta^{L-i}. \tag{14.15}$$

For the cold intersections of Lemma 16, this implies

$$|\Delta Y_i| \le C\varepsilon_i \theta_1^{L-i}, \qquad 0 \le i \le L, \tag{14.16}$$

with a fixed $\theta_1 < 1$. Long-dwell averaging and (14.4) *remain valid for these half-line trajectories.*

Proof. Recenter the half-line contraction at i using the weights $r^{-|j-i|}$ for $j \ge 0$. Prescribe Y_i, solve its recurrence backwards to zero and forwards thereafter, and retain the stable boundary x_0. The same contraction constant works uniformly in i. A change of that boundary is a forcing at index zero with weighted size $r^{-i}|\Delta x_0|$. Code differences beyond L have weighted size $Cr^{-(L-i)}$. A change in Y_i has uniformly bounded effect at i by the local regularity proof of Lemma 11. The contraction estimate proves (14.15).

It follows that $D_i = |\Delta Y_i|$ satisfies

$$D_{i+1} \le (1 + C\varepsilon_i) D_i + C\varepsilon_i \theta^i |\Delta x_0| + C\varepsilon_i \theta^{L-i}. \tag{14.17}$$

For the cold intersections, $D_0 + |\Delta x_0| \le C\varepsilon_0 \theta^L$ by (14.13), where ε_0 is the parameter at the late starting section. The initial-center contribution to variation of constants is bounded by

$$C\varepsilon_0 \theta^L \exp\left(C \sum_{j<i} \varepsilon_j \right).$$

The stable-boundary contribution is bounded by

$$\begin{aligned} C|\Delta x_0| \sum_{j<i} \varepsilon_j \theta^j \exp\left(C \sum_{m=j+1}^{i-1} \varepsilon_m \right) \\ \le C\varepsilon_0 |\Delta x_0| \exp\left(C \sum_{m<i} \varepsilon_m \right). \end{aligned}$$

The ratio $\varepsilon_0/\varepsilon_i$ and the exponential factors grow at most as $e^{C\delta i}$. Decreasing δ and enlarging θ slightly, both displayed expressions are at most $C\varepsilon_i \theta_1^{L-i}$. The last forcing term in (14.17) is exactly the sum estimated in Lemma 13. This proves (14.16).

To check averaging, compare one half-line graph at index i with a two-sided graph having the same center there and the same future code. The retained stable boundary at zero changes its fast coordinates by at most $C\theta^i$, because both boundary values lie in a bounded box. Thus (14.3) gains only the term $C\varepsilon_i \theta^i$. Its sum over any dwell beginning at m is at most $C\varepsilon_m$. All the errors in Lemma 14 are therefore unchanged in order. In particular the cold argument does not replace a half-line graph by a two-sided graph without accounting for its stable boundary. ∎

15 Entry into a strict-force region

At the exact axial selected state, the axial averaged force is zero. The next lemma supplies the finite preparation needed before the recurrent controller can use two strict force signs.

Lemma 18 (A finite inward maneuver)**.** *Starting from the averaged state $(s_a, 0, H_*)$, a finite concatenation of an excursion branch and an axial branch reaches a state $(s_*, 0, H_e)$ with $s_* < s_a$. The whole path stays in a small selected neighborhood of s_a, and its energy loss can be made arbitrarily small. There is a compact rectangle about (s_*, H_e) on which*

$$b_+(s, H) \geq b_* > 0, \qquad b_-(s, H) \leq -b_* < 0. \tag{15.1}$$

For the cold-coded family of Lemma 16, this maneuver can be realized with $O(\varepsilon_N^*)$ error in its endpoint center and logarithmic clock.

Proof. Put $\xi = s_a - s$. The explicit potential gives, on the axial branch,

$$b_-(s, H) = -\kappa(s, H)\xi, \qquad \kappa(s, H) = 2C\langle V\rangle_{L_-} > 0. \tag{15.2}$$

On the excursion branch,

$$b_+(s, H) = \eta\langle T\rangle_{L_+} - 2C\xi\langle V\rangle_{L_+}. \tag{15.3}$$

At $\xi = 0$ this is strictly positive. Choose a small compact neighborhood of (s_a, H_*) where κ has positive lower and finite upper bounds and b_+ remains positive for $0 \leq \xi \leq \xi_0$.

Follow G_+ for a fixed small slow time $\tau > 0$. If $F = b_+(s_a, H_*) > 0$ and $v = \xi' = -k$, Taylor's theorem gives

$$v_1 = F\tau + O(\tau^2), \qquad \xi_1 = \frac{F}{2}\tau^2 + O(\tau^3), \qquad H_1 = H_* - \frac{F^2}{2}\tau^2 + O(\tau^3). \tag{15.4}$$

In particular $\xi_1, v_1 > 0$ for small τ. Now follow G_-. As long as $v \geq 0$, its equations are

$$\xi' = v, \qquad v' = -av - \kappa(s, H)\xi \leq -av - \kappa_{\min}\xi_1 < 0. \tag{15.5}$$

Thus v reaches zero in finite time, no later than $v_1/(\kappa_{\min}\xi_1)$. Before that event,

$$\xi_1 \leq \xi \leq \xi_1 + v_1/a = O(\tau). \tag{15.6}$$

Both averaged branch fields satisfy $E' = -ak^2$. During the axial part,

$$\int v^2\, d\sigma \leq v_1 \int v\, d\sigma = v_1(\xi_* - \xi_1) \leq v_1^2/a.$$

Together with the initial pulse, this bounds the total loss by $O(\tau^2)$, and the kinetic energy is also $O(\tau^2)$. Taking τ small keeps H in the chosen positive interval and (15.6) inside

the chosen selected neighborhood. These same estimates rule out a first exit before the stopping time. At the stopping point $s_* = s_a - \xi_* < s_a$ and $k = 0$. Equations (15.2)–(15.3) then give (15.1) on a smaller compact rectangle.

Once τ is fixed, the total slow duration of the maneuver is finite. Lemma 14, including the half-line correction in Lemma 17, approximates each constant branch over that interval with error $O(\varepsilon_N^*)$. Choose the cycle count by following a constant tail and taking the section nearest the desired slow duration. Replacing that tail by the next branch changes the old endpoint by $O(\varepsilon_N^*)$ according to (14.16). Clock rounding and bounded connecting flights have the same order. The initial cold intersection differs from the reference center by $O(\varepsilon_N^*)$. Finite-time Gronwall thus realizes the entire maneuver with the claimed error, while preserving its strict selected and energy margins. ∎

16 Arbitrary slow target tracking from cold data

Theorem 10 (Arbitrary slow cold target tracking)**.** *For the potential of Proposition 11.2, fix a sufficiently small $q_* > 0$, the transverse cold family*

$$s(0) = s_a, \qquad q(0) = (q_*, \zeta),$$

and the finite entry of Lemma 18, with averaged endpoint $(s_*, 0, H_e)$. There is a fixed open interval $J \ni s_*$ whose closure is strictly inside the selected part of the strict-force rectangle. Let $Q : [0, \infty) \to J$ be C^3, with bounded first three derivatives, $Q(0) = s_*$ and $Q'(0) = 0$. Put

$$S(u) = Q\left(\log \frac{M+u}{M}\right), \qquad h(u) = \frac{1}{M+u}. \tag{16.1}$$

For all sufficiently large M, there is a regular cold solution from the fixed family above and a finite $t_* > 0$ such that, with $u = \log(t/t_*)$, on its tail

$$|s(t) - S(u)| + |ts_t(t) - S'(u)| \le Ch(u), \qquad \int_0^\infty k(u)^2 \, \mathrm{d}u \le \frac{C_1}{M}. \tag{16.2}$$

The rescaled normal energy stays in a compact positive interval and $q(t) \to 0$. The entire selected trajectory remains in the fixed reserved strip, and the whole cold normal bound (12.10) *holds. In particular $\omega(s) = \omega(Q)$. The initial cold parameter may be chosen arbitrarily close to the axial reference by increasing its preparatory residence.*

Proof. We give the choice of the itinerary and the compact-region bootstrap separately. All auxiliary code leaves below are defined by the extended maps. Only retained preparation and control prefixes, and their limit, are asserted to remain in the physical region.

The target and the averaged feedback. Fix the averaged turning point $(s_*, 0, H_e)$ and a compact strict-force rectangle using Lemma 18. Choose J with $\overline{J}$ inside its selected

interior. All these choices precede the choice of Q. The target (16.1) satisfies

$$\begin{gathered} S(0) = s_*, \qquad S'(0) = 0, \\ |S'(u)| \le C_Q h(u), \quad |S''(u)| \le C_Q h(u)^2, \quad |S'''(u)| \le C_Q h(u)^3. \end{gathered} \tag{16.3}$$

Indeed, if $v = \log((M+u)/M)$, its first three derivatives are $Q'(v)/(M+u)$, $(Q''(v) - Q'(v))/(M+u)^2$, and $(Q'''(v) - 3Q''(v) + 2Q'(v))/(M+u)^3$.

A later rewrite of the entry clock may put the initial target argument slightly below zero. Extend Q there by its cubic Taylor polynomial at zero, on a sufficiently short interval. The extension is C^3, remains in J, and the logarithm is defined for $u > -M$, so the composite is defined on the required short negative interval. Since $Q'(0) = 0$, an initial clock change $O(\varepsilon_N^*)$ causes a change already covered by the entry error. For targets flat to all orders at zero, the constant extension $Q(v) = s_*$ for $v < 0$ is available instead. Neither extension changes the positive-time target or its omega-limit set.

At a control section write

$$e_s = s - S(u), \qquad e_k = k - S'(u), \qquad e = (e_s, e_k).$$

The desired averaged selected acceleration is $S''(u) - 2e_k - e_s$. It is obtained by prescribing the mean force

$$b_{\mathrm{des}} = -ak - S''(u) + 2e_k + e_s. \tag{16.4}$$

For sufficiently small $|k| + |e_s| + |e_k|$, the strict bounds (15.1) put b_{des} inside $[b_-, b_+]$, a fixed positive distance from both endpoints. Consequently

$$\beta = \frac{b_{\mathrm{des}} - b_-}{b_+ - b_-} \tag{16.5}$$

lies in a fixed compact subinterval of $(0,1)$. Choose the next two-block word by Lemma 14, using this fraction and intended slow duration $h = h(u)$. Its averaged velocity equation is exactly $k' = S'' - 2e_k - e_s$.

The discrete tracking estimate. The endpoint estimates (14.4), Taylor's theorem for the target, and (14.16) give

$$e_{\mathrm{new}} = (I + hA)e + O(h^2 + \varepsilon), \qquad u_{\mathrm{new}} = u + h + O(h^2 + \varepsilon), \qquad A = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}. \tag{16.6}$$

The prefix estimate is needed here: choosing a new future changes the old endpoint center and clock by $O(\varepsilon)$, and that change is included in (16.6). The positive definite matrix

$$Q = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \quad \text{satisfies} \quad A^{\mathsf{T}}Q + QA = -I. \tag{16.7}$$

Hence there exist $c_0, h_0 > 0$ such that

$$\|(I + hA)e\|_Q \le (1 - c_0 h)\|e\|_Q, \qquad 0 < h \le h_0. \tag{16.8}$$

Choose M sufficiently large. Then choose the late axial residence so long that the parameter at the end of the finite entry obeys

$$\varepsilon(u) \le c_1 h(u)^2 \tag{16.9}$$

for the small fixed constant needed in (16.6). This persists for all future u: the exact relation is $\varepsilon(u) = \varepsilon_* e^{-au}$, and, when $M \ge 2/a$, $e^{-au}(M+u)^2$ is nonincreasing for $u \ge 0$. A fixed small enlargement covers an initial interval $u = O(\varepsilon_N^*)$ as well.

There is a constant B_0, independent of M and of the starting time, such that

$$\|e_j\|_Q \le B_0 h(u_j) \tag{16.10}$$

is invariant under (16.6). Indeed, the right-hand side of (16.8) applied to $B_0 h$ and the local error give

$$\|e_{j+1}\|_Q \le B_0 h - (c_0 B_0 - C) h^2.$$

On the other hand, $h(u_{j+1}) = h - h^3 + O(h^4)$ by (16.6) and (16.9). Choose $B_0 > 2C/c_0$ and then take M large. The negative h^2 margin dominates the $O(h^3)$ change of the barrier. Increasing M also makes $|k| + |e_s| + |e_k|$ small enough for (16.5) at every step.

During a word, bounded acceleration and its slow duration $O(h)$ give

$$|s - S(u)| + |k - S'(u)| \le Ch(u), \qquad |k| \le C_2 h(u). \tag{16.11}$$

The same bounds cover the short transition flights. Also $u_{j+1} - u_j \ge h(u_j)/2$ for large M and a sufficiently small starting parameter. The sequence of planned endpoint clocks therefore tends to infinity: if it were bounded, its positive increments would be bounded below.

The cold-anchored choice of infinitely many words. Fix M before the late residence N. Choose a first finite-entry code as in Lemma 18, and let t_* be its endpoint physical time. This clock origin is fixed once and for all. It is not reset when the future is changed. Subsequent rewrites move the entry endpoint clock by $O(\varepsilon_N^*)$ and its center by the same order. Thus the initial tracking error is $O(\varepsilon_N^*)$, with initial $u = O(\varepsilon_N^*)$, which lies inside (16.10) for N large enough.

At stage j, let n_j be the end of the fixed prefix. Complete its future by a constant tail and take its actual cold intersection from Lemma 16. Evaluate that trajectory's center and clock at n_j, compute (16.4)–(16.5), and append the two chosen dwell blocks and their connecting flights. Keep all symbols through n_j unchanged and again use a constant completion beyond the new endpoint n_{j+1}. The new cold intersection need not have the old center at n_j, but (14.16) bounds the change by $C\varepsilon_{n_j}$. Therefore the actual new endpoint satisfies (16.6). This proves (16.10) by induction for the successively constructed endpoints.

The increasing fixed prefixes define one infinite future code ω^*. The cold intersection map is continuous under common-prefix convergence by (14.14) and the scalar contraction

in Lemma 16. Its cold parameters therefore converge to the cold parameter for ω^*. For a fixed word, compare ω^* directly with the stage at which that word was fixed. Equation (14.16) controls the whole preceding prefix, including its clock, with exponentially localized error. Thus (16.11) holds for the limiting code on every word. There is no sum of unbounded errors from successive rewrites: each such comparison is made directly with that one finite stage. Smooth dependence of the regular cold solution on each fixed physical-time interval identifies this limiting trajectory with one regular cold solution of (12.4).

The physical-region and energy bootstrap. Reserve a positive distance from the averaged entry state to the energy boundaries, and a positive distance from the target range to the selected boundaries. During the fixed early cold prefix, smooth dependence keeps the solution near the axial reference. The recut cone and bounded-flight estimates in Lemma 15 keep the remaining cold residence inside its reserved neighborhood. The finite entry has the strict bounds proved in Lemma 18.

For a retained controlled prefix, up to a possible first exit from the physical rectangle, (16.11) and (12.6) give

$$E(\sigma_*) - E(\sigma) = a \int_0^u k(v)^2 \, dv \le aC_2^2 \int_0^u (M+v)^{-2} \, dv \le \frac{aC_2^2}{M}. \tag{16.12}$$

The $O(\varepsilon_N^*)$ initial clock shift changes this bound only by a harmless error. Take M so large that this loss and $k^2/2$ are less than a fixed fraction of the reserved energy margin. Since $H = E - k^2/2$, H cannot reach an energy boundary. The selected estimate in (16.11) and the bound $|k| \le C_2/M$ similarly exclude the selected and velocity boundaries. The fast orbit stays in the fixed return-flow tubes by construction. Hence no first exit is possible.

For a later finite stage, compare each earlier word directly with the stage that fixed it, using (14.16); the same bounds apply. They consequently pass to the limiting code. The bounded return-flow tubes and the positive lower return-time bound give a solution for every future ρ. The initial finite cold preparation is already an actual regular solution. This establishes the entire physical orbit in the valley strip. Lemma 8 now applies for all physical time.

Finally the normal Hamiltonian remains in a compact positive interval, so Z and Z_ρ are bounded. Therefore $q = t^{-a}Z \to 0$. The change $v = \log((M+u)/M)$ is increasing and unbounded, so $\omega(S) = \omega(Q)$. The tracking error tends to zero, hence $\omega(s) = \omega(Q)$. In particular a nondegenerate target cluster set gives nonconvergence. Equation (16.12) proves the squared-velocity bound in (16.2). The cold parameter can be made arbitrarily close to the reference parameter by increasing the initial axial residence: the recut preimages in Lemma 15 shrink to the reference cold point. ∎

17 Exact convex globalization on a compact collar

The local valley will be extended without perturbing the constructed physical trajectory. The extension is quadratic outside a compact ball; in particular it has a globally Lipschitz gradient.

Lemma 19. *For $\delta > 0$, define*

$$\phi_\delta(r) = \begin{cases} |r|, & |r| \geq \delta, \\ \dfrac{3\delta}{8} + \dfrac{3r^2}{4\delta} - \dfrac{r^4}{8\delta^3}, & |r| \leq \delta. \end{cases} \tag{17.1}$$

This function is convex, semialgebraic, and C^2, with $|\phi_\delta'| \leq 1$ and $\phi_\delta \geq |\cdot|$. The inequality is strict on $(-\delta, \delta)$. Consequently

$$\mathcal{M}_\delta(u, v) = \frac{u + v + \phi_\delta(u - v)}{2} \tag{17.2}$$

is a convex semialgebraic function of class C^2. It is nondecreasing in each argument and agrees with $\max\{u, v\}$ when $|u - v| \geq \delta$.

Proof. The value and first two derivatives of the polynomial piece agree with those of $|r|$ at both endpoints. Inside the smoothing interval,

$$\phi_\delta'(r) = \frac{3r}{2\delta} - \frac{r^3}{2\delta^3}, \qquad \phi_\delta''(r) = \frac{3}{2\delta}\left(1 - \frac{r^2}{\delta^2}\right) \geq 0.$$

Thus $|\phi_\delta'| \leq 1$. Writing $u = |r|/\delta \in [0, 1]$ gives

$$\phi_\delta(r) - |r| = \frac{\delta}{8}(1 - u)^3(u + 3),$$

which proves the remaining inequalities. The Hessian of $\mathcal{M}_\delta$ is

$$\frac{\phi_\delta''(u - v)}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \geq 0,$$

and its two partial derivatives are $(1 + \phi_\delta')/2$ and $(1 - \phi_\delta')/2$. This proves convexity and monotonicity; the exact maximum property follows from (17.1). ∎

Proposition 17.1 (Exact compact-collar extension)**.** Write $X = (s, q) \in \mathbb{R} \times \mathbb{R}^2$. Suppose P is convex, nonnegative, C^2, and semialgebraic on a neighborhood of the closed ball $\overline{B}_{R_2}$, and that its zero set there is exactly $q = 0$. Given $0 < R_1 < R_2$, there is a finite, convex, nonnegative, coercive, semialgebraic C^2 function $f : \mathbb{R}^3 \to \mathbb{R}$ such that:

1. $f = P$ on $\overline{B}_{R_1}$;
2. ∇f is globally Lipschitz;

3. for some $r_0 \in (R_1, R_2)$,

$$\operatorname*{argmin}_{\mathbb{R}^3} f = \{(s,0) : |s| \le r_0\}, \qquad \min f = 0. \tag{17.3}$$

The extension depends only on P and the chosen radii, not on the eventual choice of a trajectory lying in B_{R_1}.

Proof. Fix $\delta_g > 0$, let $M_P = \max_{|X|=R_2} P(X)$, and choose $L_g > 0$ so that

$$L_g(R_2^2 - R_1^2) > M_P + 4\delta_g. \tag{17.4}$$

Set

$$B_g = L_g R_1^2 + 2\delta_g, \qquad Q(X) = L_g|X|^2 - B_g. \tag{17.5}$$

Then $Q \le -2\delta_g$ on $\overline{B}_{R_1}$, whereas $Q > P + 2\delta_g$ on $|X| = R_2$. By continuity, $Q > P + \delta_g$ on an open collar of that sphere contained in the domain of P.

Define

$$f(X) = \begin{cases} \mathcal{M}_{\delta_g}(P(X), Q(X)), & |X| < R_2, \\ Q(X), & |X| \ge R_2. \end{cases} \tag{17.6}$$

On the outer collar the first expression is exactly Q. Thus the two pieces agree on a neighborhood of the gluing sphere, and f is globally C^2 and semialgebraic. Composition of a convex coordinatewise nondecreasing function with convex functions proves convexity of the interior piece. Outside the ball, $f = Q$ is convex. The collar agreement therefore gives a positive semidefinite Hessian everywhere, so f is globally convex.

On the inner ball, $P \ge 0$ and $Q \le -2\delta_g$, hence $P - Q \ge 2\delta_g$ and $f = P$ exactly. Nonnegativity inside the outer ball follows from $\mathcal{M}_{\delta_g}(P, Q) \ge P \ge 0$. On its boundary $Q > 2\delta_g$, and the radial quadratic is increasing with $|X|$, so $f > 0$ outside the ball as well. Coercivity follows from the quadratic expression there.

The Hessian of the inner expression is continuous on a compact set and therefore bounded, while $D^2 f = 2L_g I$ outside. If L_f is a common operator-norm bound, integration along a line segment gives

$$|\nabla f(X) - \nabla f(Y)| \le L_f |X - Y| \qquad (X, Y \in \mathbb{R}^3).$$

This proves item 2.

It remains to identify every minimizer, including the endpoints of the segment. If $|X| < R_2$ and $f(X) = 0$, the inequality $f \ge P \ge 0$ forces $P(X) = 0$, hence $q = 0$. At such a point put $v = Q(s, 0)$. Lemma 19 gives

$$\mathcal{M}_{\delta_g}(0, v) = 0 \quad \Longleftrightarrow \quad v \le -\delta_g.$$

Indeed equality holds on that range by the exact maximum formula; for $-\delta_g < v < \delta_g$, the strict inequality $\phi_{\delta_g}(v) > |v|$ makes the value positive, and for $v \ge \delta_g$ it equals $v > 0$. There are no zeros outside the outer ball. Consequently

$$f(s, q) = 0 \quad \Longleftrightarrow \quad q = 0, \quad L_g s^2 - B_g \le -\delta_g.$$

Thus (17.3) holds with

$$r_0^2 = \frac{B_g - \delta_g}{L_g} = R_1^2 + \frac{\delta_g}{L_g}. \tag{17.7}$$

This is strictly greater than R_1^2 and, by (17.4), strictly less than R_2^2. The segment is therefore nondegenerate and lies entirely in the region where P is defined. Its points attain value zero, which is the global minimum. All parameters have been selected independently of any particular orbit. ∎

17.1 Preserving the complete regular cold orbit

Apply Proposition 17.1 to the valley of Proposition 11.2. Reserve a selected-coordinate bound $S_1 < S$ for the fixed early cold prefix, the recut axial preparation, the finite inward entry, and the subsequent controlled tail. Before choosing the limiting cold parameter, restrict the transverse cold family so that $|q(0)| \le Q_0$ and choose

$$S_1^2 + Q_0^2 < R_1^2 < R_2^2 < S^2. \tag{17.8}$$

The reference normal amplitude can be made arbitrarily small, and homogeneity preserves the frozen network at its resulting positive energy, so these strict inequalities impose no restriction on the construction. The closed outer ball lies in the convex strip.

By the exact cold normal estimate in Lemma 8, $|q(t)| \le |q(0)| \le Q_0$ throughout every retained physical prefix and throughout the final orbit. Together with $|s(t)| \le S_1$, this gives

$$|X(t)|^2 \le S_1^2 + Q_0^2 < R_1^2 \qquad (t \ge 0).$$

The objective (17.6) and all its parameters are therefore fixed before the final cold datum is selected, and the complete physical trajectory stays where $f = P$ and $\nabla f = \nabla P$ exactly. Auxiliary future completions used to define coding leaves need not satisfy this confinement; only the retained physical prefixes and their limit are used here.

For completeness, there is no uniqueness ambiguity at the singular initial time. A regular cold solution for a globally Lipschitz gradient satisfies

$$X(t) = X_0 - \int_0^t u^{-\alpha} \int_0^u r^{\alpha} \nabla f(X(r))\, \mathrm{d}r\, \mathrm{d}u. \tag{17.9}$$

For two solutions with the same X_0, the sup norm of their difference on $[0,t]$ is at most $L_f t^2/[2(\alpha+1)]$ times itself. This proves local uniqueness for small t; ordinary positive-time uniqueness continues it. The same integral equation gives existence near zero and $X_t(0) = 0$. Global Lipschitz continuity prevents finite-time escape on positive-time intervals. Hence the preserved local-valley orbit is precisely the unique global regular cold solution for f with that initial position.

18 One fixed global objective and every continuum target

We now fix the objective before choosing any target or limiting cold datum. The exact globalization in Section 17 is applied to a collar reserved for the entire cold preparation family, not only for one eventual tail.

Proposition 18.1 (A fixed global scalar controller)**.** There are a finite even $p > 4$, $\alpha = 1 + 2/p$, a function $f_0 : \mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}$, a nondegenerate compact interval I, and an open interval $J \Subset \operatorname{int} I$ with the following properties. The function f_0 is nonnegative, coercive, convex, semialgebraic and C^2, with $\operatorname{Lip}(\nabla f_0) \le 1$ and

$$\operatorname{argmin} f_0 = I \times \{(0,0)\}.$$

There are fixed $s_a, q_* > 0$ and $s_* \in J$ such that every C^3 curve $Q : [0,\infty) \to J$ with bounded first three derivatives, $Q(0) = s_*$ and $Q'(0) = 0$ has the following realization. For sufficiently large M, put

$$S(u) = Q\left(\log \frac{M+u}{M}\right).$$

Some regular cold solution for the fixed f_0, from

$$s(0) = s_a, \qquad q(0) = (q_*, \zeta), \qquad X_t(0) = 0, \tag{18.1}$$

satisfies, for a finite $c \in \mathbb{R}$ and all sufficiently large t,

$$|s(t) - S(\log t - c)| + |ts_t(t) - S'(\log t - c)| \le \frac{C}{M + \log t - c}, \qquad q(t) \to 0. \tag{18.2}$$

Its entire trajectory lies in one fixed bounded collar. Only ζ changes with the target.

Proof. Choose the perturbed ellipse and then a sufficiently large finite even $p > 4$ as in Proposition 11.2. The billiard perturbation is fixed before the steepness is chosen. Fix the linked axial and nonaxial periodic branches, the force-separating polynomial T, and $c_0 > 0$ denoting the separation constant called c in (11.11).

Fix a strip width $S > 0$ and then K so large that $s_a = c_0/(2K) < S/8$. Choose $\eta > 0$ sufficiently small and set $C = K\eta$. The local potential is the one fixed finite expression

$$P(s,q) = (1 + Cs^2)V(q) + \eta s\big(T(q) - c_0 V(q)\big).$$

It has all the convexity, coercivity and uniform-network properties of Proposition 11.2 on a neighborhood of the closed strip.

Reserve $S_1 = S/2$ for the selected coordinate. The cold graph neighborhoods lie around s_a inside $|s| < S/4$. The finite inward maneuver can be chosen to stay there, with its strict-force neighborhood inside $|s| < S/3$. Choose $Q_0 < S/8$ and set

$$R_1 = \frac{3S}{4}, \qquad R_2 = \frac{7S}{8}. \tag{18.3}$$

Then $S_1^2 + Q_0^2 < R_1^2 < R_2^2 < S^2$. Choose a positive reference energy H_* so small that

$$2q_* < Q_0, \qquad q_* = \frac{\sqrt{2H_*}}{a}.$$

Homogeneity transports the frozen network to a compact positive band about H_*. Restrict the original transverse cold family (18.1) so that $|q(0)| \le 2q_* < Q_0$. The member $\zeta = 0$ is the exact axial reference.

Next fix the averaged finite inward maneuver from $(s_a, 0, H_*)$ to $(s_*, 0, H_e)$ in Lemma 18. Choose its pulse short enough to retain the selected margins and a positive residual-energy margin. Fix the braking duration, its strict-force rectangle, and $J \ni s_*$ with $\overline{J}$ strictly inside that rectangle and inside $|s| < S/3$. None of these choices depends on Q.

Now fix the extension in Proposition 17.1 with the radii (18.3). Explicitly, take $\delta_g > 0$ and $L_g > 0$ so large that

$$L_g(R_2^2 - R_1^2) > \max_{|X|=R_2} P(X) + 4\delta_g, \qquad B_g = L_g R_1^2 + 2\delta_g.$$

Formula (17.6) defines a single global $f_0 = P$ throughout $\overline{B}_{R_1}$, with all the asserted regularity and convexity properties and minimizer interval

$$I = \left[-\sqrt{\frac{B_g - \delta_g}{L_g}}, \sqrt{\frac{B_g - \delta_g}{L_g}} \right]. \tag{18.4}$$

It remains to verify that the same extension works for every target.

Given Q, choose M large enough for its tracking tube and the loss C_Q/M to fit the already fixed selected and energy margins. Next choose the axial residence index N so large that the cold graph slopes, finite-entry error, return rounding and $\varepsilon \le c_1(M+u)^{-2}$ meet the required tolerances. Theorem 10 gives the target orbit through genuine cold intersections. Its fixed early prefix stays close to the axial reference, its recut cold residence is controlled by Lemma 15, and its entry and feedback prefixes remain inside $|s| < S_1$. These are the all-time selected bounds, including the preparatory residence.

Every retained physical prefix and the limiting cold orbit satisfy Lemma 8:

$$|q(t)| \le |q(0)| < Q_0.$$

Thus $|X(t)|^2 < S_1^2 + Q_0^2 < R_1^2$ from $t = 0$ onward. The fixed extension agrees with P and its gradient along every such trajectory. Auxiliary future completions only define code leaves and need not lie in this collar; the retained physical prefixes and their limit do.

For completeness, uniqueness and regularity at the singular endpoint follow from the Volterra equation

$$X(t) = X_0 - \int_0^t r^{-\alpha} \int_0^r v^\alpha \nabla f_0(X(v))\, \mathrm{d}v\, \mathrm{d}r. \tag{18.5}$$

On $[0,T]$ its contraction constant in the uniform norm is at most $L_f T^2/[2(\alpha+1)]$, where $L_f = \mathrm{Lip}(\nabla f_0)$. Its fixed point has $X_t(0) = 0$ and $X_{tt}(0) = -\nabla f_0(X_0)/(\alpha+1)$. Ordinary positive-time uniqueness continues the solution, and global Lipschitz continuity prevents finite-time blowup on every interval away from zero. Hence the preserved orbit is the unique global regular cold solution for the fixed objective.

The tracking statement follows from (16.2), with $c = \log t_*$. If $L_f > 1$, replace f_0 once and for all by λf_0, where $0 < \lambda \le 1/L_f$. The corresponding trajectory is $\widetilde{X}(t) = X(\sqrt{\lambda}\, t)$; direct substitution shows that it solves the equation for λf_0 with the same α and cold position. This common fixed time rescaling only replaces c by $c - \frac{1}{2}\log\lambda$. It preserves every omega-limit set and all geometrical properties. The normalized scalar objective is fixed before any target is selected. ∎

18.1 Synchronization and proof of the universal theorem

Proof of Theorem 1. Take the fixed normalized scalar objective f_0, the intervals I, J, and the fixed cold family of Proposition 18.1. For every $m \ge 1$, define

$$F_m(s_1, q_1, \dots, s_m, q_m) = \sum_{j=1}^{m} f_0(s_j, q_j), \qquad q_j \in \mathbb{R}^2. \tag{18.6}$$

All regularity, nonnegativity, convexity and semialgebraicity assertions follow from those of f_0. Coercivity follows because an unbounded product sequence has an unbounded block, whose nonnegative summand tends to infinity along the corresponding subsequence. More explicitly, the sublevel set $\{F_m \le R\}$ is contained in the product of the compact sets $\{f_0 \le R\}$. The zero set is exactly $I^m \times \{0\}$. Moreover

$$|\nabla F_m(X) - \nabla F_m(Y)|^2 = \sum_{j=1}^{m} |\nabla f_0(X_j) - \nabla f_0(Y_j)|^2 \le \sum_{j=1}^{m} |X_j - Y_j|^2.$$

Thus the gradient Lipschitz constant is at most one, independently of m. The same finite degree and damping exponent work for every product.

Let $K \subset J^m$ be any continuum. Apply Lemma 4 in the convex open box J^m from $s_e = (s_*, \dots, s_*)$. Write its vector target as

$$S(u) = (S_1(u), \dots, S_m(u)) = Q\left(\log \frac{M+u}{M}\right).$$

All components obey Proposition 18.1. Choose one common M sufficiently large for their finitely many derivative and margin bounds. Independently choose each scalar cold itinerary and its transverse parameter ζ_j. Since the equation for F_m splits into independent blocks at the same physical time, their product is one regular cold solution with

$$s_j(t) - S_j(\log t - c_j) \longrightarrow 0, \qquad q_j(t) \longrightarrow 0$$

for finite entry-clock constants c_j.

It is essential to compare the joint trajectory with one common vector target, rather than merely multiply its coordinate omega-limit sets. Fix any finite reference c_*, put $u = \log t - c_*$ and $d_j = c_j - c_*$. For large u, the derivative bound gives

$$|S_j(u - d_j) - S_j(u)| \le \frac{C_j |d_j|}{M + u - |d_j|} \longrightarrow 0. \tag{18.7}$$

Consequently

$$|s(t) - S(\log t - c_*)| \longrightarrow 0.$$

Since the vector curve S has exactly the cluster set K, the joint positional omega-limit set of the product solution is $K \times \{0\}$. Each block remains in the fixed scalar collar, so the product trajectory is bounded. All selected cold coordinates equal s_a and all first normal cold coordinates equal q_*. Only the m transverse parameters ζ_j depend on K. This proves every fixed-objective and fixed-family assertion. ∎

Corollary 18.1 (A regular cold interval counterexample)**.** *There is a convex, coercive, nonnegative, semialgebraic C^2 function on $\mathbb{R}^3$, with globally Lipschitz gradient and a nondegenerate compact minimizer segment, for which a bounded regular cold solution at some $\alpha = 1 + 2/p \in (1, 3/2)$ fails to converge. Its normal coordinates tend to zero and its omega-limit set is a nondegenerate closed interval of minimizers.*

Proof. Take $m = 1$ and any nondegenerate closed interval $K \Subset J$ in Theorem 1. For an explicit slow target with two extreme cluster values, choose $d > 0$ so that $[s_*, s_* + 2d] \subset J$ and let

$$Q(v) = s_* + d(1 - \cos v), \qquad S(u) = s_* + d\left[1 - \cos\left(\log \frac{M + u}{M}\right)\right]. \tag{18.8}$$

This Q has bounded first three derivatives and $Q'(0) = 0$. For every $r \in [s_*, s_* + 2d]$, choose $\vartheta \in [0, \pi]$ with $r = s_* + d(1 - \cos\vartheta)$ and put

$$u_j = M(e^{\vartheta + 2\pi j} - 1).$$

Then $u_j \to \infty$, $S(u_j) = r$, and the cold tracking estimate gives a subsequence converging to $(r, 0, 0)$. Conversely the target lies in this interval and the tracking and normal errors vanish. Thus the omega-limit set is exactly the interval. ∎

For $m \ge 2$, the target may be a circle, a connected graph, a compact region, a non-locally-connected continuum, or a singleton, provided it lies in J^m. Its complexity is encoded by the cold initial datum. The objective itself remains one finite semialgebraic expression. The construction uses $2m$ normal coordinates; no optimal dimension claim is made.

19 Persistent cold drift with zero residual energy

The positive-energy controller in Section 16 reserves a fixed normal-energy band. We now keep the same objective and cold family, but replace that controller by one which exhausts the rescaled energy. The normal dynamics is normalized to a nondegenerate energy shell, while the logarithm of the actual energy is retained among the center coordinates.

Theorem 11 (A cold interval limit set at zero residual energy)**.** *The fixed three-dimensional objective $f_0 = F_1$ of Proposition 18.1 admits a bounded regular cold solution from the family* (18.1) *such that*

$$E(\sigma) \longrightarrow 0, \qquad H(\sigma) \longrightarrow 0, \qquad \omega(X) = [s_e, s_e + 2d] \times \{(0,0)\} \tag{19.1}$$

for some $d > 0$ and a compact interval $[s_e, s_e + 2d] \subset \operatorname{int} I$. The entire trajectory stays in the collar where f_0 is exactly homogeneous of degree p in the normal variables, up to the fixed multiplicative normalization of the objective. Here $p > 4$ is the same finite even degree and $\alpha = 1 + 2/p$ is the same critical damping as before. In addition,

$$E(\sigma) \asymp \frac{1}{1 + \sigma - \sigma_e} \quad (\sigma \geq \sigma_e), \tag{19.2}$$

where σ_e is a sufficiently late fixed entry time and $\asymp$ means two positive constant bounds.

The theorem asserts one nondegenerate interval limit set at zero residual energy. It does not add a zero-energy requirement to the arbitrary-continuum conclusion of Theorem 1. Its geometric inputs are Proposition 11.2, the exact cold reference and unstable transverse variation in Lemma 9, and the fixed extension of Proposition 17.1. The estimates which depended on a positive energy band are proved anew below.

19.1 Exact energy normalization and the two slow clocks

We first work before the harmless final constant scaling of the objective, so its local expression is P. Set

$$\begin{gathered} \gamma = \frac{p-2}{2p}, \qquad E = H + \frac{k^2}{2} > 0, \qquad v = \frac{k}{E}, \quad z = E^{-1/p} Z, \quad w = E^{-1/2} W, \\ r = \log E, \qquad \delta = \varepsilon E^{-\gamma}, \qquad \ell = \log \delta, \qquad \theta_\rho = E^\gamma, \qquad \tau_\sigma = E. \end{gathered} \tag{19.3}$$

The selected coordinate is scalar in this section. The clock $\sigma = \log t$ remains the physical logarithmic time; τ will parametrize the selected target.

Lemma 20 (Exact normalized equations)**.** *In the variables* (19.3)*, the critical system is*

$$\begin{aligned} z_\theta &= w + \frac{a}{p}\delta E v^2 z, & w_\theta &= -P_z(s,z) + \frac{a}{2}\delta E v^2 w, \\ s_\theta &= \delta E v, & v_\theta &= \delta[-av - P_s(s,z) + aEv^3], \\ r_\theta &= -a\delta E v^2, & \tau_\theta &= \delta E, \\ \ell_\theta &= -a\delta(1 - \gamma E v^2), & E &= e^r. \end{aligned} \tag{19.4}$$

It preserves the constraint

$$J := \frac{|w|^2}{2} + P(s,z) = 1 - \frac{Ev^2}{2}, \qquad \sigma = -\frac{\ell + \gamma r}{a}, \qquad \sigma_\theta = \delta. \tag{19.5}$$

Conversely, the inverse transformation on this constraint reconstructs the exact critical equations.

Proof. From $E_\sigma = -ak^2$ and $k = Ev$ one has $E_\rho = -a\varepsilon E^2 v^2$ and $E_\theta = -a\delta E^2 v^2$. Differentiating $E^{-1/p}Z$, using $Z_\rho = W$ and $1/2 - 1/p = \gamma$, gives the first equation. Differentiating $E^{-1/2}W$ and using the degree-$(p-1)$ homogeneity of P_Z gives the second. Similarly, the degree-p homogeneity of P_s gives the equation for $v = k/E$. The remaining equations follow by differentiating the two logarithms and the clocks.

For an independent check of the constraint, put $\mathcal{C} = |w|^2/2 + P(s,z) + Ev^2/2$. Euler's identity $z \cdot P_z = pP$ and (19.4) give

$$\mathcal{C}_\theta = a\delta E v^2 (\mathcal{C} - 1).$$

Thus $\mathcal{C} = 1$ is invariant. The logarithmic identity in (19.5) follows from $\delta = \varepsilon E^{-\gamma}$; differentiation gives $\sigma_\theta = \delta$. On the constraint, reversing all changes gives (4.1); in particular $\varepsilon = \delta E^\gamma$ obeys $\varepsilon_\rho = -a\varepsilon^2$. ∎

Choose a broad compact selected preparation collar K_{prep} containing s_a and the finite inward maneuver, and numbers $E_{\max}, V_{\mathrm{prep}} > 0$ with

$$\begin{gathered} s \in K_{\mathrm{prep}}, \qquad |v| \le V_{\mathrm{prep}}, \qquad r \le \log E_{\max}, \\ E_{\max} V_{\mathrm{prep}}^2 \le \eta_0 < 1, \qquad \gamma\eta_0 < \tfrac{1}{2}. \end{gathered} \tag{19.6}$$

All coefficients of (19.4) and their required derivatives are uniformly bounded there, apart from the displayed small factor δ. Indeed every positive-order r derivative of e^r equals e^r, while $J \in [1 - \eta_0/2, 1]$. Thus the direction $r \to -\infty$ does not collapse the normalized normal shell. The smaller selected and velocity collars used for feedback will be chosen only after entry. They need not contain the preparation.

For each branch $c \in \{-, +\}$, homogeneity transports its unit-energy orbit by

$$z_J(\theta) = J^{1/p} z_1(J^\gamma \theta), \qquad w_J(\theta) = J^{1/2} w_1(J^\gamma \theta). \tag{19.7}$$

Consequently, with $\beta_c(s) = b_c(s,1)$,

$$b_c(s,J) = J\beta_c(s), \qquad T_c(s,J) = J^{-\gamma} T_c(s,1). \tag{19.8}$$

The near-unit band in (19.6) has uniform periods, hyperbolicity, transverse connecting flights and section charts by Proposition 11.2. For positive δ, the vector field on any such finite normalized return flight is a uniformly C^2 perturbation of order δ. No inverse power of E enters this estimate.

19.2 Uniform coding on the negative logarithmic-energy half-line

Lemma 21 (Normalized coding and cold prefix locality)**.** *The cross-form, coded-leaf, local regularity and half-line constructions of Lemmas 10–17 apply to* (19.4) *on* (19.6)*, with center*

$$Y = (s, v, r, \tau, \ell),$$

and constants uniform for $r \le \log E_{\max}$. A common future prefix through index L gives, at its genuine cold intersections,

$$|\Delta Y_i| \le C\delta_i \vartheta_1^{L-i}, \qquad 0 \le i \le L, \tag{19.9}$$

for a fixed $0 < \vartheta_1 < 1$. In particular it controls both τ and the actual σ, and it changes E_i by a relative $O(\delta_i \vartheta_1^{L-i})$ amount.

Proof. We verify the hypotheses and the changes needed in the earlier proofs. Use frozen section charts at (s, J), where $J = 1 - e^r v^2/2$. The uniform near-unit shell and the finite-flight perturbation give cross forms

$$\begin{aligned} x_{i+1} &= f_{cd}(x_i, y_{i+1}, Y_i, \delta_i), \\ y_i &= g_{cd}(x_i, y_{i+1}, Y_i, \delta_i), \\ Y_{i+1} &= Y_i + \delta_i \Phi_{cd}(x_i, y_{i+1}, Y_i, \delta_i). \end{aligned} \tag{19.10}$$

After the fixed local iterates in Lemma 10, the joint fast derivative of (f, g) has norm at most $\lambda < 1$ and the image has a uniform interior margin. All derivatives needed through order two are uniformly bounded. The τ coordinate is additive: the normal and other center equations do not depend on it.

For the auxiliary coding maps, smoothly clamp s and v to slightly larger preparation collars. Replace e^r above a finite upper level by a bounded smooth function agreeing with e^r throughout the working region. No cutoff is made as $r \to -\infty$. The resulting J can be clamped to a slightly larger compact positive band for evaluating frozen charts. To treat ℓ on the whole line, use

$$\delta_{\text{eff}} = \chi(e^\ell),$$

where $\chi(u) = u$ below the physical cutoff and is a smooth positive cap above it. Choose the cap so its first two ℓ derivatives are $O(\delta_{\max})$ and $|\partial_\ell \log \delta_{\text{eff}}|$ is bounded. The extended ℓ drift can be kept between two fixed negative multiples of δ_{eff}. These extensions leave (19.4) unchanged on the physical region. The center increments are globally bounded and uniformly Lipschitz, including the unbounded r, τ, ℓ directions. For sufficiently small $\delta_{\max}$ each center map is $I + g$ with $\mathrm{Lip}(g) < 1$, hence a global diffeomorphism: its inverse solves $x = y - g(x)$ by contraction.

For a proposed bounded fast sequence $U = (x_i, y_i)$, anchor Y at index zero and solve its forward and backward recurrences. With the weight $R^{-|i|}$, the center response satisfies

$$\|\Delta Y\|_R \le \frac{C\delta_{\max}}{R - 1 - C\delta_{\max}} \|\Delta U\|_R. \tag{19.11}$$

This is the geometric sum of $D_{i+1} \le (1 + C\delta_{\max})D_i + C\delta_{\max}|\Delta U_i|$ and its backward counterpart. Thus the fast fixed-point contraction constant is at most

$$\lambda R + \frac{C\delta_{\max}}{R - 1 - C\delta_{\max}} < 1 \tag{19.12}$$

after first choosing $1 < R < 1/\lambda$ and then reducing $\delta_{\max}$. Only uniform derivative bounds enter this argument; boundedness of the center coordinate itself is unnecessary.

The local C^1 proof also remains uniform. Choose $1 < R_1$ and $R_1^2 < R_2 < 1/\lambda$. The change produced by an anchor increment h is bounded at index i by $C|h|R_1^{|i|}$. The uniform second-derivative bound makes the Taylor remainder after the Jacobi equations at most $C|h|^2 R_1^{2|i|}$; the center remainders have an extra $\delta_{\max}$ factor. Its R_2 norm is $O(|h|^2)$. The contraction inverse in that norm gives the local derivative and its continuity exactly as in (13.8). This does not assert differentiability of the nonlinear sequence operator in a single exponential norm.

For locality, the bounded logarithmic derivative of the cap and the $O(\delta_{\max})$ center increments give

$$\delta_{\mathrm{eff},j} \le C\delta_{\mathrm{eff},i} e^{C\delta_{\max}|j-i|}. \tag{19.13}$$

At a physical anchor $\delta_{\mathrm{eff},i} = \delta_i$. The local center drift is bounded by this expression times $C|j - i|$. Weighted geometric summation therefore makes the frozen-graph defect $O(\delta_i)$, and gives an $O(\delta_i)$ derivative in ℓ. Changes of symbols at distance L have weighted size $O(\vartheta^L)$ for some $\vartheta < 1$. These are the local estimates of Lemma 12, now uniform in r.

On retained physical flights, there is also the exact identity

$$(\delta^{-1})_\theta = a(1 - \gamma E v^2). \tag{19.14}$$

By (19.6) and the upper and lower flight-time bounds, $\delta_i \asymp (\delta_0^{-1} + i)^{-1}$. This supplies the same local ratio bounds on physical prefixes.

For half-line graphs, the stable boundary x_0 is retained. Recentring the weighted contraction at index i gives

$$|\Delta U_i| \le C|\Delta Y_i| + C\vartheta^i|\Delta x_0| + C\vartheta^{L-i}. \tag{19.15}$$

The boundary term is the forcing at index zero, with its actual exponential weight; it cannot be discarded. The cold-intersection construction verified just below gives

$$|\Delta Y_0| + |\Delta x_0| \le C\delta_0\vartheta^L. \tag{19.16}$$

Combining (19.15) with the center recurrence gives

$$D_{i+1} \le (1 + C\delta_i)D_i + C\delta_i\vartheta^i|\Delta x_0| + C\delta_i\vartheta^{L-i}.$$

In its variation-of-constants sum, the ratios δ_j/δ_i and the growth factors are bounded by $Ce^{C\delta_{\max}(i-j)}$. Reducing $\delta_{\max}$ and enlarging ϑ slightly to $\vartheta_1 < 1$ sums the geometric forcing to $C\delta_i\vartheta_1^{L-i}$. The initial-center and stable-boundary contributions have the same bound by (19.16). This proves (19.9) and closes the small-difference bootstrap. Formula (19.5) controls σ, and $E = e^r$ converts its r estimate into the claimed relative energy estimate.

It remains to justify the cold intersections used here. Fix the original reference amplitude and write $E_* = H_* = a^2q_*^2/2 > 0$. Lemma 9 gives its exact axial orbit with $k = 0, E = E_*$. In normalized variables it has $v = 0, J = 1, \theta = E_*^\gamma\rho$ and $\delta = 1/(a\theta)$. The first selected, energy and section-time variations in the cold transverse parameter vanish; its unstable normal variation is nonzero. The change to (s, v, r, ℓ) is smooth near this fixed positive E_*, so these assertions persist.

At a late axial return N, the recut cold curve is a graph over a fixed unstable interval with stable and center C^1 slopes $O(\delta_N^*)$. Indeed the transformed version of (14.10) is

$$D_{j+1} \le \kappa_0 D_j + C\delta_j^*, \qquad \kappa_0 < 1,$$

and $\delta_j^* \asymp (1 + j)^{-1}$. Splitting the geometric convolution at $j = N/2$ bounds it by $C\delta_N^*$, while the first part is exponentially small. The covering and recutting argument of Lemma 15 then gives the fixed unstable interval. Constants may depend on the fixed E_*; no uniformity as $E_* \downarrow 0$ is needed.

Assign one common value of τ on this late preparation section, so its cold-curve slope is zero there. Propagate it by $\tau_\sigma = E$ thereafter. Its initial additive constant will be chosen so the nominal finite entry ends at $\tau = 0$. This is a choice of clock origin, not another physical initial condition. Each future center-stable graph has uniformly bounded derivative. Substitution of the C^1-small cold curve into its scalar intersection equation is a contraction, as in Lemma 16. A common future prefix changes the unstable intersection coordinate by $O(\vartheta^L)$ and its stable and center coordinates by $O(\delta_N^*\vartheta^L)$. This proves (19.16). Thus each auxiliary future leaf meets a genuine regular cold datum at the preparation section. Agreement with the unmodified physical equations is asserted for the retained prefixes and their limit after the first-exit argument below, not for arbitrary extended future completions. ∎

Lemma 22 (Normalized dwell estimates with both clocks)**.** *Define, on a constant branch c,*

$$G_c(s, v, r, \tau) = \big(Ev, -av - J\beta_c(s) + aEv^3, -aEv^2, E\big). \tag{19.17}$$

For a two-block word of intended logarithmic duration $0 < h \le h_0$, with δ/h sufficiently small and mixing fraction β bounded away from zero and one, its genuine cold-coded

endpoint satisfies

$$\begin{aligned}\Delta\sigma &= h + O(h^2+\delta),\\ (s,v,r,\tau)_{\mathrm{new}} &= (s,v,r,\tau) + h[\beta G_+ + (1-\beta)G_-] + O(h^2+\delta).\end{aligned} \tag{19.18}$$

The constants are uniform on (19.6)*, including* $r \to -\infty$*. Bounded connecting flights, integer rounding and subsequent future-prefix changes are included in this error.*

Proof. Along a dwell occupying indices $m, \dots, m+n$, the local frozen comparison, symbolic locality and stable-boundary memory give, per return,

$$\Delta(s,v,r,\tau) = \delta_i T_c(s_i,J_i) G_c(s_i,v_i,r_i,\tau_i) + R_i, \qquad \Delta\sigma = \delta_i T_c(s_i,J_i) + r_i^{\mathrm{err}},$$

where

$$|R_i| + |r_i^{\mathrm{err}}| \le C\big[\delta_i^2 + \delta_i(\vartheta^{i-m} + \vartheta^{m+n-i} + \vartheta^i)\big]. \tag{19.19}$$

The leading term in v uses exactly the mean force (19.8); the others are already constant along a frozen normal period. The physical clock uses the exact identity $\sigma_\theta = \delta$. The last geometric term is the half-line stable-boundary contribution.

On a bounded σ interval, δ changes by a bounded relative factor. The squared-mesh sum is at most $C\delta_{\mathrm{start}} \sum \delta_i$, while every geometric boundary layer sums to $C\delta_{\mathrm{start}}$. This proves an $O(\delta_{\mathrm{start}})$ finite-window averaging error. For a short intended interval of length h, choose the two dwell counts by rounding their desired durations divided by the initial δT_c. The rounding and connecting flights cost $O(\delta)$. The center, period and relative parameter changes over that interval are $O(h)$, and their accumulated defect is $O(h^2)$. This proves (19.18) simultaneously for both clocks and all four center variables. Lemma 21 changes a previously fixed endpoint by $O(\delta)$ when its future is changed. The force fraction and the mesh chosen below are uniformly smooth in these coordinates, so this change is absorbed by the same error bound. ∎

19.3 Cold entry with a relative energy bound

The reference amplitude is kept fixed before any late coding. Retain the $E_* = H_*$ and $q_* = \sqrt{2E_*}/a$ already chosen in Proposition 18.1, so $2q_* < Q_0$. Restrict the same transverse cold family to $|\zeta| < q_*/2$. Its initial total rescaled energies are at most $5E_*/4$, and we may set $E_{\max} = 2E_*$. Choose $V_{\mathrm{prep}} > 0$ satisfying (19.6). We now choose a finite maneuver within this broad collar, before selecting the much smaller feedback collar.

Lemma 23 (A finite entry with controlled relative loss)**.** *A fixed averaged maneuver from* $(s_a, 0, E_*)$ *ends at* $(s_e, 0, E_e)$ *with* $s_e < s_a$*,* $E_e > 0$*, and a compact selected neighborhood* K_s *of* s_e *on which*

$$\beta_+(s) \ge \beta_* > 0, \qquad \beta_-(s) \le -\beta_* < 0. \tag{19.20}$$

It stays strictly inside the broad preparation collar and has $|k|/E < V_{\mathrm{prep}}/2$*. Its genuine cold-coded realizations have endpoint error at most* $C_{\mathrm{entry}}\delta_N^*$ *in* (s,v,r,τ,σ)*, uniformly under future completions, where* C_{entry} *is finite after the maneuver has been fixed.*

Proof. The force identities (15.2)–(15.3), or homogeneity applied to them, give

$$b_-(s,H) = -H\kappa_0(s)(s_a - s), \qquad b_+(s,H) = H\beta_+(s),$$

where κ_0 is smooth and positive near s_a and $\beta_+(s_a) > 0$. Put $\xi = s_a - s$ and $u = \xi_\sigma = -k$. Follow the excursion branch for a short fixed time $d_0 > 0$. Taylor expansion and the work identity give

$$\begin{gathered} 0 < u_1 \le CE_* d_0, \qquad cE_* d_0^2 \le \xi_1 \le CE_* d_0^2, \\ 0 \le E_* - E_1 \le CE_*^2 d_0^3. \end{gathered} \tag{19.21}$$

Indeed $u_\sigma(0) = E_*\beta_+(s_a) > 0$ and $\xi_1 = E_*\beta_+(s_a)d_0^2/2 + O(E_* d_0^3)$; the energy loss is $a\int_0^{d_0} u^2\,\mathrm{d}\sigma$. Along this pulse $|k|/E = O(d_0)$.

Switch to the axial branch and continue until the first zero of u. As long as $u \ge 0$,

$$\xi_\sigma = u, \qquad u_\sigma = -au - H\kappa_0(s)\xi \le -au. \tag{19.22}$$

The additional selected displacement is at most u_1/a, and

$$a\int u^2\,\mathrm{d}\sigma \le au_1 \int u\,\mathrm{d}\sigma \le u_1^2 \le CE_*^2 d_0^2. \tag{19.23}$$

Also $u^2/2 \le CE_*^2 d_0^2$. Choosing $E_* d_0^2$ sufficiently small gives $H = E - u^2/2 \ge E_*/2$ by a first-exit argument. Consequently, while $u \ge 0$,

$$u_\sigma \le -\frac{E_*}{2}\kappa_{0,\min}\xi_1 < 0.$$

It reaches zero in finite time, at most $C/(E_* d_0)$. At that time,

$$0 < \xi_1 \le s_a - s_e \le CE_* d_0, \qquad E_e \ge E_* - CE_*^2 d_0^2. \tag{19.24}$$

The relative loss is $O(E_* d_0^2)$ and $|k|/E = O(d_0)$ throughout. Taking d_0 smaller puts the whole maneuver strictly inside K_{prep} and $|v| < V_{\mathrm{prep}}/2$. At its endpoint the axial force is strictly negative and the excursion force remains positive. Continuity gives (19.20) on a sufficiently small K_s.

The entry duration may be large, but it is fixed once E_*, d_0 are fixed. The finite-window averaging in Lemma 22, followed by smooth changes of coordinates on its positive energy band, gives an error $C_{\mathrm{entry}}\delta_N^*$. Factors such as $1/E_*$ are finite and are absorbed into this constant. The same argument propagates τ, whose additive origin is chosen so the averaged endpoint has $\tau = 0$. The actual physical clock is controlled by (19.5). Lemma 21 makes the estimate uniform under later completions. No estimate uniform as $E_* \downarrow 0$ after fixing the maneuver is asserted. ∎

The original global extension reserved the broad selected collar $|s| < S_1$ and all cold preparation neighborhoods. The present shorter pulse and its braking segment can be chosen within those same bounds. Its later K_s need not coincide with the old controller's interval J. The objective and its extension do not change.

19.4 An adaptive controller in the internal clock

Choose a control velocity bound $0 < V < V_{\mathrm{prep}}$ and a nonconstant periodic target

$$S(\tau) = s_e + d(1 - \cos(\omega\tau)), \qquad d > 0, \quad \omega > 0, \tag{19.25}$$

whose image lies strictly inside K_s. The parameters d, ω may be reduced, but are then fixed. Put

$$e_s = s - S(\tau), \qquad e_v = v - S'(\tau), \qquad e = (e_s, e_v).$$

For the normalized averaged force prescribe

$$B_{\mathrm{des}} = -av + aEv^3 - ES''(\tau) + E(2e_v + e_s). \tag{19.26}$$

With this force, the averaged equations in σ give exactly

$$e_\sigma = EAe, \qquad A = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix}. \tag{19.27}$$

Indeed $s_\sigma = Ev$, $\tau_\sigma = E$, and the selected velocity equation is $v_\sigma = -av - B_{\mathrm{des}} + aEv^3$.

By (19.20), a sufficiently small V, small target derivatives and a fixed small error neighborhood make B_{des} lie a fixed positive distance inside $[J\beta_-(s), J\beta_+(s)]$. Thus

$$\beta = \frac{B_{\mathrm{des}} - J\beta_-(s)}{J(\beta_+(s) - \beta_-(s))} \tag{19.28}$$

is separated from zero and one. This mixing is implemented by the actual connected branches of Proposition 11.2; membership in a frozen force hull alone would not implement it.

Let

$$Q = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \qquad A^TQ + QA = -I.$$

There is $c_0 > 0$ such that, for all sufficiently small $\lambda > 0$,

$$\|(I + \lambda A)e\|_Q \le (1 - c_0\lambda)\|e\|_Q. \tag{19.29}$$

Here $\|e\|_Q^2 = e^TQe$; the estimate follows by expanding that square and absorbing the $\lambda^2 A^TQA$ term. Decrease V so that, in addition to the force margin,

$$aV^2 < c_0/8, \qquad (\gamma + 4)E_{\max}V^2 < 1/2. \tag{19.30}$$

These choices leave a nonconstant target available.

Lemma 24 (A shrinking tracking tube)**.** *There are $B > 0$ and arbitrarily small fixed $\kappa > 0$ such that the two-block controller with intended mesh*

$$h = \kappa E^2 \tag{19.31}$$

preserves the endpoint bound

$$\|e\|_Q \le B\kappa E \tag{19.32}$$

through the stagewise cold-prefix construction, provided entry errors are sufficiently small and

$$\delta \le \eta\kappa^2 E^4 \tag{19.33}$$

at entry, with a sufficiently small fixed $\eta > 0$. All retained words, and their limiting cold orbit, obey an enlarged bound

$$|e_s| + |e_v| \le C\kappa E \tag{19.34}$$

throughout each word. The selected, velocity and force margins can be reserved for this enlarged bound.

Proof. For a two-block word chosen by (19.28), Lemma 22 and Taylor expansion of S at the actual new τ give

$$e_{\mathrm{new}} = (I + EhA)e + O(h^2 + \delta). \tag{19.35}$$

All target derivatives used here are bounded. The phase coordinate is advanced by the true word; it is not reset when the future code changes. The stabilizing step is $Eh = \kappa E^3$, whereas $h^2/(Eh) = \kappa E$. This determines the energy-dependent mesh.

On every actual retained prefix with $|v| \le V$,

$$\frac{\mathrm{d}}{\mathrm{d}\sigma} \log \frac{\delta}{E^4} = -a[1 - (\gamma + 4)Ev^2] \le -a/2. \tag{19.36}$$

Thus (19.33) improves along each such trajectory. Choose a strict entry version of it uniformly over cold completions. This is possible by Lemmas 21 and 23. Subsequent recodings require no summation of changes in the barrier: each retained genuine prefix starts with the same strict uniform entry margin and satisfies (19.36) separately.

At an endpoint satisfying (19.32), estimates (19.29), (19.35) and (19.33) bound the new error by

$$B\kappa E - (c_0 B - C_1)\kappa^2 E^4, \tag{19.37}$$

where C_1 is independent of E and of small κ. To compare with the new tube radius, note that the exact energy equation and decreasing E give, on the same true word,

$$E_{\mathrm{new}} \ge E - aV^2E^2\Delta\sigma, \qquad \Delta\sigma = h + O(h^2 + \delta).$$

A future rewrite changes r by $O(\delta)$, hence E by $O(E\delta)$. Therefore, in the stagewise endpoint comparison,

$$B\kappa E_{\mathrm{new}} \ge B\kappa E - (aV^2 + o(1))B\kappa^2 E^4 - CB\kappa E\delta. \tag{19.38}$$

The $o(1)$ is uniform for $E \le E_{\max}$ as $\kappa \downarrow 0$. The last term is at most $CB\eta\kappa^3 E^5$. Choose B so that $c_0 B - C_1 > c_0 B/2$, and then reduce κ. Inequality (19.30) makes (19.37) no larger than (19.38). This proves the endpoint induction, including the $O(\delta)$ recoding defect.

The barrier also gives $\delta/h \le \eta\kappa E^2$. Thus, after reducing κ, each intended block has many normalized returns, its rounding error is harmless, and $\Delta\sigma = h + O(h^2)$ lies between $h/2$ and $2h$. On a word, the exact σ derivatives of (s, v, r, τ) are bounded uniformly in the broad collar. Its duration is $O(h)$, so the endpoint-to-interior change in e is $O(h) = O(\kappa E^2)$; the energy remains comparable during the word. This proves (19.34) for finite retained words, with a fixed enlarged constant.

For the limiting code, compare any word directly with the stage that first fixed it. The center and phase change is $O(\delta)$ and the relative energy change is $O(\delta)$ by (19.9). By (19.33), these are smaller than any fixed small fraction of κE after reducing κ. Hence final endpoints obey, for example, $2B\kappa E$, rather than requiring exactly the finite-stage constant B. Enlarging the constant in (19.34) absorbs this slack and the within-word change. Reserve all selected, velocity and force margins for this enlarged tube before choosing κ. A first-exit argument then closes the assumptions $s \in K_s$ and $|v| \le V$ used in the proof, both for retained physical prefixes and for their limit. ∎

19.5 Parameter order and one genuine infinite cold code

We make explicit the order in which the preceding choices are used. The local potential, global extension, broad selected and normal collars, and positive reference amplitude q_* are already fixed by Proposition 18.1. Choose $E_{\max} = 2E_*$, the broad preparation velocity bound and the finite maneuver of Lemma 23. This fixes s_e, E_e, K_s, β_* and its duration. Next choose the smaller V, target S and error neighborhood, leaving strict margins in (19.20), (19.28) and (19.30). Fix the uniform coding constants on the preparation collars, then B and a sufficiently small κ as in Lemma 24.

One further smallness condition on κ will prove energy exhaustion without assuming it. If $T = 2\pi/\omega$ is the target period, put

$$M_2 = \int_0^T |S'(u)|^2 \, \mathrm{d}u > 0. \tag{19.39}$$

Using the enlarged constant C of (19.34), reduce κ so that

$$\eta_v := C\kappa E_{\max}, \qquad T\eta_v^2 < M_2/4. \tag{19.40}$$

This does not require $E \to 0$.

Only now choose the axial residence N. Since $\delta_N^* \to 0$, it can make

$$C_{\mathrm{entry}}\delta_N^* \ll \kappa E_e, \qquad \delta_{\mathrm{entry}} \ll \kappa^2 (E_e/2)^4, \tag{19.41}$$

uniformly under future completions. The actual entry energy lies between $E_e/2$ and $3E_e/2$, its errors fit (19.32), and its barrier has a strict margin. All choices involving E_* and the possibly long finite entry were made before N; no interchange between a small reference energy and a late residence is needed.

At each control stage, complete the current finite prefix by a constant future and take its genuine cold intersection. Evaluate its center and the single propagated phase τ, then compute (19.26), (19.28) and (19.31). Append the two dwell blocks and bounded connecting flights, keeping the entire previous word unchanged. The replacement cold intersection changes its old endpoint only by $O(\delta)$, including r, τ, ℓ, so the estimates of Lemma 24 apply.

The successive prefixes define one infinite code. Their initial transverse cold parameters converge by common-prefix continuity from Lemma 21. For every fixed retained word, compare the infinite code directly with the finite stage that fixed it. Estimate (19.9) controls the entire previous prefix with the slack already allowed in (19.34). No sum over all future rewrites is used. Continuous dependence of the regular cold solution on each fixed physical-time interval, using (18.5), identifies this limit with one genuine cold solution, rather than an arbitrarily prescribed positive-time state.

There is no finite-time exhaustion of the normalizing energy. On every retained control prefix and its limit,

$$(E^{-1})_\sigma = av^2 \le aV^2, \qquad E(\sigma) \ge \frac{1}{E(\sigma_e)^{-1} + aV^2(\sigma - \sigma_e)} > 0 \tag{19.42}$$

on finite σ intervals. In particular r cannot reach $-\infty$ in finite time. Nor can the infinitely many planned words accumulate at a finite σ: otherwise (19.42) would bound E below there, and their increments, at least $\kappa E^2/2$, would have a fixed positive lower bound. Equivalently, the normalized return bounds and (19.14) give $\delta_i \asymp i^{-1}$ and a divergent sum of actual σ increments. Thus $\sigma \to \infty$. Finally (19.42) implies

$$\tau(\sigma) - \tau(\sigma_e) = \int_{\sigma_e}^{\sigma} E(u)\,\mathrm{d}u \longrightarrow \infty. \tag{19.43}$$

The target phase cannot stop at a finite value.

19.6 Exact exhaustion and the interval omega-limit set

We now prove the asserted limiting energy; it has not been assumed in the tracking estimates. For every complete late target period, the inequality $|x + y|^2 \ge |x|^2/2 - |y|^2$, (19.34) and (19.40) give

$$\int_{nT}^{(n+1)T} v^2\,\mathrm{d}\tau \ge \frac{1}{2}\int_{nT}^{(n+1)T} |S'|^2\,\mathrm{d}\tau - \int_{nT}^{(n+1)T} |v - S'|^2\,\mathrm{d}\tau \ge M_2/4. \tag{19.44}$$

The exact energy equation in the internal clock is

$$E_\tau = -aEv^2, \qquad (\log E)_\tau = -av^2. \tag{19.45}$$

Together with (19.43), (19.44) forces $E \to 0$. This is repeated positive dissipation over an unbounded clock, not a choice of an exact total energy budget. The enlarged tracking

tube now gives

$$s - S(\tau) \longrightarrow 0, \qquad v - S'(\tau) \longrightarrow 0, \qquad H = E - \tfrac{1}{2}E^2v^2 = E(1+O(E)) \longrightarrow 0. \tag{19.46}$$

Also $k = Ev \to 0$.

For completeness we derive the two-sided rate. Put $E_n = E(nT)$ on late full periods. Equations (19.44)–(19.45) and $|v| \le V$ give

$$e^{-aV^2T}E_n \le E_{n+1} \le r_0E_n, \qquad r_0 = e^{-aM_2/4} < 1. \tag{19.47}$$

Within each period, E differs from its initial value by at most the fixed factor e^{aV^2T}. Since $\sigma_\tau = 1/E$,

$$\sigma(nT) - \sigma(n_0T) \le C\sum_{j=n_0}^{n-1} E_j^{-1} \le CE_n^{-1}\sum_{j=n_0}^{n-1} r_0^{n-j} \le CE_n^{-1}.$$

This proves the upper reciprocal-σ bound. The lower bound is (19.42). Bounded within-period ratios extend both bounds to every sufficiently large τ, and adjusting constants covers the finite interval from entry. This is (19.2).

The constraint (19.5) and coercivity keep z, w uniformly bounded. Hence

$$Z = E^{1/p}z \longrightarrow 0, \qquad W = E^{1/2}w \longrightarrow 0, \qquad q = t^{-a}Z \longrightarrow 0.$$

The clock τ is continuous, strictly increasing and unbounded. The image and omega-limit set of (19.25) are the full interval $[s_e, s_e + 2d]$. The tracking limit (19.46) therefore proves exactly the omega-limit assertion (19.1).

19.7 Preserving the fixed global objective and physical rates

It remains to verify the original cold and global requirements for this new controller. The fixed early cold prefix and recut residence have the selected bounds from Lemma 15. Lemma 23 uses the same broad selected collar, and the target and enlarged tube lie strictly inside it. Thus all retained physical prefixes and their limiting orbit satisfy $|s| < S_1$ from the original construction.

Lemma 8 applies independently of the controller. While the cold solution lies in the nonnegative valley, it gives

$$E(0) = \frac{a^2|q(0)|^2}{2}, \qquad |W(\rho)| \le a|q(0)|, \qquad |q(t)| \le |q(0)| < Q_0. \tag{19.48}$$

Here $E(0)$ denotes the limit at the cold endpoint. For the last inequality, integrate $Z_\rho = W$ from $Z(0) = 0$ and use $q = Z/(a\rho)$. The reserved radii satisfy $S_1^2 + Q_0^2 < R_1^2 < R_2^2 < S^2$. Thus the complete trajectory stays in B_{R_1}, where the already fixed extension is exactly P. The first-exit argument and uniqueness of (18.5) identify it with the global regular cold orbit of that extension. Its convexity, semialgebraicity, C^2 regularity, coercivity, compact minimizer interval and global gradient Lipschitz bound remain those of Proposition 18.1.

If the extension was finally multiplied by $\lambda > 0$, use the same cold position and $\widetilde{X}(t) = X(\sqrt{\lambda}\,t)$. Let $t' = \sqrt{\lambda}\,t$. The new rescaled normal position is $\widetilde{Z}(t) = \lambda^{-a/2}Z(t')$, while $\widetilde{W}(t) = W(t')$ and $\widetilde{k}(t) = k(t')$. Since $ap = 2$,

$$\lambda P(s, \lambda^{-a/2}Z) = P(s, Z).$$

Thus $\widetilde{E}(t) = E(t')$ exactly. Zero residual energy, the interval omega-limit set and the reciprocal-logarithmic rate are all preserved by the fixed scaling which makes $\mathrm{Lip}(\nabla f_0) \le 1$. This completes the proof of Theorem 11.

Corollary 19.1 (Mechanical energy on the zero-energy drift)**.** *The orbit in Theorem 11 satisfies*

$$\frac{|X_t(t)|^2}{2} + f_0(X(t)) = \Theta\!\left(\frac{1}{t^2 \log t}\right), \qquad f_0(X(t)) = O\!\left(\frac{1}{t^2 \log t}\right). \tag{19.49}$$

Its selected physical length and its time-weighted physical selected-force integral are infinite:

$$\int_{t_e}^{\infty} |s_t(t)|\,\mathrm{d}t = \infty, \qquad \int_{t_e}^{\infty} t\,|\partial_s f_0(X(t))|\,\mathrm{d}t = \infty. \tag{19.50}$$

Proof. Before the fixed objective normalization, (2.10) reads

$$E_{\mathrm{phys}}(t) = t^{-2}\left[E - a\varepsilon Z \cdot W + \frac{a^2}{2}\varepsilon^2 |Z|^2\right].$$

Since $Z = E^{1/p}z$ and $W = E^{1/2}w$, the cross term divided by E is $O(\varepsilon E^{-\gamma}) = O(\delta)$, and the last term divided by E is $O(\delta^2)$. The barrier implies $\delta \to 0$. Therefore

$$E_{\mathrm{phys}}(t) = t^{-2}E(\log t)(1 + o(1)) = \Theta\!\left(\frac{1}{t^2 \log t}\right).$$

The potential is at most $t^{-2}E$, which gives its upper bound. No pointwise lower bound for the oscillating normal potential is asserted. Multiplying the objective by λ multiplies the old physical energy at $\sqrt{\lambda}\,t$ by λ, preserving these orders. Finally this is an excited zero-energy tail, so both divergence assertions follow from Theorem 5. Directly, $\int |k|\,\mathrm{d}\sigma = \int |v|\,\mathrm{d}\tau = \infty$ by (19.44) and bounded v; (6.5) then rules out an absolutely integrable rescaled selected force. Homogeneity gives precisely the weight t in (19.50), as in (4.3). ∎

The time weight in (19.50) is indispensable. The relative force bound and (19.2) give

$$|\partial_s f_0(X(t))| = O\!\left(\frac{1}{t^2 \log t}\right), \qquad \int_{t_e}^{\infty} |\partial_s f_0(X(t))|\,\mathrm{d}t < \infty,$$

although the weighted integral diverges. Also $\tau_\sigma = E$ and (19.2) give, after an additive shift, $\tau = \Theta(\log \sigma) = \Theta(\log\log t)$. The selected oscillation therefore proceeds on a slower unbounded clock even as the instantaneous tangential force tends to zero.

The construction uses branch-specific force means, but it assigns no single canonical action to the switching normal trajectory. Energy-changing adiabatic itineraries and the need for uniform cross-form and return-time control already occur in Gelfreich–Turaev [10]; homogeneous energy normalization also appears in Pereira–Turaev [12]. Those results concern different energy-transport settings. The argument here verifies the shrinking-energy normalization, the autonomous dissipative selected equation, both clocks and the singular cold intersection for this particular convex objective.

20 The structural boundary and remaining questions

The exact energy identity separates the strict and critical regimes. For $\alpha > 1 + 2/p$, normal kinetic dissipation and the homogeneous virial give integrability of the normal potential. Uniform coercivity transfers this to absolute selected-force integrability. The completed-square estimate then gives finite full physical length. These conclusions require actual compact selected confinement and exact homogeneity, but no convexity.

At $\alpha = 1 + 2/p$, normal dissipation disappears from the rescaled total energy. A necessary zero-force barycenter condition remains, but it does not characterize accessible motion. Scalar, multiplicative and separable mode actions force point selection in the jointly convex classes proved here. The phase-volume theorem identifies a broader sufficient structure: a verified differentiable force corrector determines its mean through microcanonical volume differentiation, while joint convexity yields a convex fixed-volume energy. The exact corrected-log identity excludes excited zero-energy tails under that hypothesis, and the integrable-error distance argument gives point convergence even with many effective minimizers. The rotating nonseparable quartic verifies this mechanism beyond fixed-coordinate multiplicative shapes. Excited trajectories selected by these critical mechanisms have infinite normal length despite point convergence.

For zero-residual-energy tails without such a corrector, the normalized energy clock retains a nondegenerate effective description. The joint-measure theorem proves conditional frozen invariance, velocity stationarity and an exact dissipation law with a possible velocity-variance term. Smooth first integrals are concentrated on transported levels and restrict the available force hull. Singleton force means give qualitative closure; in a jointly convex family they force attraction to the effective minimizer set and slower decay than the reciprocal-logarithmic rescaled energy scale. Flat phase-volume minima have identical frozen potentials and degenerate tangential derivatives. Neither qualitative closure nor this rigidity supplies an integrable off-set force or a positive limiting phase volume by itself.

The accessible network permits a different outcome. One fixed global objective in each dimension $3m$ realizes every continuum in a fixed minimizer sub-box from a fixed m-parameter cold family. Zero residual rescaled normal energy does not restore point

selection in general: the same fixed three-dimensional objective F_1 has a genuine regular cold trajectory with a nondegenerate interval omega-limit set, $H \to 0$ and $E(\log t) \asymp 1/\log t$. Its mechanical energy is of order $1/(t^2 \log t)$. Uniform energy-normalized coding, a retained logarithmic energy coordinate, both slow clocks and an energy-dependent mesh allow the construction to reach this asymptotic regime. Its periodic energy-time selected limit rules out any universal single-valued autonomous velocity law in the selected position alone. The general estimate $k = O(H)$ also shows why exact exhaustion is compatible with dissipation: an excited zero-energy tail has infinite selected length and a divergent absolute rescaled selected-force integral.

These statements have distinct quantifiers. The strict-damping theorem applies to every exactly homogeneous confined tail of the stated degree. The general effective law supplies necessary limits of actual trajectories, not realization of arbitrary invariant-measure choices. The corrector theorem assumes a differentiable solution with controlled fixed-state parameter derivatives; equality of invariant means is weaker. The global critical constructions use one sufficiently large finite even degree supplied by the steep-potential argument. The zero-energy result realizes one interval, while the arbitrary-continuum theorem does not prescribe zero residual energy. The rate crossover at $(p+2)/(p-2)$ is not a second convergence threshold. No classification below $1+2/p$, robustness under arbitrary asymptotically homogeneous remainders, or automatic capture into a local valley is proved.

Several questions remain precise. For a general common-mean family, the gap between qualitative closure and a controlled corrector remains open: the arguments do not decide whether every excited confined tail has positive residual energy or whether attraction to a nontrivial effective minimizing set implies point convergence. The rigidity estimate identifies a smaller tangential force near that set but not its summability. For nonunique force means, smooth first-integral transport need not exhaust the accessibility restrictions between invariant components, and the possibility of a genuine positive velocity-variance defect is not settled here. Hyperbolicity is a sufficient source of switching and cold accessibility, but its necessity for persistent drift is not established. Finally, the product construction uses two normal coordinates for each selected coordinate; universal cold realization with fewer normal coordinates is a separate geometric and dynamical problem.

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